Ministry Approved
2013

Compulsory Education Textbook

Mathematics

SHUXUE

Grade 3 Volume 2

北京师范大学出版社

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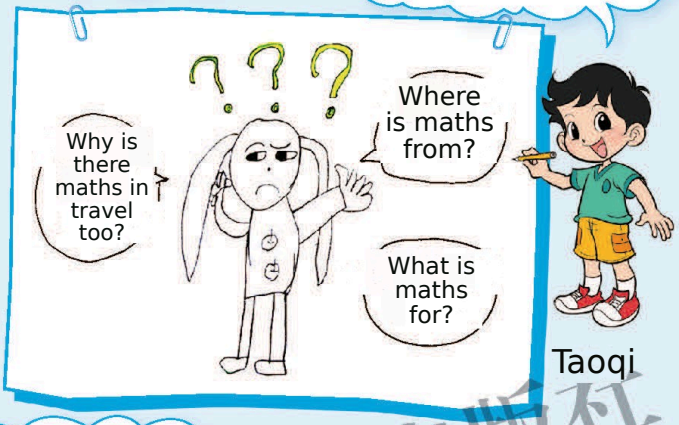
Dear children,

What questions about maths would you like to think about?

Wise Old Man

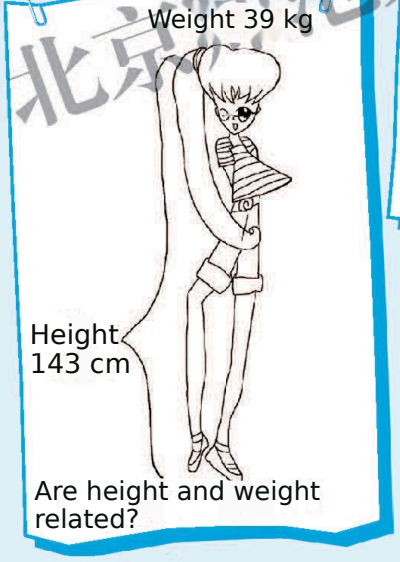
The students asked all kinds of questions...

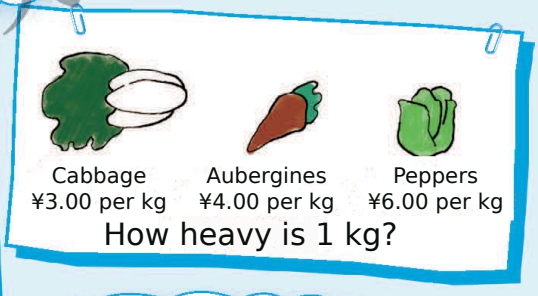


Taoqi

I think maths comes from life, and there are many maths problems in life too.

Xiaoxiao





Please ask a question too.

Clever Dog

Children, I hope you will think actively and find questions of your own. I'm sure you can become a little expert at finding and thinking about questions.

Your friends, the editors

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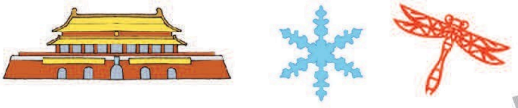

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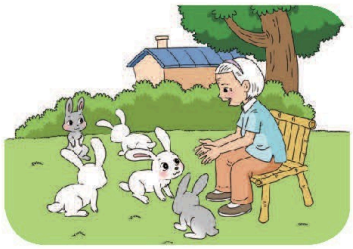

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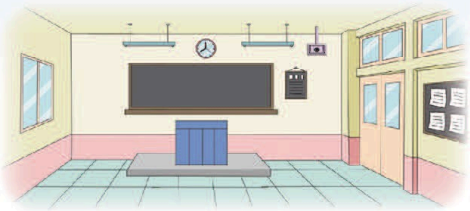

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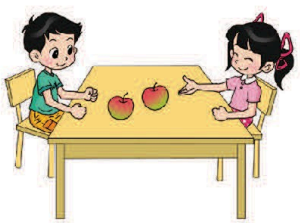

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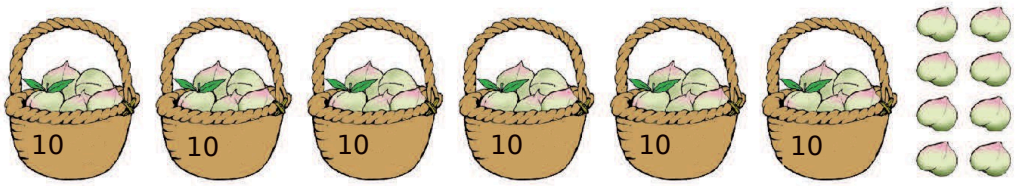
Unit 1 Division

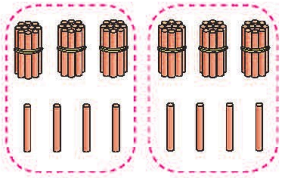
Sharing Peaches



These peaches are shared equally among 2 monkeys. How many does each monkey get? Share them and calculate.

□ ○ □ = □ ( )

Use sticks instead of peaches to share: first share 6 bundles, then...



$60 \div 2 = 30$

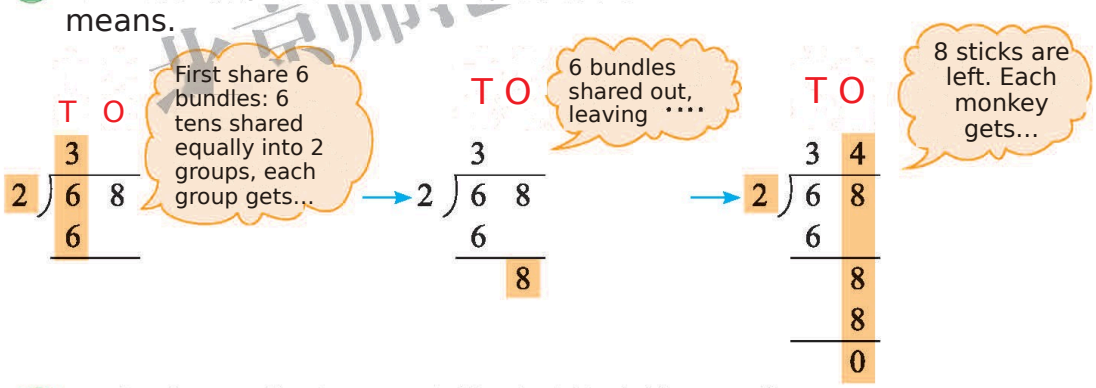
$8 \div 2 = 4$

$30 + 4 = 34$

A number sentence can show the process of sharing.

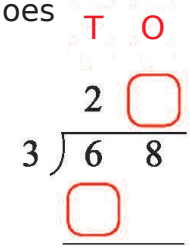
Answer: _____

Look and say what each step of the column method means.



1 more monkey comes along. Now 68 peaches are shared equally among 3 monkeys. How many does each monkey get, and how many are left?

□ ○ □ = □ ( ) ····· □ ( )



Answer: _____

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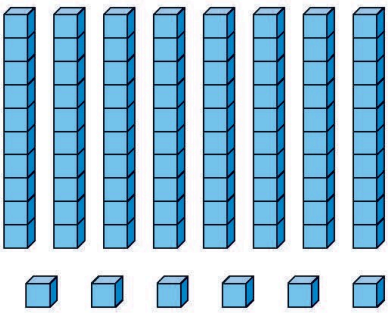
Practice

1. The kindergarten has 86 blocks.



(1) Share these blocks equally between 2 big classes. How many blocks does each class get? Share them and calculate.

(2) Calculate with the column method. Then use how the blocks were shared to explain each step.

(3) If these blocks are shared equally among 4 small classes, how many blocks does each class get, and how many are left?

2. Calculate.

$2\overline{)48}$	$2\overline{)89}$	$3\overline{)69}$	$5\overline{)55}$
$6\overline{)67}$	$2\overline{)84}$	$4\overline{)85}$	$3\overline{)96}$

3. Fill in.

4. Taoqi loves taking photos, and Xiaoxiao helps him organise the album. How many pages can be filled, and how many photos are left?

5. Fill in.

( ) ×2=28	( ) ×3=33
( ) ×2=64	( ) ×4=84
( ) ×8=88	( ) ×7=77

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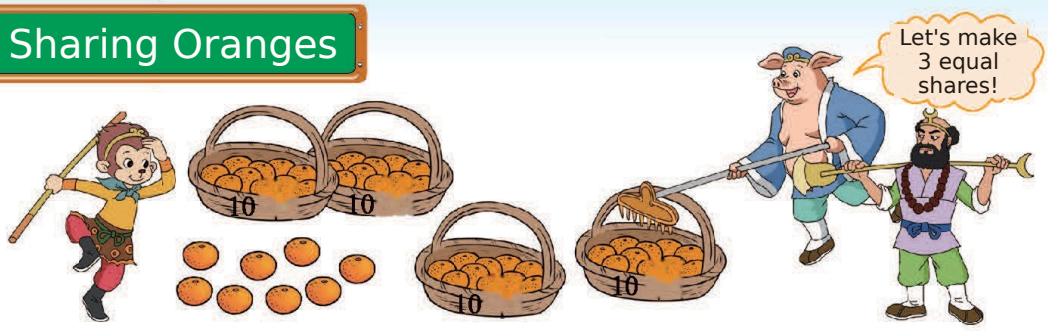
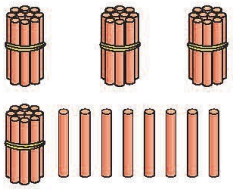
Sharing Oranges



Use sticks for the oranges to help them share and calculate. How many does each person get?

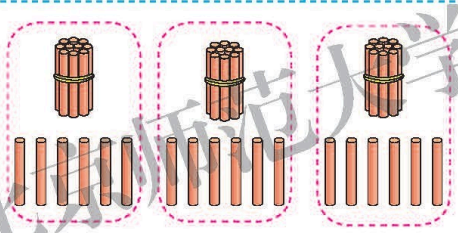
□ ○ □ = □ ( )

First give each person 1 bundle, leaving...



1 bundle has 10 sticks. 18 sticks in all for 3 people, so each person...

Each gets 16 sticks.



Number sentences show the sharing:

30 ÷ 3 = 10
18 ÷ 3 = 6
10 + 6 = 16

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Answer: _____

Can you show the sharing and the result with the column method?

$$3\overline{)48} \rightarrow \begin{array}{r} 1 \\ 3\overline{)48} \\ 3 \\ \hline 1 \end{array} \rightarrow \begin{array}{r} 1 \\ 3\overline{)48} \\ 3 \\ \hline 18 \end{array} \rightarrow \begin{array}{r} 16 \\ 3\overline{)48} \\ 3 \\ \hline 18 \\ 18 \\ \hline 0 \end{array}$$

Use the sharing to explain what each step means.

Calculate and think

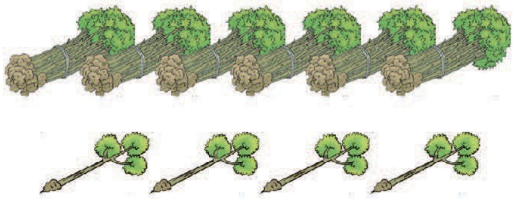
$2\overline{)38}$ $4\overline{)52}$ $3\overline{)72}$

Why do we start dividing from the highest place?

I tried it. Starting from the lowest place is sometimes a lot of trouble!

Practice

1.



(1) 64 saplings are shared equally among 4 groups. How many does each group get? Share them and calculate.

(2) Calculate with the column method, and use the sharing to explain what each step means.

2.

$3\overline{)81}$ $6\overline{)90}$ $7\overline{)98}$ $6\overline{)84}$ $4\overline{)96}$

3.

75 red stars in all, put in ___ equal rows, with ___ stars in each row.

96 orange dots in all, ___ in each row, making ___ rows.

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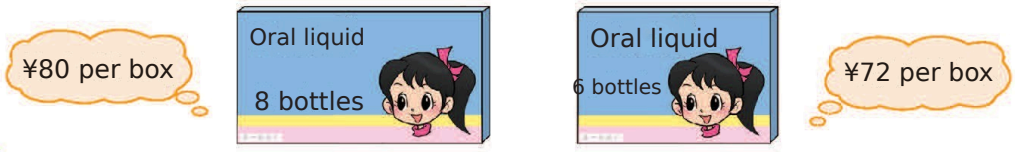
4. Two pharmacies sell the same oral medicine. Which pharmacy is cheaper?

Kangquan Pharmacy Yimin Pharmacy



5. Guess how many beads there might be.

Digits in the Quotient

The railway from Beijing to Siping is 888 km long, and the bullet train takes about 6 hours.

Have you heard of "high-speed rail"? China's high-speed railways hold many "world records". Find out about this new "world champion" and share with a partner!

How many km does it travel per hour on average? First estimate how many digits the quotient has, then calculate.

□○□=□ ( )

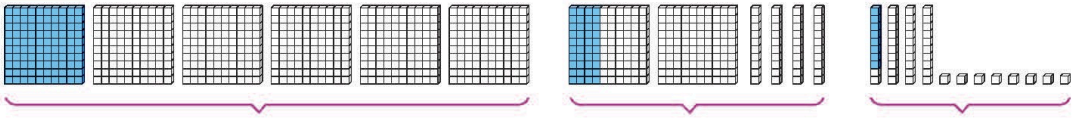
$600 \div 6 = 100$,

888÷6 must be greater than 100, so the quotient is certainly a three-digit number...

$$\begin{array}{r} 148 \\ 6\overline{)888} \\ 6 \\ \hline 28 \\ 24 \\ \hline 48 \\ 48 \\ \hline 0 \end{array}$$

Answer: _____ .

Use the picture below to explain what each step of the column method means.



$600 \div 6 = 100$ $240 \div 6 = 40$ $48 \div 6 = 8$

$$100 + 40 + 8 = 148$$

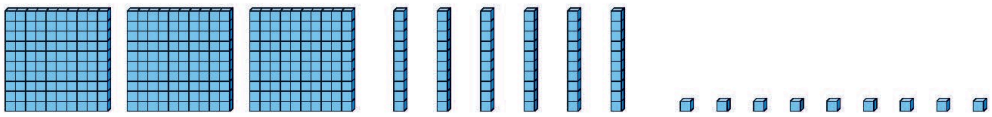
First estimate how many digits the quotient has, then calculate.

$565 \div 5 =$ $456 \div 3 =$ $784 \div 7 =$

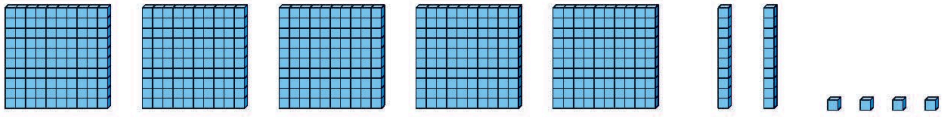
Practice

1. Share out the blocks and use the column method. Use the sharing to explain each step.

$369 \div 3 =$



$524 \div 4 =$



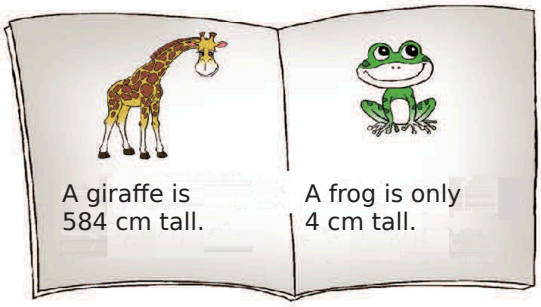
2. First estimate how many digits the quotient has, then calculate.

$348 \div 2 =$	$636 \div 4 =$	$885 \div 5 =$
$868 \div 7 =$	$396 \div 3 =$	$896 \div 8 =$

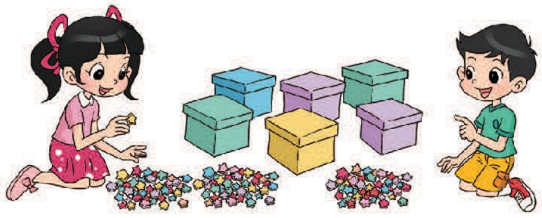
3. (1) How many books per class?

685 books

(2) How many times as tall as the frog is the giraffe?

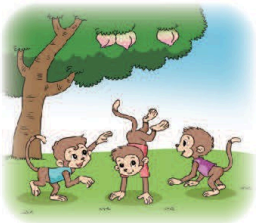


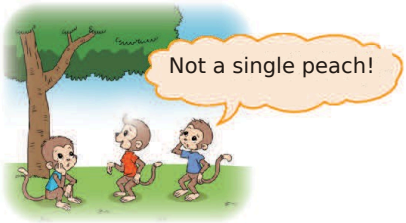
4. Xiaoxiao folded 375 lucky stars and Taoqi folded 309. They want to share the stars equally among 6 gift boxes to give to 6 kindergarten classes. How many stars go in each box?



Monkey Trouble

Talk about it and think

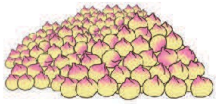




$6 \div 3 = \square$ $3 \div 3 = \square$ $0 \div 3 = \square$

0 divided by any number other than 0 is 0.

How many peaches does each monkey get on average? Explain your thinking.



306 peaches

$\square \bigcirc \square = \square$ ( )

0 divided by 3 equals 0

$300 \div 3 = 100$
$6 \div 3 = 2$
$100 + 2 = 102$

$$\begin{array}{r} 1 \\ 3\overline{)306} \\ 3 \\ \hline 0 \end{array}$$

What now?

$$\begin{array}{r} 102 \\ 3\overline{)306} \\ 3 \\ \hline 06 \\ 6 \\ \hline 0 \end{array}$$

Write it like this:

$$\begin{array}{r} 102 \\ 3\overline{)306} \\ 3 \\ \hline 6 \\ 6 \\ \hline 0 \end{array}$$

Answer: _____ °

Calculate and think

0 divided by 6 equals 0

$$\begin{array}{r} 14 \\ 6\overline{)840} \\ 6 \\ \hline 24 \\ 24 \\ \hline 0 \end{array}$$

What do we do next?

$$\begin{array}{r} 140 \\ 6\overline{)840} \\ 6 \\ \hline 24 \\ 24 \\ \hline 0 \\ 0 \\ \hline 0 \end{array}$$

Write it like this:

$$\begin{array}{r} 140 \\ 6\overline{)840} \\ 6 \\ \hline 24 \\ 24 \\ \hline 0 \end{array}$$

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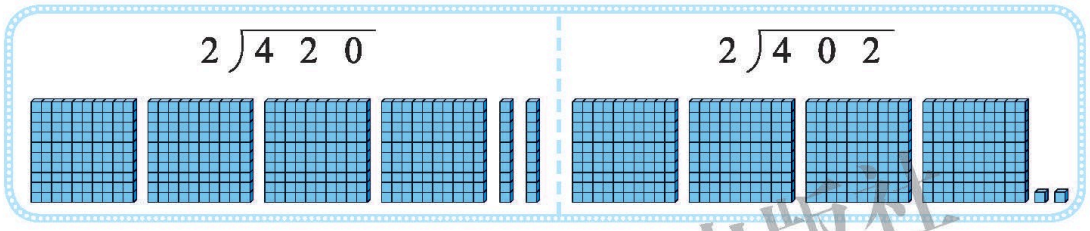
Practice

1. How many fish does each kitten get when shared equally?

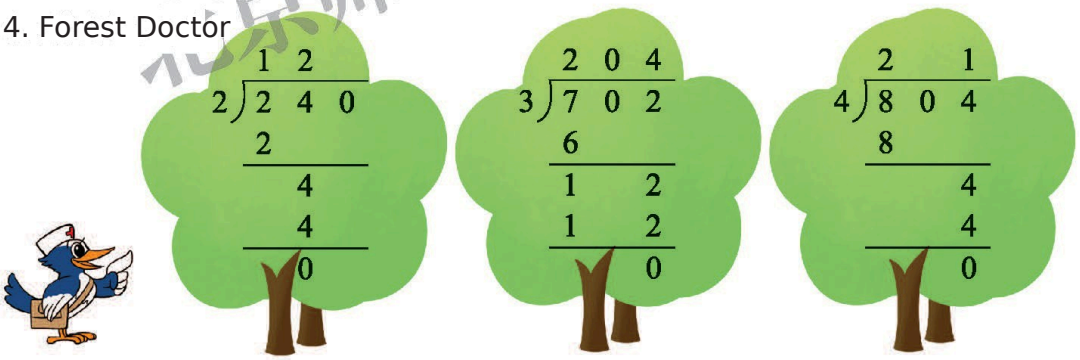
2. Circle and calculate.



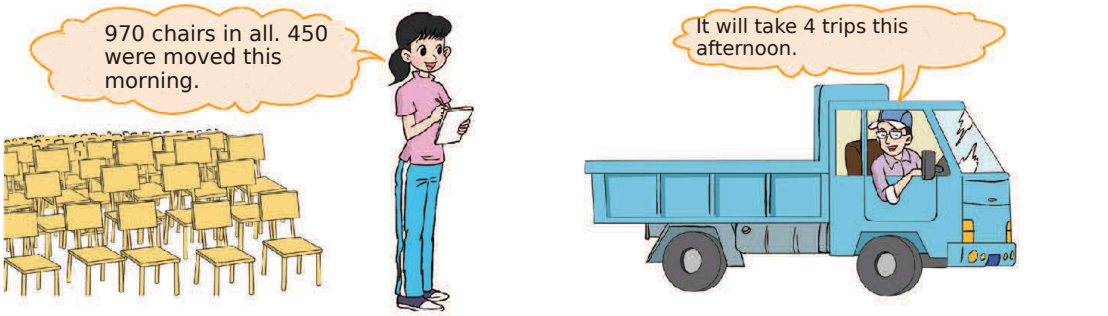
3. Calculate.

420÷3= 880÷4= 780÷6= 309÷3=

4. Forest Doctor



5. On average, how many chairs are moved on each trip in the afternoon?



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Saving

How much did each class make on average? Explain how you calculated.

□○□=□ ( )

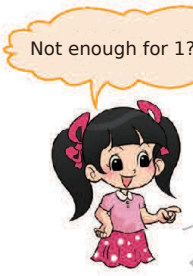
$900 \div 3 = 300$

$12 \div 3 = 4$

$300 + 4 = 304$

$$\begin{array}{r} 3 \\ 3\overline{)912} \\ 9 \\ \hline 1 \end{array}$$

Can you help Taoqi solve the problem?



$$\begin{array}{r} 304 \\ 3\overline{)912} \\ 9 \\ \hline 1 \\ 0 \\ \hline 12 \\ 12 \\ \hline 0 \end{array}$$

Write it like this:

$$\begin{array}{r} 304 \\ 3\overline{)912} \\ 9 \\ \hline 12 \\ 12 \\ \hline 0 \end{array}$$

Answer: _____ 。

The school used the money from selling recyclables to buy 522 pens and gave them to the children in 4 classes of its sister rural primary school. How many pens does each class get on average, and how many are left?

□○□=□ ( ) ····· □ ( )

$$\begin{array}{r} 130 \\ 4\overline{)522} \\ 4 \\ \hline 12 \\ 12 \\ \hline 2 \\ 0 \\ \hline 2 \end{array}$$

Write it like this:

$$\begin{array}{r} 130 \\ 4\overline{)522} \\ 4 \\ \hline 12 \\ 12 \\ \hline 2 \end{array}$$

Answer: _____ 。

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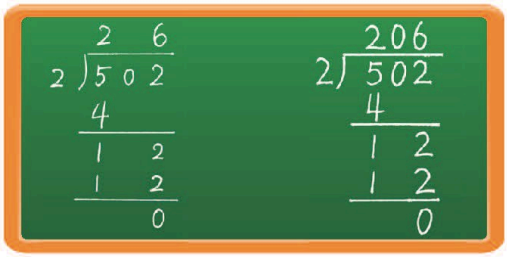
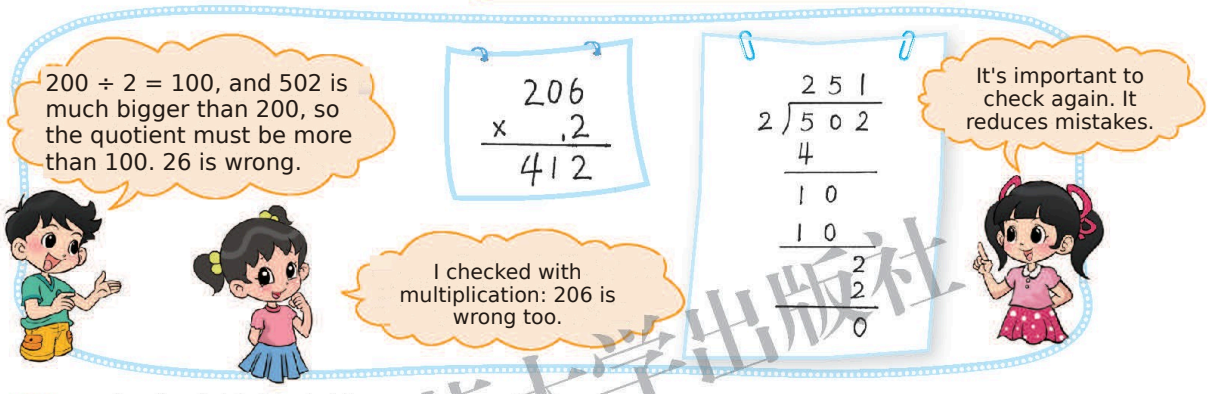
Try It

Are their calculations correct? Think and explain.

$502 \div 2 = \square$





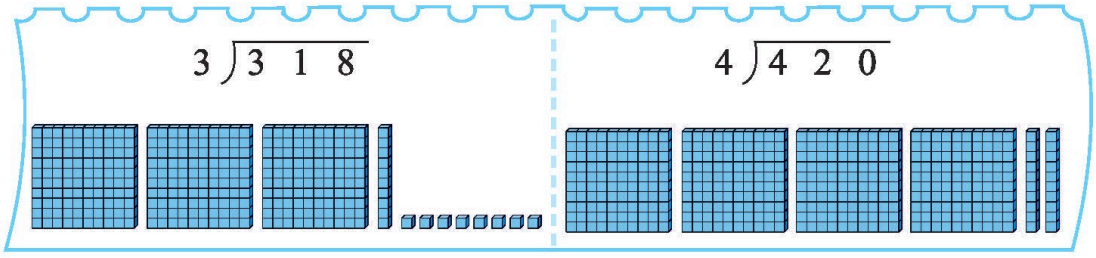
Use the column method to calculate, then check.

$315 \div 3 =$ $960 \div 8 =$ $805 \div 5 =$

Talk with a partner about what each step of the column method means.

Practice

1. Circle and calculate.



2. Lele's family took 412 photos on their trip and put them equally into 4 albums. How many photos are in each album?

How did you work it out? Tell a partner.

3.

$624 \div 6 =$ $704 \div 5 =$ $846 \div 7 =$

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4. Forest Doctor

$$\begin{array}{r} 34 \\ 2\overline{)680} \\ 6 \\ \hline 8 \\ 8 \\ \hline 0 \end{array} \qquad \begin{array}{r} 24 \\ 4\overline{)816} \\ 8 \\ \hline 16 \\ 16 \\ \hline 0 \end{array} \qquad \begin{array}{r} 101 \\ 7\overline{)727} \\ 7 \\ \hline 7 \\ 7 \\ \hline 0 \end{array}$$

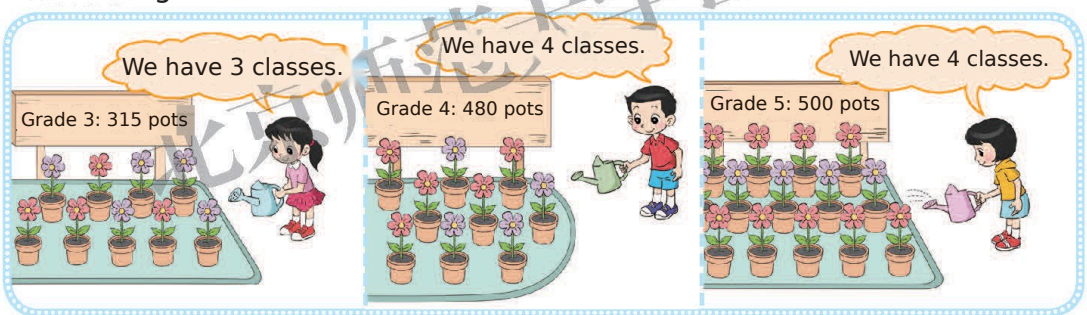
5. Calculate first, then check.

$621 \div 3 =$	$432 \div 4 =$	$814 \div 2 =$
$905 \div 5 =$	$630 \div 6 =$	$604 \div 4 =$

6. Fill in.

Dividend	606	832	720		
Divisor	6	8	3	6	4
Quotient				105	190

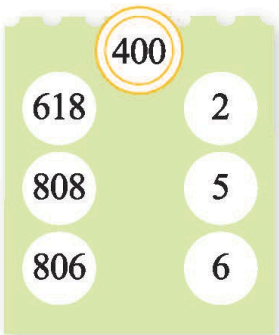
7. Watering Flowers

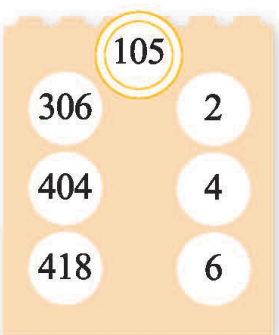


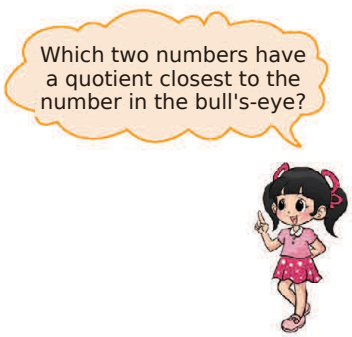
(1) On average, how many pots does each Grade 3 class water?

(2) Ask another division question and try to answer it.

8. Estimate, calculate, and match.







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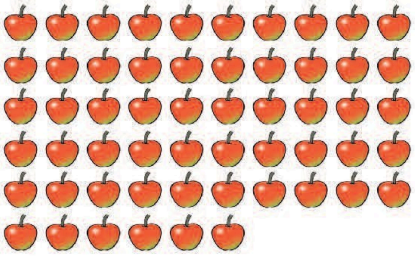
Exercise 1

1. (1) How many apples does each person get on average? Circle, then write a number sentence.



(2) 216 pine cones are put equally into 2 baskets. How many go in each basket? Circle first, then write a number sentence.

2. Fill in the largest possible number.

$(\quad)\times 6<58$	$(\quad)\times 7<50$	$5\times(\quad)<26$
$9\times(\quad)<65$	$(\quad)\times 8<70$	$(\quad)\times 7<34$

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3. First estimate how many digits the quotient has, then calculate.

$72\div 3=$	$402\div 2=$	$812\div 4=$
$456\div 4=$	$612\div 2=$	$525\div 5=$

4. Calculate and check.

$734\div 2=$	$636\div 6=$	$565\div 5=$	$780\div 6=$

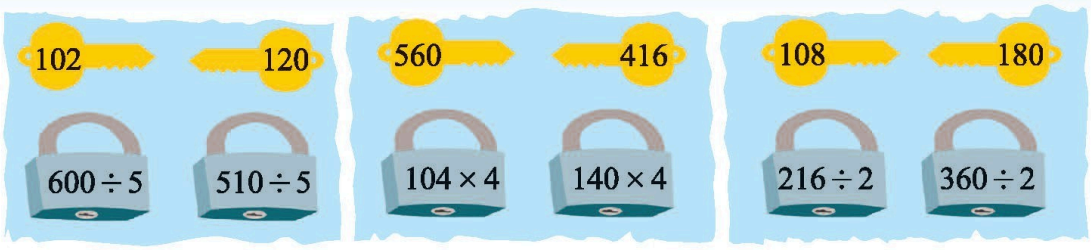
5. Forest Doctor

$$\begin{array}{r} 1\ 7 \\ 5\overline{)\,8\ 0\ 5} \\ 5 \\ \hline 3\ 5 \\ 3\ 5 \\ \hline 0 \end{array}$$

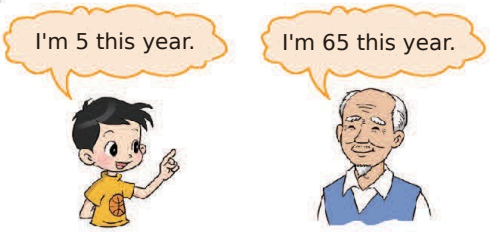
$$\begin{array}{r} 1\ 2\ 0 \\ 3\overline{)\,3\ 7\ 0} \\ 3 \\ \hline 7 \\ 6 \\ \hline 1 \end{array}$$

$$\begin{array}{r} 1\ 5\ 7 \\ 6\overline{)\,9\ 4\ 9} \\ 6 \\ \hline 3\ 4 \\ 3\ 0 \\ \hline 4\ 9 \\ 4\ 2 \\ \hline 7 \end{array}$$

6. Unlocking Game



7.



(1) This year, 's age is how many times 's age?

(2) Next year, 's age is how many times 's age?

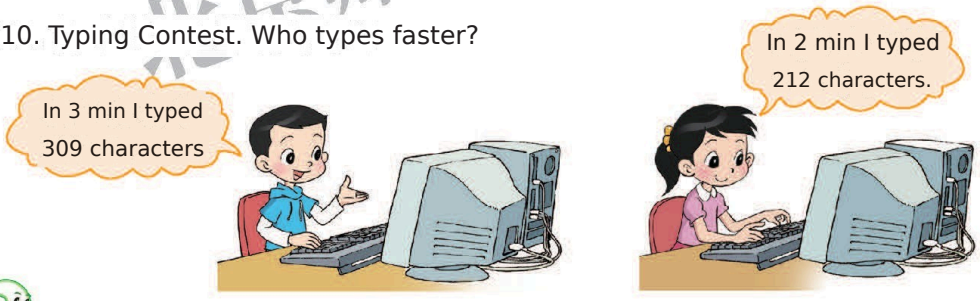
8. 540 pupils from Chunfeng Primary School are in a group gymnastics display, in 4 formations. On average, how many pupils are in each formation? What if there are 5 formations?

9. The PE monitor accidentally stained the purchase record. Can you work out how much each tube of shuttlecocks costs?

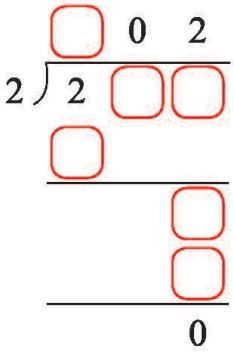
Basketballs	¥65 each	1 ball
Shuttlecocks	¥? per tube	3 tubes

¥137 in total.

10. Typing Contest. Who types faster?



11 Think and fill in.



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Stamps

With 5 stamps on each page, estimate about how many pages these stamps will fill.

Fewer than 300 stamps, so surely fewer than 60 pages are needed.

200 stamps need 40 pages, so 285 stamps need more than 40 pages.

Calculate how many pages can be filled.

□○□=□ ( )

Not enough hundreds for a 1, so write 0.

$$\begin{array}{r} 057 \\ 5\overline{)285} \\ 0 \\ \hline 28 \\ 25 \\ \hline 35 \\ 35 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 57 \\ 5\overline{)285} \\ 25 \\ \hline 35 \\ 35 \\ \hline 0 \end{array}$$

If the hundreds aren't enough for a 1, look at the first two digits. You can write it like this.

Oh, they really mean the same thing!

Answer: _____ .

With 8 stamps per page, how many full pages? How many stamps left?

□○□=□ ( ) ····· □ ( )

$$8\overline{)285}$$

Using the situation, explain how to check a division with a remainder.

Check:

□□ × 8 + □ = □□□

Answer: _____ .

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Practice

1.

(1) Estimate: about how many skipping ropes can be cut from this rope?

(2) How many skipping ropes can be cut?

(3) If each rope is 5 m long, how many can be cut and how many metres are left? Write a number sentence, calculate and check.

2. Taoqi is a military fan. He has 144 different rocket launch pictures. He plans to show them to his classmates. Each display board can hold 9 pictures.

(1) How many display boards are needed?

(2) Using the situation, explain what each step means.

3. First estimate how many digits the quotient has, then calculate.

$355 \div 5 =$	$164 \div 4 =$	$522 \div 6 =$	$217 \div 7 =$
$156 \div 4 =$	$846 \div 6 =$	$364 \div 7 =$	$906 \div 6 =$

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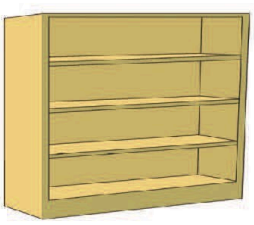
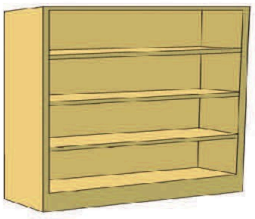
4. In ◯ write ">", "<" or "=".

$96 \div 8$ ◯ $96 \div 6$ $120 \div 4$ ◯ $240 \div 8$

$90 \div 5$ ◯ $80 \div 5$ $156 \div 3$ ◯ $745 \div 5$

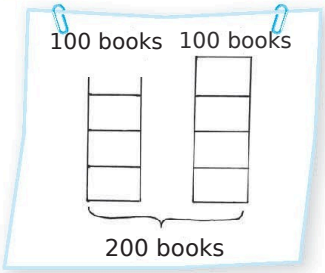
5. Ask a maths question and try to answer it.

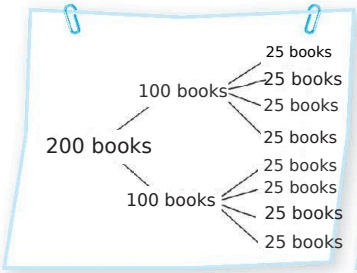
New Books

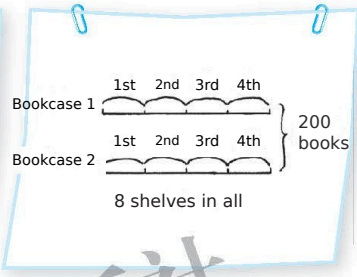



200 books

On average, how many books go on each shelf of each bookcase? Draw a diagram first.







Write a number sentence and calculate, then talk about it with a partner.

First I find how many books go on each bookcase, then...

$200 \div 2 = 100$ (books)

$100 \div 4 = 25$ (books)

$200 \div 2 \div 4$

$= \boxed{100} \div 4$

$= \boxed{25}$ (books)

I first find how many shelves the two bookcases have in all, then...

$200 \div (2 \times 4)$

$= 200 \div \boxed{8}$

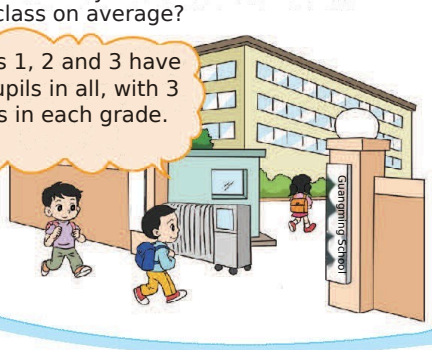
$= ____$ (books)

Answer: _____。

First write a number sentence and calculate, then explain your thinking.

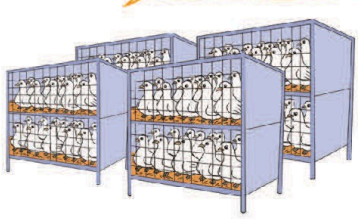
How many students are there in each class on average?

Grades 1, 2 and 3 have 270 pupils in all, with 3 classes in each grade.



On average, how many pigeons are on each level of each cage?

96 pigeons in all.



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Practice

1. The library bought 4 sets of the Encyclopaedia for ¥384 in total. On average, how much does each book cost? Draw a diagram first, then write a number sentence and calculate.

8 books per set

2.

(1) How many cases? Circle and calculate.

6 balls per box

4 boxes per case

(2) How many per team on average?

736 players in all.

The "Youth Cup" football tournament has 8 groups, and each group has 4 teams.

3. Calculate.

$756 \div 7 \div 6$ $\quad$ $624 \div 2 \div 3$

$801 \div (3 \times 3)$ $\quad$ $624 \div (2 \times 3)$

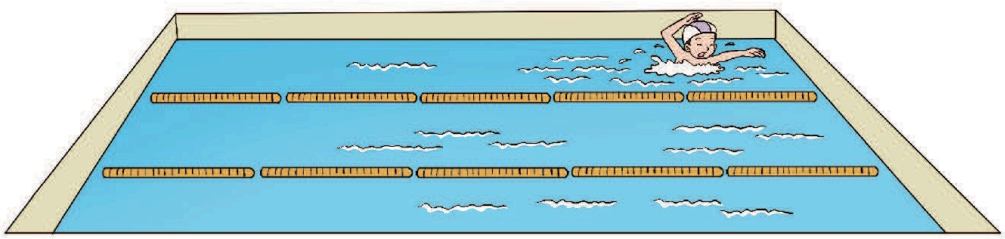
4. Join up using the rule.

48×2 $\quad$ 32×5 $\quad$ $96 \div 3$

$160 \div 2$ $\quad$ $240 \div 5$ $\quad$ 80×3

The answer to each problem is the first number of the next one!

5. Qisi went swimming. He swam to the end of the lane and back 3 times, 150 m in total. How long is the lane of this pool?



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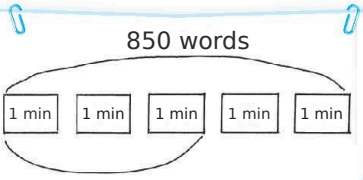
Story Time

In Chinese class, students take turns telling stories. Taoqi chose a story of 850 words.

Practising before class took me 5 min.

The class agreed each story should take no more than 3 min.

How many words can Taoqi tell in 3 min? Find a way to show the information in the problem.



5 min	850 words
3 min	? words

Can you write a number sentence to solve the problem?

First find how many words he can tell in 1 min, then...

850÷5=170 (words)

170×3=510 (words)

Another number sentence:

850 ÷ 5 × 3

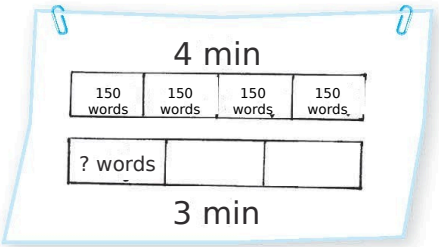
= 170 × 3

=

Taoqi could cut some of the story, or tell it faster.

Answer: _____ .

Xiaoxiao took 4 min to tell a story, 150 words per minute on average. To finish in 3 min, how many words should she tell per minute?



150 × 4 ÷ 3

= 600 ÷ 3

=

First work out how many words there are in total, then...

Answer: _____ .

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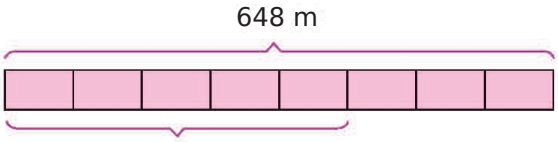
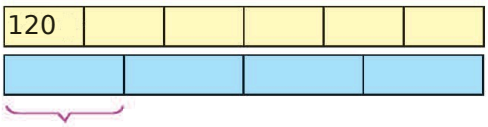
Practice

1. Discuss, then write a number sentence and solve.





2. Draw and calculate.

(1) A toy factory made 630 toy cars in 3 days. At this rate, how many can it make in 5 days?

(2) A toy factory has to make a batch of toy planes. It plans to make 180 planes a day and finish in 5 days. In fact the job took only 4 days. On average, how many planes were made each day?

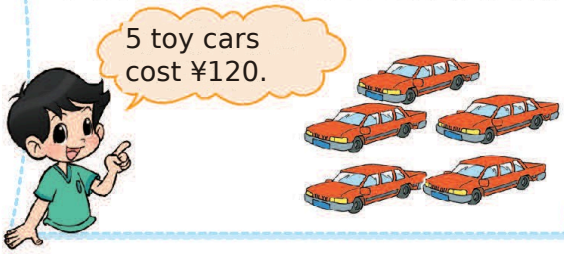
3.

$279 \div 3 \div 3$ $\quad$ $735 \div 5 \div 7$

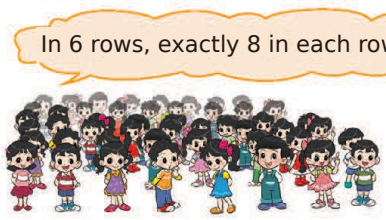
$960 \div 6 \times 4$ $\quad$ $800 \div (2 \times 4)$

4.

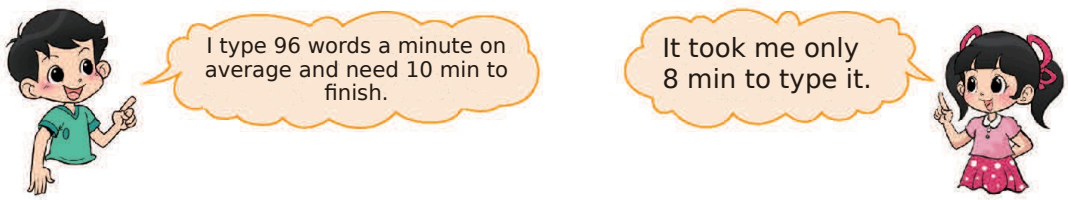
(1) How much do 9 of these toy cars cost?



(2) In 4 rows, how many are in each row?



5. They typed the same text. How many words a minute did Xiaoxiao type on average?



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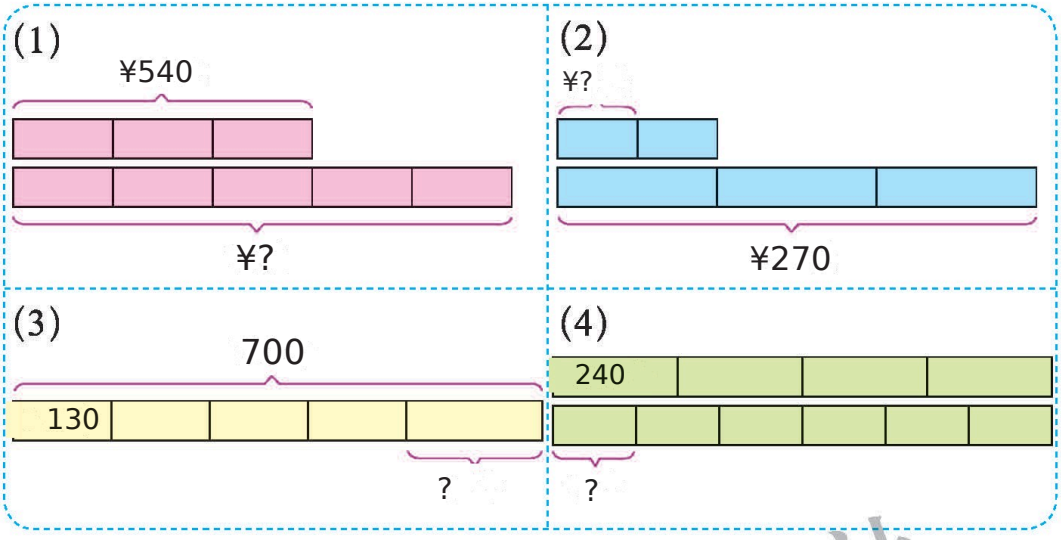
Exercise 2

1. Talk about it, then write a number sentence and calculate.



2. First estimate how many digits the quotient has, then calculate.

$582 \div 4 =$	$330 \div 8 =$	$411 \div 6 =$
$725 \div 6 =$	$424 \div 7 =$	$914 \div 3 =$
$515 \div 5 =$	$960 \div 6 =$	$243 \div 3 =$

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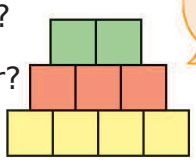
3. In the picture, 3 small red squares stand for 120.

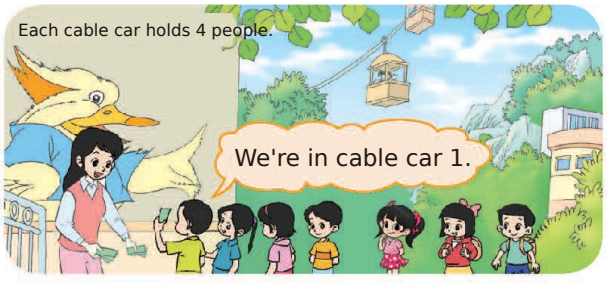
What do the 2 green squares stand for?

What do the 4 yellow squares stand for?



Each small square stands for the same number.

4.



(1) Xiaoxiao is number 5 in line. She should take cable car number ( ).

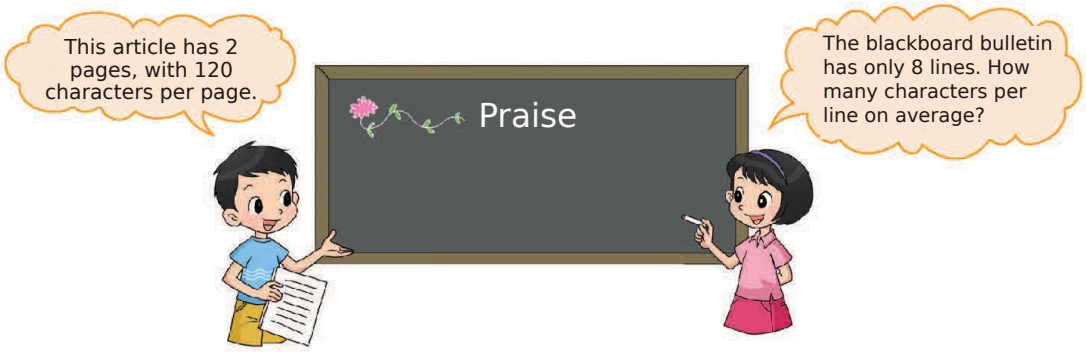
(2) Qisi is number 15 in line. He should take cable car number ( ).

(3) Taoqi is number 52 in line. He should take cable car number ( ).

5.

$721 \div 7 \times 5$	$198 \div 9 \div 2$	$16 \times 8 \div 4$
$60 + 124 \div 2$	$200 - 700 \div 5$	$720 \div (3 \times 2)$

6.



7. The table shows the number of pupils in each grade at Xinhua Primary School.

Grade	1	2	3	4	5	6
Pupils	95	100	98	105	97	99

(1) Each grade takes 9 buses to the Natural History Museum. How many pupils are on each bus on average?

(2) The museum allows at most 200 visitors per group. How should the groups be arranged?

8. Buy prizes with ¥100.

(1) How much do 10 pens cost?

(2) How many ballpoint pens can you buy with the money left?

(3) With ¥100 for prizes, what else could you buy? Write two buying plans.

Unit 2 Motion of Figures

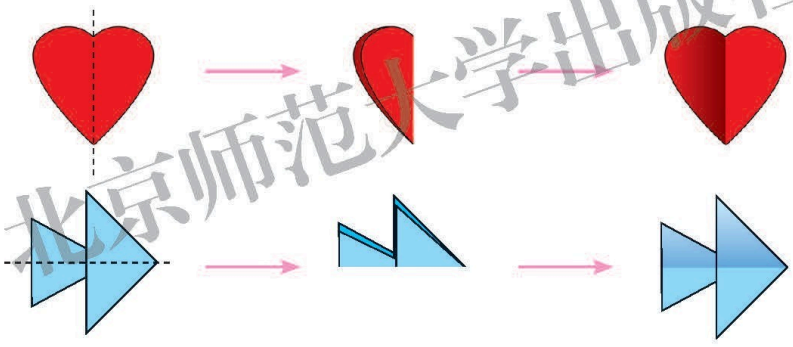
Line Symmetry (1)

Look at these figures. What do they have in common?

Both halves of each figure are the same.

How do you know both halves are the same?

Fold and look at Figure 1 on Appendix page 1.



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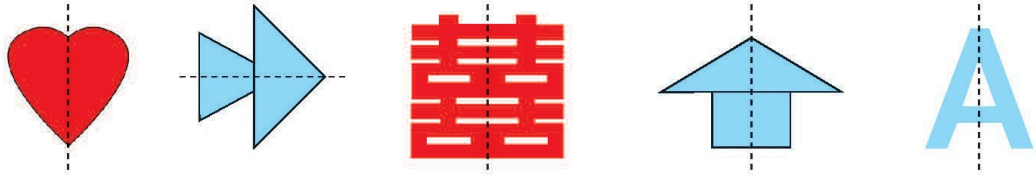
Look and say.



Line of symmetry

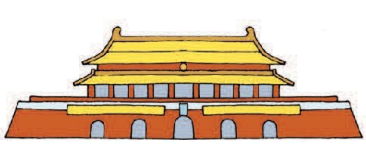
Figures like these are line-symmetric.

They match when folded in half.

How can we find the line of symmetry of a line-symmetric figure?

Practice

1. Fold and look at Figure 2 on Appendix page 1. Which of the figures below are line-symmetric?



2. Which figures below are line-symmetric? Tick (✓) them.

3. Find out which of these digits, Chinese characters and letters are line-symmetric.

0 1 2 3 4 5 6 7 8 9 10

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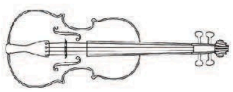
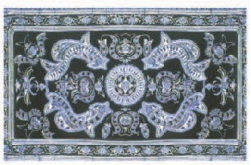
4. Fold in half to find the lines of symmetry of the figures below (Figure 3 on Appendix page 1).



5. Many patterns in daily life are line-symmetric. Find some and talk about them with a partner.

Ginkgo leaf

Cable-stayed bridge


Bouyei batik

Beijing Opera mask

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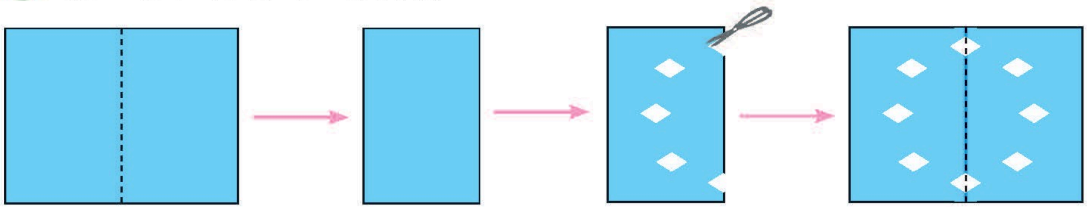
Line Symmetry (2)

Try it. What do you notice?

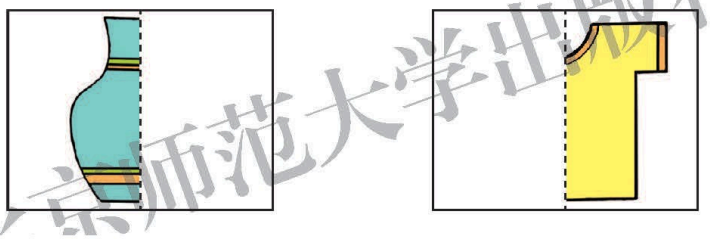


Fold in half　　Cut out a pattern　　Unfold

To make a line-symmetric figure, first fold the paper in half.

After folding, you only need to make half of the figure...

Each figure below is half of a line-symmetric figure. What is the whole figure? Try it with Figure 4 on Appendix page 1.



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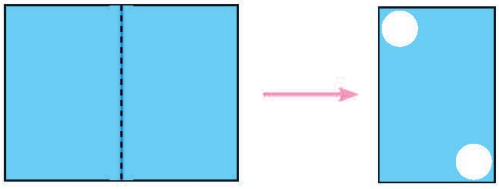
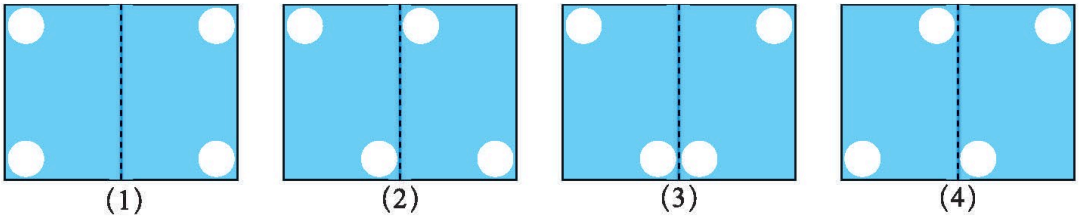
Fold a sheet of paper in half and cut out two circles. Which one do you get when you unfold it? Think, then try it.





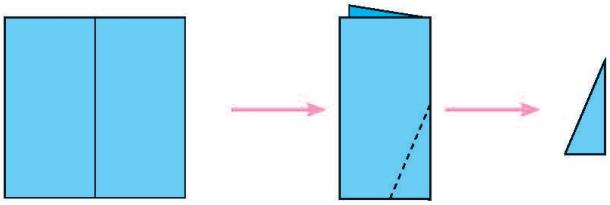
(1)　(2)　(3)　(4)

The unfolded figure should be line-symmetric.

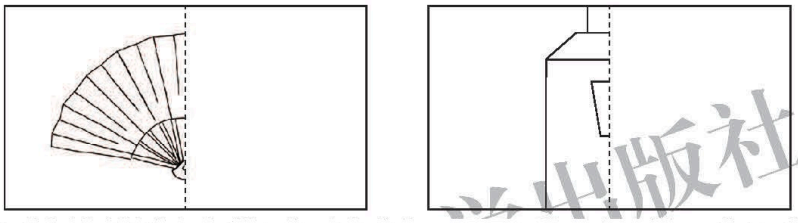
The lower circle is near the line of symmetry, and the upper circle is far from it...

Practice

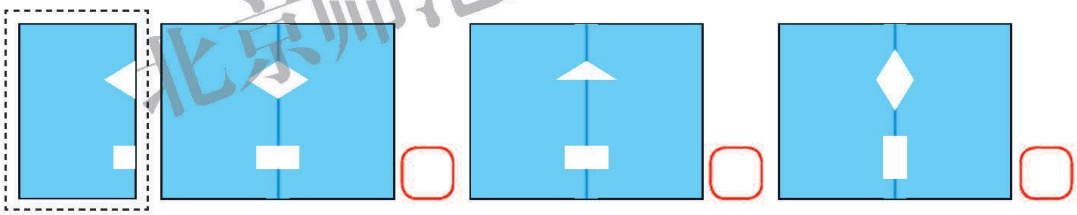
1. As shown, fold a square sheet of paper in half and cut off a triangle. What figure do you get when you unfold it? Try it and see.



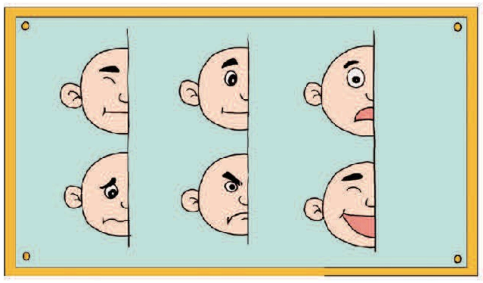
2. Each figure below is half of a line-symmetric figure. What is the whole figure? Try it with Figure 1 on Appendix page 2.



3. Taoqi cut two holes in a folded sheet of paper. Which one will you get when you unfold it? Think, try it, and tick (✓) the answer.



4. In the children's gallery, half of each face is drawn. Guess which face is happy and which is angry.



5. Below is Xiaoxiao's paper cutting. Make a line-symmetric paper cutting of your own and show it in the class display.

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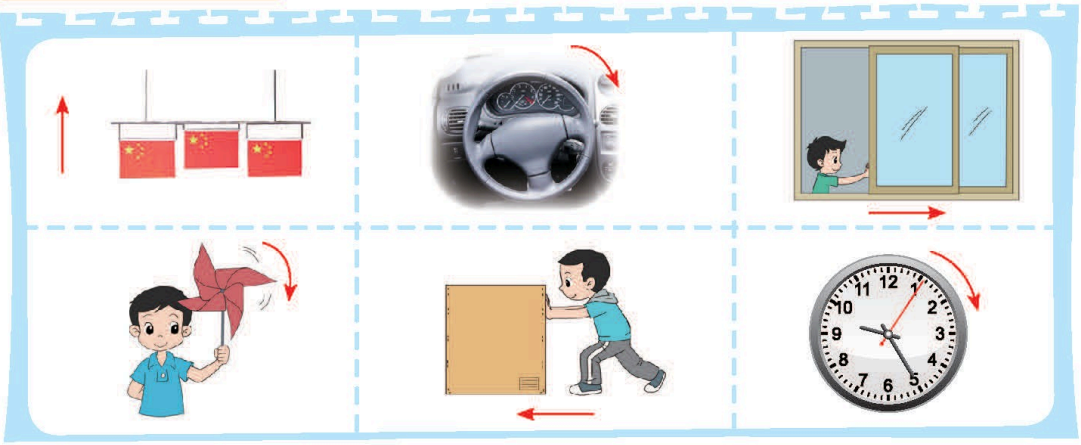
Translation and Rotation



How does each one move? Can you sort them into two groups by how they move?

The steering wheel, pinwheel and clock hands are all rotating.

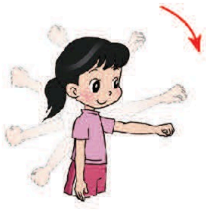
The flags, sliding window and box are all...

Let's learn.

These are examples of translation

These are examples of rotation

Try making a movement that shows translation or rotation.



What other examples of translation and rotation have you seen? Talk about them with your partner.

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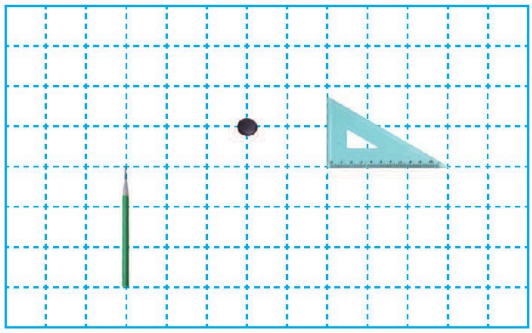
Try It

Move and trace.

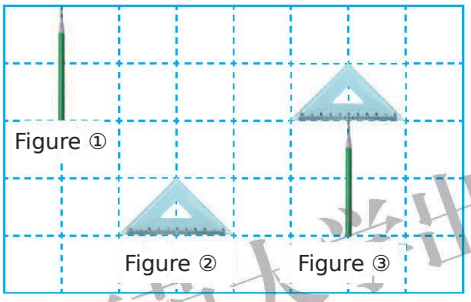
Move the counter down 4 squares.

Move the pencil right 3 squares.

Move the set square left 2 squares.

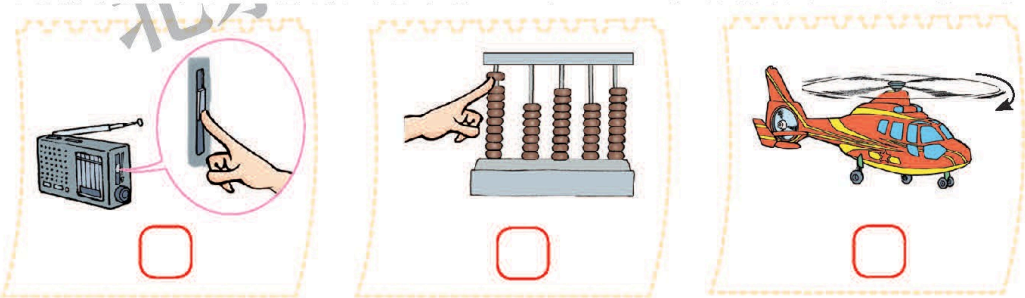


Say how the pencil and the set square can be translated to their places in Figure ③.



Practice

1. How does each object move? Draw "△" for translation and "○" for rotation.



2. Colour in.

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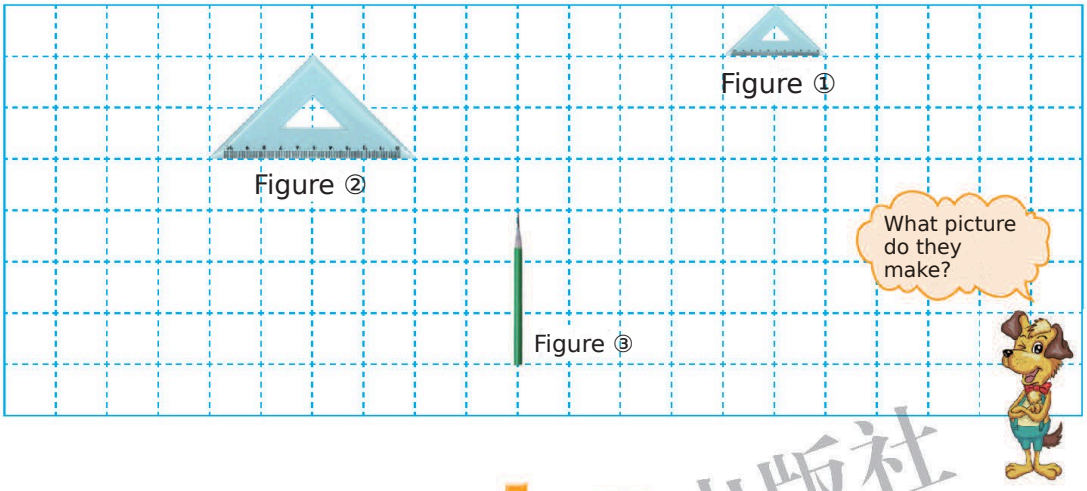
3. Move and trace.

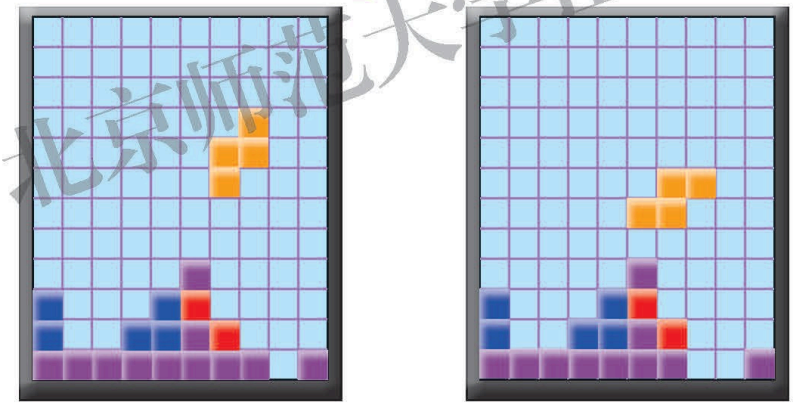
(1) Translate Figure ① left 5 squares;

(2) Translate Figure ② right 4 squares;

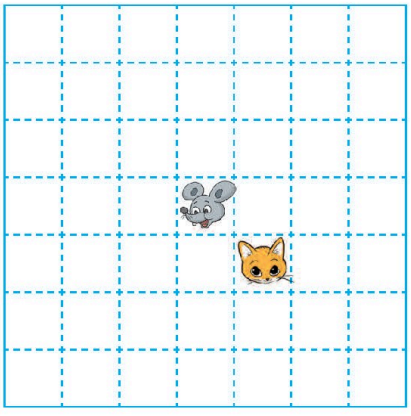
(3) Translate Figure ③ up 1 square.



4. To fill the bottom row, say how and each need to be translated.



5. On the right is an animal board game. Toss a coin to decide who goes first. Each turn you may translate only 1 square. See if the cat can catch the mouse. Cut out Figure 2 on Appendix page 2 and play with a partner.



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Unit 3 Multiplication

Patterns

Calculate and say how you calculated.

$5 \times 1 =$	$3 \times 2 =$	$12 \times 4 =$
$5 \times 10 =$	$3 \times 20 =$	$12 \times 40 =$
$50 \times 10 =$	$30 \times 20 =$	$120 \times 40 =$

50×10 is 10 groups of 50, which is 500

120×40 is...

Look at the number sentences above. What do you notice?

I notice the two rows below are related to the first row.

$12 \times 4 = 48$
$12 \times 40 = 480$
$120 \times 40 = 4800$

Using what you found, write more number sentences and work them out.

$6 \times 3 =$ $15 \times 4 =$ $18 \times 2 =$

Using $16 \times 3 = 48$, can you write the answers below straight away?

$16 \times 30 =$ $160 \times 3 =$

$160 \times 30 =$ $16 \times 300 =$

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Practice

1. Calculate, then write another group of number sentences like these.

$3 \times 8 =$ $30 \times 8 =$ $30 \times 80 =$	$17 \times 5 =$ $17 \times 50 =$ $170 \times 50 =$	$6 \times 9 =$

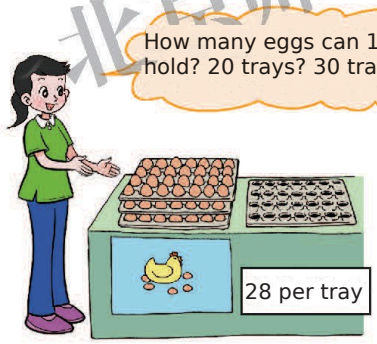
2. Using $24 \times 20 = 480$, write the answers below straight away.

$240 \times 20 =$ $240 \times 2 =$ $24 \times 200 =$

3. Match the number sentences with the same result.

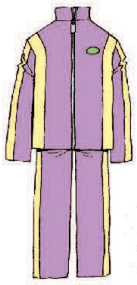
70×22	70×220	16×200	16×20
700×22	7×220	160×2	160×20

4.



Trays	1	10	20	30
Eggs				

5. Buying Tracksuits



Large tracksuit: ¥70 each

Small tracksuit: ¥60 each

(1) The gymnastics team needs 13 large and 16 small tracksuits. How much will each size cost in total?

(2) Teacher Zhang brought ¥2000 to buy the tracksuits. Is that enough?

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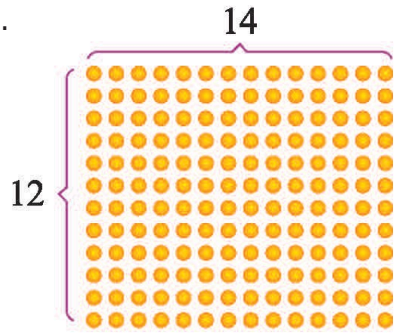
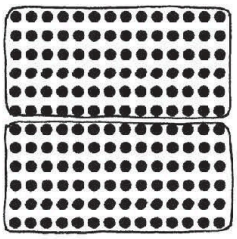
Formation Drill (1)

The school is holding a formation drill. There are 12 rows in total, with 14 students in each row.

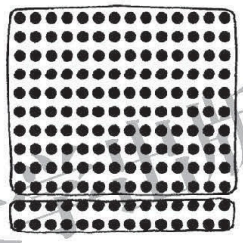


How many students take part in the formation drill? Circle and calculate.

□ ○ □ = □ ( )



14×12
$= 14 \times 6 \times 2$
$= 84 \times 2$
$= 168$



$14 \times 10 = 140$
$14 \times 2 = 28$
$140 + 28 = 168$

$10 \times 10 = 100$
$10 \times 4 = 40$
$10 \times 2 = 20$
$2 \times 4 = 8$
$100 + 40 + 20 + 8 = 168$

Answer: _____

Can you understand the method below? Explain it using the dot diagram.

14×12

×	10	4
10	100	40
2	20	8

$100 + 40 + 20 + 8 = 168$

Calculate.

15×11 23×12

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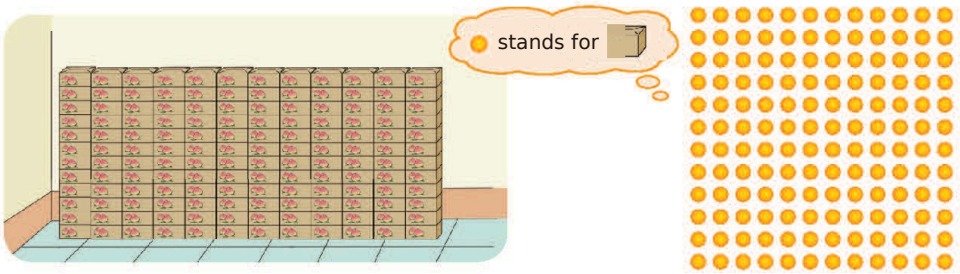
Practice

1. How many boxes of fruit are there in total? Circle and calculate.



2. Fill in and calculate.

23×13

×	20	3
10		
3		

14×11

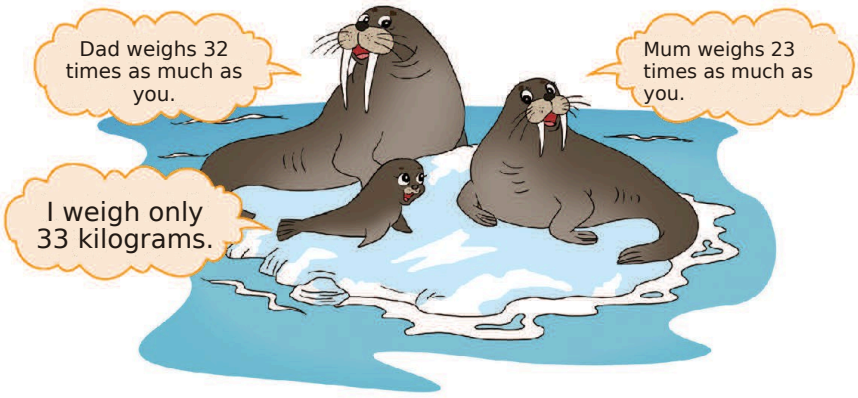
×	10	4
10		
1		

3. Calculate.

11×11　　12×11　　13×13　　12×13

4. How much altogether?

5. How many kilograms do Father Walrus and Mother Walrus each weigh?



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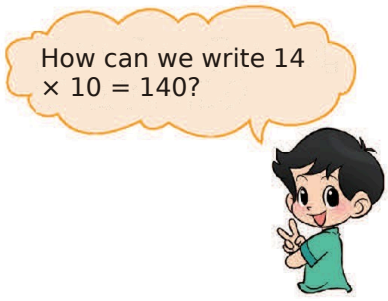
Formation Drill (2)

Can you work out 14×12 using the column method?

$$\begin{array}{r} 1\ 4 \\ \times\ 1\ 2 \\ \hline 2\ 8 \end{array}$$



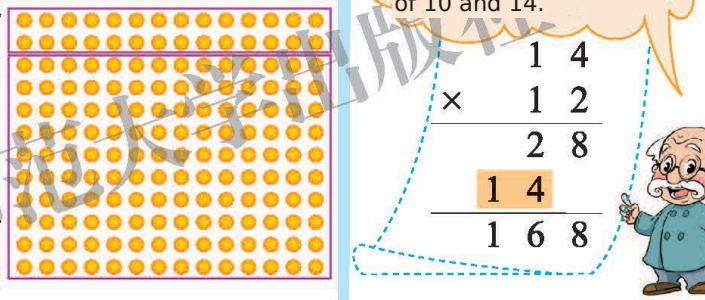
Look, think and explain each step of the column method.

$$\begin{array}{r} 1\ 4 \\ \times\ 1\ 2 \\ \hline 2\ 8 \\ +\ 1\ 4\ 0 \\ \hline 1\ 6\ 8 \end{array}$$

14×2

14×10

28+140



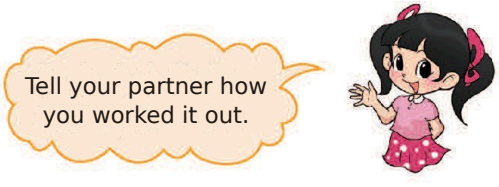
$$\begin{array}{r} 1\ 4 \\ \times\ 1\ 2 \\ \hline 2\ 8 \\ 1\ 4\ \ \\ \hline 1\ 6\ 8 \end{array}$$

Calculate.

$$\begin{array}{r} 2\ 1 \\ \times\ 2\ 3 \\ \hline \end{array}$$

$$\begin{array}{r} 3\ 4 \\ \times\ 1\ 2 \\ \hline \end{array}$$

$$\begin{array}{r} 3\ 3 \\ \times\ 1\ 3 \\ \hline \end{array}$$



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Practice

1. The students are doing a gymnastics display, with 13 in each row and 12 rows. How many students are there in total?

(1) Use the column method.

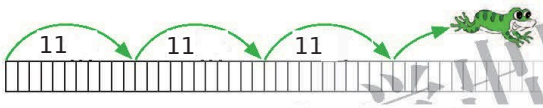
(2) Circle, and explain each step of the column method.

2.

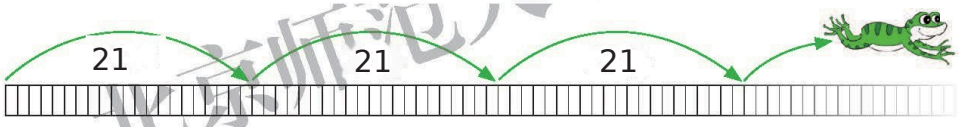
3 2	3 4	2 2	2 1
× 1 3	× 2 1	× 4 3	× 2 1

3. Count and calculate.

(1) The little frog jumps 11 squares each time. How many squares in 18 jumps?

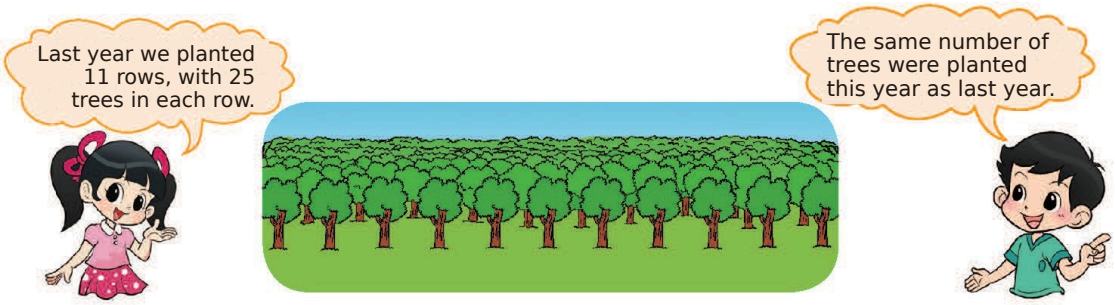


(2) The big frog jumps 21 squares each time. How many squares in 14 jumps?

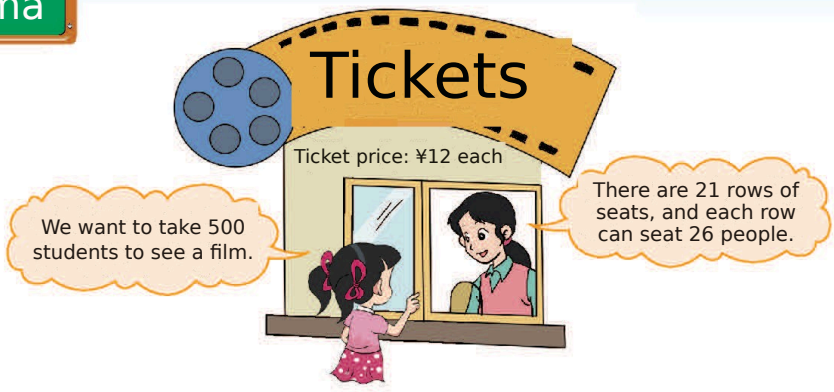


4. 32 teams take part in the football World Cup, with 23 players in each team. How many players are there in total?

5. How many trees are there in the “Young Pioneers Forest” now?



Cinema



Estimate. Are there enough seats in the cinema?

Treat 21 rows as 20 rows; with 26 people in each row, it can seat 520 people. That is enough.

21→20
26→25
20×25=500. That's enough.

Calculate. How many seats does the cinema have?

□○□=□ ( )

20 x 26 = 520
1 x 26 = 26
520 + 26 = 546

×	20	1
20	400	20
6	120	6

400+120+20+6=546

Use the column method...

$$\begin{array}{r} 26 \\ \times\ 21 \\ \hline 26 \\ 520 \\ \hline 546 \end{array}$$

Answer: _____

Class 3(1) has 38 students. A film ticket costs ¥12. How much do the tickets cost?

□○□=□ ( )

I estimate it is more than ¥380 but less than ¥480...

$$\begin{array}{r} 38 \\ \times\ 12 \\ \hline 76 \\ 38_{1} \\ \hline 456 \end{array}$$

Answer: _____

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Practice

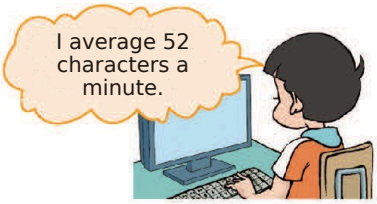
1. Guangming School has 15 classes, with an average of 38 students per class.

   (1) Does the school have 600 students? How did you estimate?

   (2) Calculate how many students the school has.

2. Can he finish typing this text in 30 minutes?

1495 characters



3. Calculate.

$32\times24=$	$26\times51=$	$42\times26=$	$74\times14=$
$18\times23=$	$28\times12=$	$38\times25=$	$56\times24=$

4. Class 3(2) has 42 students. Each buys 1 schoolbag and 1 pencil case.

¥24

¥13

(1) How much for pencil cases?

(2) How much for schoolbags?

(3) How much in total?

There are many ways to multiply. Look at the calculations below. Can you understand them?

In Taiwan, China, a "window" is used to record the result of mental calculation.

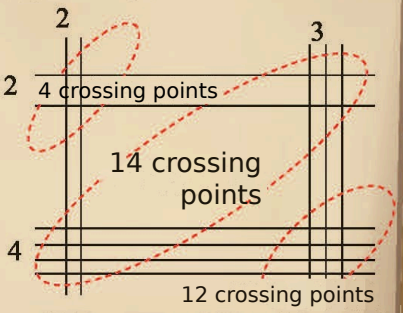
$$\begin{array}{r} 24 \\ \times\ 23 \\ \hline 12 \\ 60 \\ 80 \\ 400 \\ + \\ \hline 552 \end{array}$$

window

Ancient Indians calculated from the highest place.

$$\begin{array}{r} 24 \\ \times\ 23 \\ \hline 4 \\ 8 \\ 6 \\ +\ 12 \\ \hline 552 \end{array}$$

You can also multiply by drawing lines, as in 24×23.



Result: 400+140+12=552

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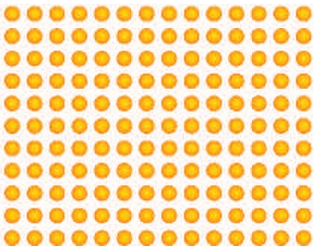
Exercise 3

1. Calculate and explain.

11×14



16×18

×	10	6
10		
8		

2. Given $12 \times 2 = 24$, write the answers below directly.

$12 \times 20 =$ $120 \times 2 =$ $120 \times 20 =$ $12 \times 200 =$

3. Explain each step of the column methods below in the ( ).

$$\begin{array}{r} 2\ 6 \\ \times\ 1\ 4 \\ \hline 1\ 0\ 4 \\ 2\ 6 \\ \hline 3\ 6\ 4 \end{array}$$

$1\ 0\ 4$ …… $(26 \times 4 = 104)$

$2\ 6$ …… ( )

$3\ 6\ 4$ …… ( )

$$\begin{array}{r} 4\ 8 \\ \times\ 3\ 7 \\ \hline 3\ 3\ 6 \\ 1\ 4\ 4 \\ \hline 1\ 7\ 7\ 6 \end{array}$$

$3\ 3\ 6$ …… ( )

$1\ 4\ 4$ …… ( )

$1\ 7\ 7\ 6$ …… ( )

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4. Choose numbers to fill in the blanks.

10 20 30 40 50 60 70 80

( ) × ( ) = 800 ( ) × ( ) = 1200

( ) × ( ) = 3500 ( ) × ( ) = 4200

( ) × ( ) = 2800 ( ) × ( ) = 2400

5. Use the column method to calculate.

36×43 19×23 65×15 77×17

53×18 28×32 69×12 34×49

6. Are these calculations correct? Estimate and explain.

$54 \times 24 = 4526$ ( ) $37 \times 84 = 318$ ( )

$12 \times 35 = 320$ ( ) $35 \times 72 = 315$ ( )

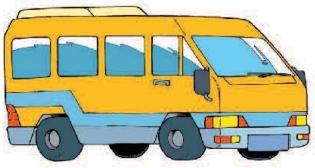
7. Forest Doctor

$$\begin{array}{r} 43 \\ \times\ 15 \\ \hline 205 \\ 43 \\ \hline 635 \end{array}$$

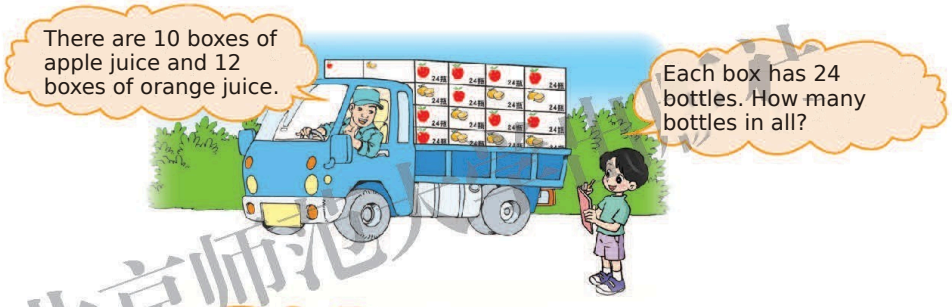
$$\begin{array}{r} 14 \\ \times\ 26 \\ \hline 84 \\ 28 \\ \hline 112 \end{array}$$

$$\begin{array}{r} 24 \\ \times\ 34 \\ \hline 96 \\ 72 \\ \hline 716 \end{array}$$

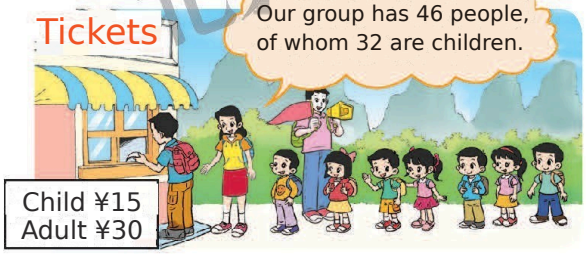
8. How many seats can 35 buses like this provide?



9.



10.



(1) How much are the children's tickets?

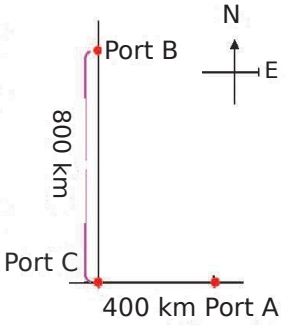
(2) How much are the adult tickets?

(3) How much do they spend in total?

11. Each ship travels about 36 km per hour.

(1) The ship Huoju sets off from Port C and travels east for 12 hours. Is it east or west of Port A? How many kilometres is it from Port A?

(2) The ship Hongqi sets off from Port C and travels north for 18 hours. About where is it? Mark it with a dot. How many kilometres is it from Port B now?



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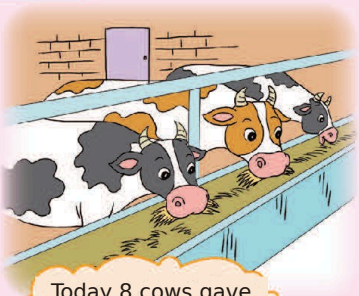
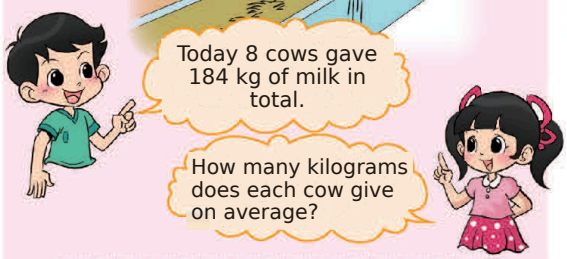
Review

What I Learned

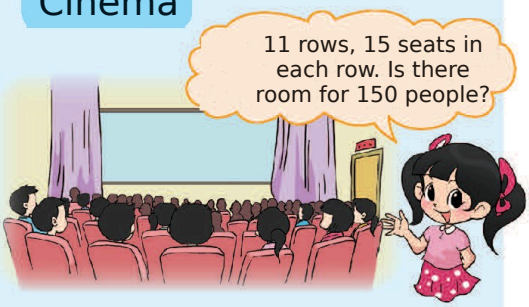
Milk Yield

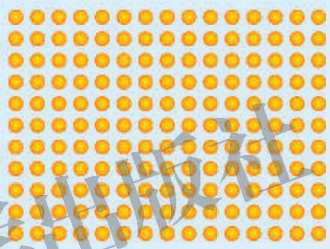




How many digits does the quotient of 184÷8 have? Can you work it out with the column method? Explain each step.

Cinema

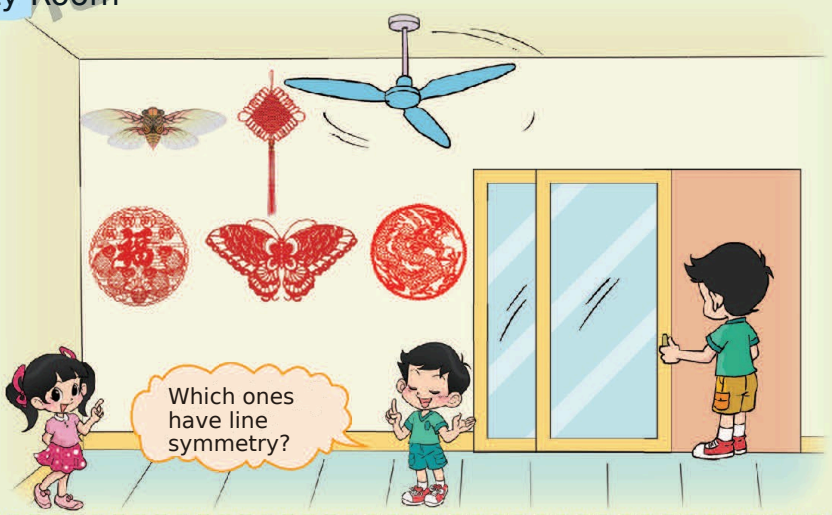




What does 15×11 mean?

Circle and calculate.

Activity Room



Look at the picture above. What kind of figure has line symmetry?

Where do you see translation and rotation in everyday life?

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My Progress

I know that many beautiful patterns have line symmetry.

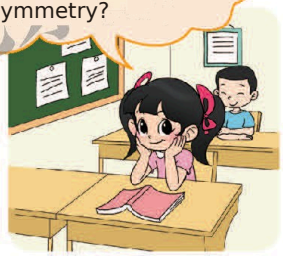
Estimate first, then calculate. I make far fewer mistakes!

I like solving problems by drawing pictures best, because pictures are so concrete.

I like using dot diagrams to multiply. They show me what each step of the column method means.

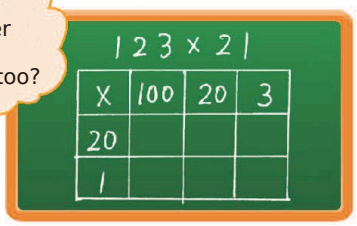
My Questions

A parallelogram looks so perfect. Why doesn't it have line symmetry?



Besides translation and rotation, how else can figures move?

Can I multiply a three-digit number by a two-digit number this way too?



Why is the column method for division different from that for addition, subtraction and multiplication?

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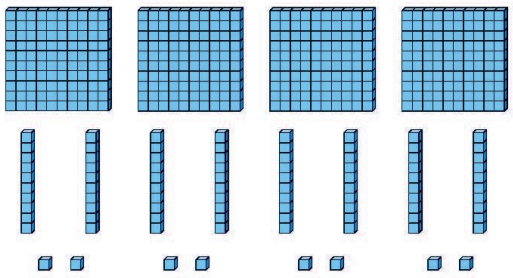
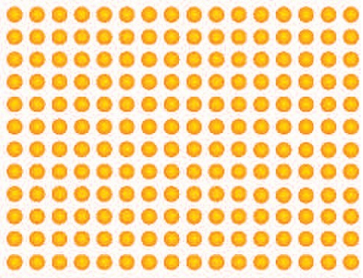
Consolidate and Apply

1. Circle and calculate.

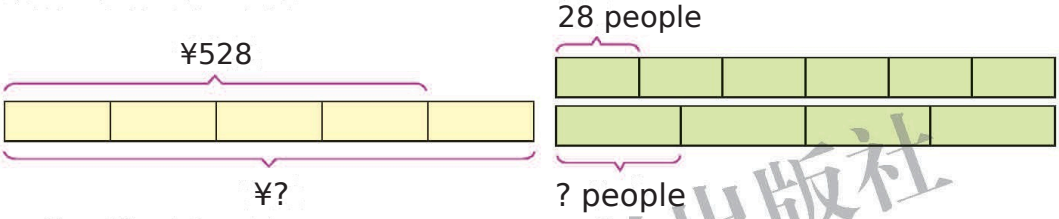
488 ÷ 4



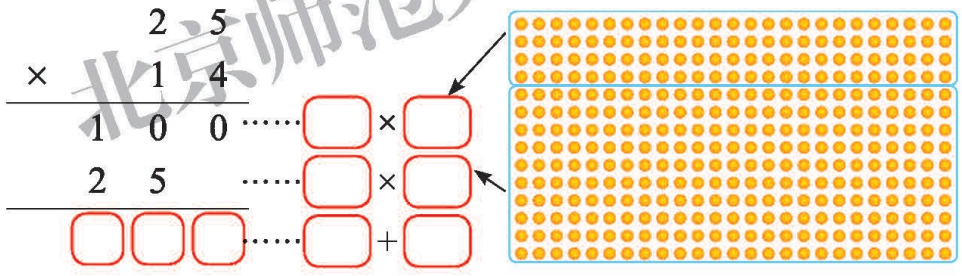
16 × 12



2. Talk about it and calculate.



3. Think and fill in.



$$\begin{array}{r} 25 \\ \times\ 14 \\ \hline 100 \cdots\cdots \square \times \square \\ 25 \cdots\cdots \square \times \square \\ \hline \square\square\square \cdots\cdots \square + \square \end{array}$$

4.

21 × 30	20 × 16	99 × 10	18 × 40
19 × 80	60 × 7	30 × 33	32 × 10

5.

84 ÷ 7	942 ÷ 3	560 ÷ 5	306 ÷ 3
603 ÷ 3	635 ÷ 9	912 ÷ 8	416 ÷ 4
15 × 28	16 × 62	76 × 14	24 × 77

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6.

$106 \times 2 \times 5$	$105 \times 2 \div 5$	$816 \div 4 \div 2$
$702 \div 2 \div 3$	$118 \times 2 \div 4$	$7 \times 280 \div 4$

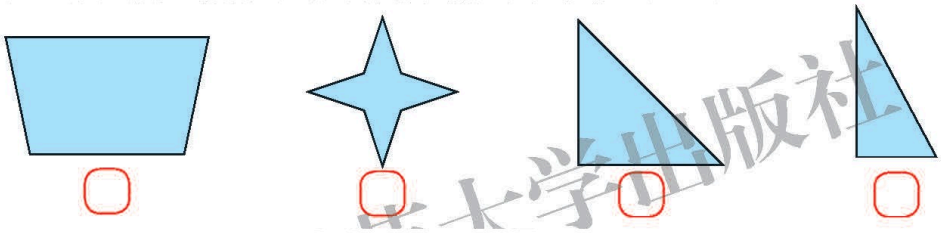
7. Forest Doctor

$$\begin{array}{r} 12 \\ 4\overline{)408} \\ -\ 4 \\ \hline 8 \\ -\ 8 \\ \hline 0 \end{array}$$

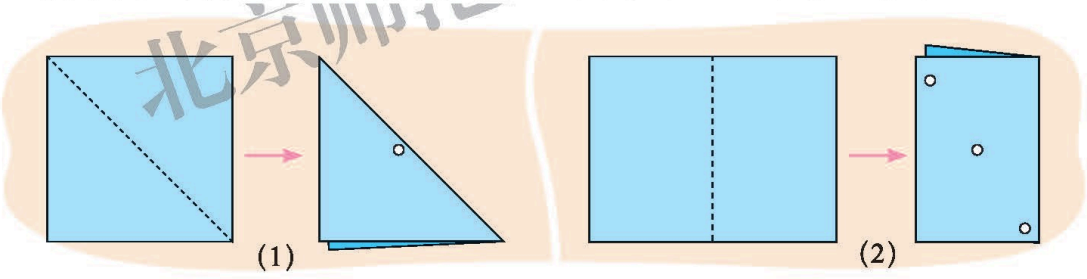
$$\begin{array}{r} 13 \\ 4\overline{)522} \\ -\ 4 \\ \hline 12 \\ -\ 12 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 24 \\ \times\ 29 \\ \hline 216 \\ 48 \\ \hline 264 \end{array}$$

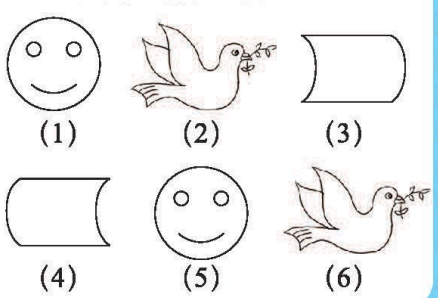
8. Look and think. Put a ✓ under each figure with line symmetry.



9. Follow the steps below. What figure do you get after unfolding? Think and try.



10. Which figures fit onto each other after a translation?



11. A group exercise display by 225 people uses different formations.

(1) In 3 rows, how many in each row?

(2) In 5 rows, how many in each row?

(3) In 9 rows, how many in each row?

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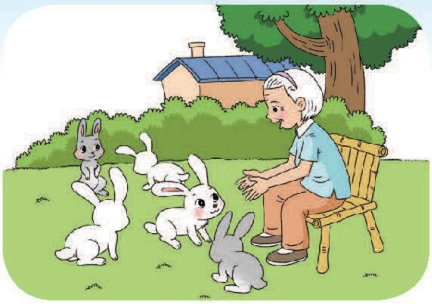
12. Grandma Li spent ¥130 on 5 rabbits. She wants to buy 15 more rabbits. How much more money does she need?



13. A book has 104 pages. Taoqi has read 19 pages. He wants to read the rest in 5 days. On average, how many pages does he read each day?

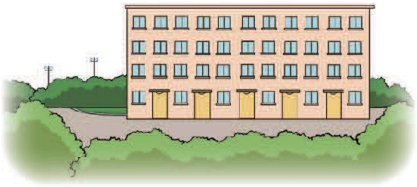
14.

The Forest Concert Hall has 76 seats.

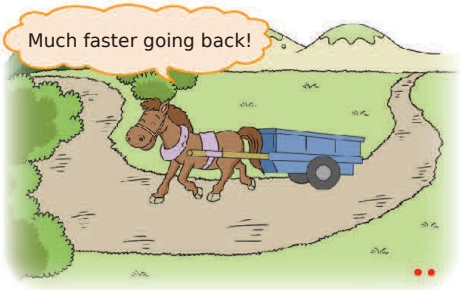
(1) 42 tickets were sold. How much was collected?

(2) The rest sell at ¥12 each. At most, how much more can be collected?

15. Xiangyang Community has 6 buildings like this with 480 households in all. On average, how many households are in each unit?

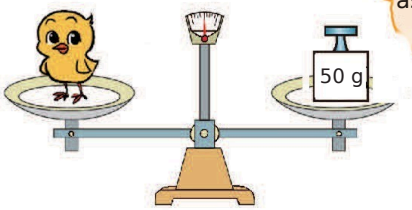


16. A pony delivers corn. It takes 8 minutes to get there and 6 minutes to come back. On the way back, how many metres does it run per minute on average?

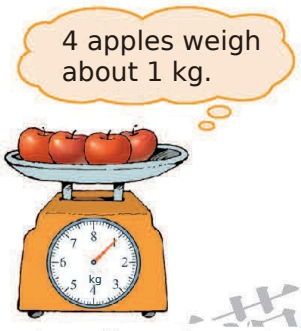


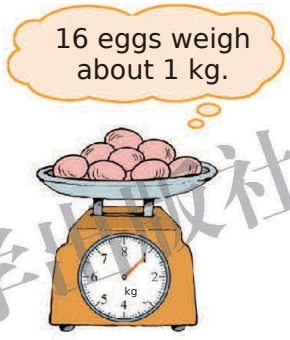
Unit 4 Kilograms, Grams and Tonnes

How Heavy?

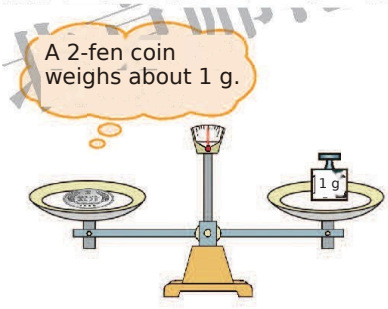


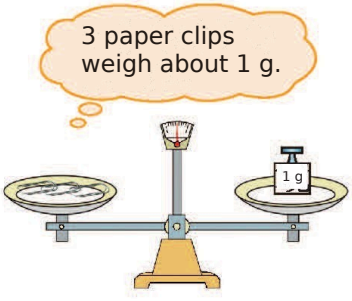
How heavy is 1 kg? Weigh it and feel it.



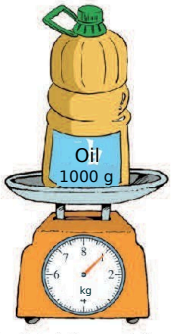


How heavy is 1 g? Weigh it and feel it.





Look and say.



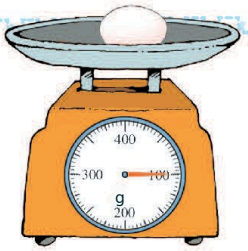
What other objects in daily life weigh 1 kg?

1 kg = 1000 g

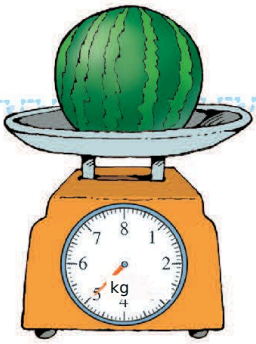
Find some objects, feel them, estimate how heavy they are, and then weigh them.

Practice

1. Fill in.



This egg weighs ( ) g



This watermelon weighs ( ) kg

Xiaoxiao weighs ( ) kg

2. Fill in the brackets with suitable units.

260 ( )

400 ( )

10 ( )

2 ( )

3. Fill in.

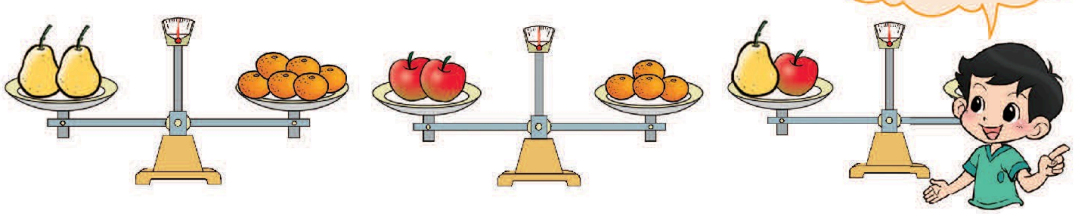
3 kg = ( ) g 4000 g = ( ) kg 8000 g = ( ) kg

4. Weigh and fill in.

	My weight	Partner's weight	My bag	A potato	A peanut
My estimate					
Measured					

5. Which is heavier, 1 kg of cotton or 1 kg of salt? Explain your thinking.

6. Look and guess.



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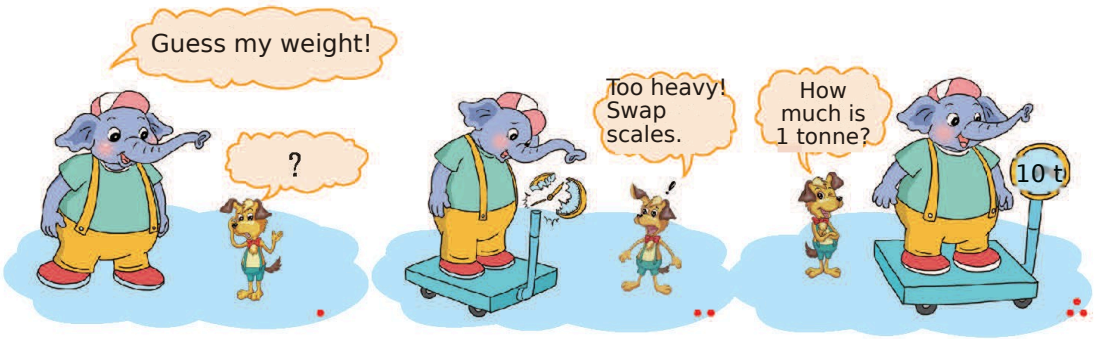
How Heavy Is 1 Tonne?

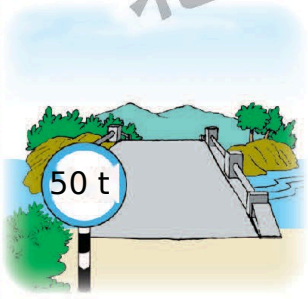


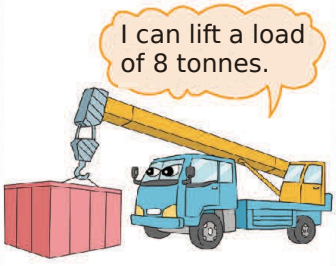
Look and say: about how heavy is 1 tonne?

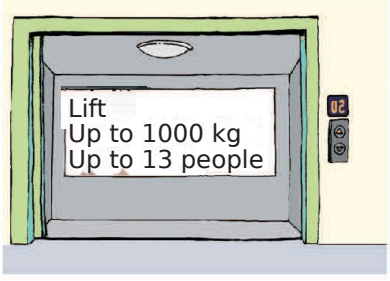
1 t = 1000 kg

Tonne can be written as "t".

Look and say: where have you seen "tonnes" in daily life?



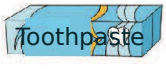




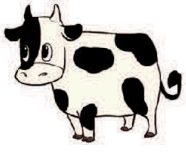
Fill in the correct unit (gram, kilogram, tonne).

A truckload of goods
Weighs 3 ( )

A bottle of cooking oil
Weighs 2 ( )



A tube of toothpaste
Weighs 50 ( )



A cow
Weighs 250 ( )

Practice

1. Read and fill in the correct unit (gram, kilogram, tonne).

It was a sunny day, and Mum and Dad took me to the zoo. First we bought some food. Dad bought 1___ of apples, 1___ of pears and a 50___ boiled egg. The supermarket had an electronic scale. I stood on it, and I already weighed 35___!

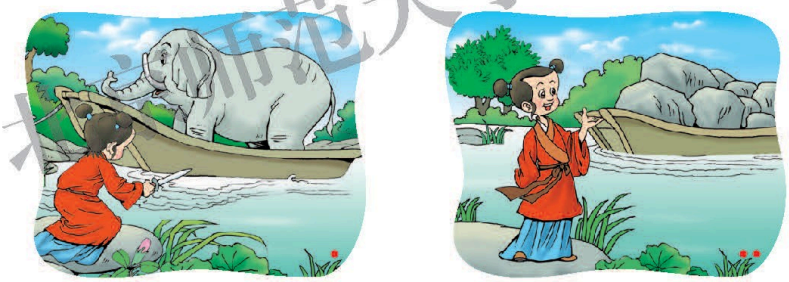
At 9 o'clock we arrived at the zoo. There were so many kinds of animals! I guessed their weights: a lion weighs about 350___, an elephant at least 2___, a talking parrot about 95___, and a big water buffalo at least 450___. After the zoo, we went home happily.

2. Fill in.

3 t = ( ) kg　　5000 g = ( ) kg

( ) t = 2000 kg　　( ) kg = 6 t

3. Maths story.



The stones were weighed in 8 batches. See the table.

Batch	1	2	3	4	5	6	7	8
Mass/kg	280	220	250	300	230	250	270	350

How heavy is this elephant? Work it out.

Did You Know?

Besides tonnes, kilograms and grams, there are many other units of mass. For example, the units commonly used in China include jin and liang:
1 jin = 500 g, 1 liang = 50 g. Britain uses pounds and ounces as units of mass.

Want to know more units of mass? Look them up!

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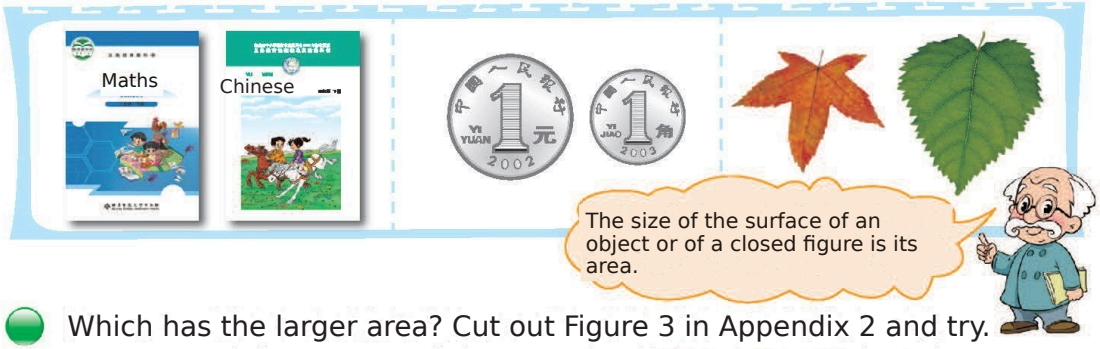
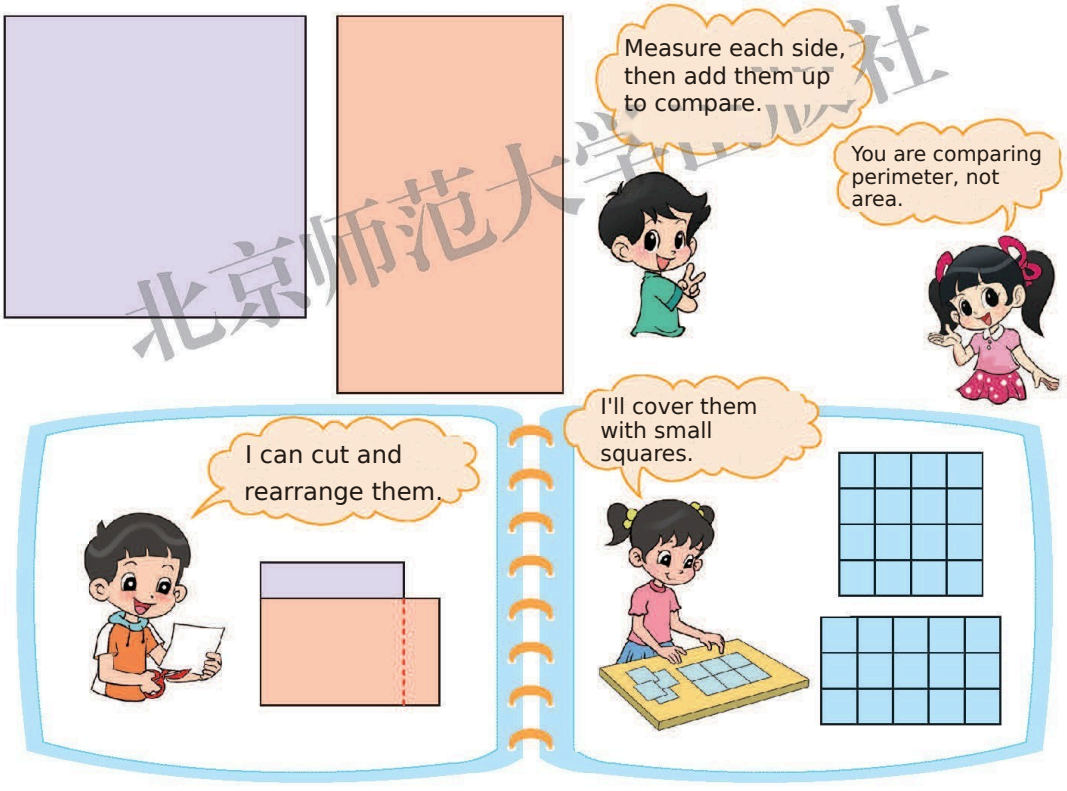
Unit 5 Area

What Is Area?

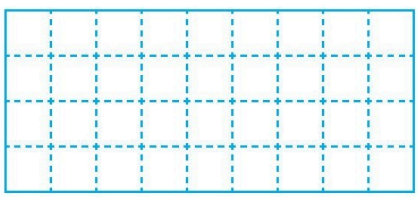
Look and compare



Which has the larger area? Cut out Figure 3 in Appendix 2 and try.



In the grid on the right, draw 3 different figures, each with an area of 7 squares.



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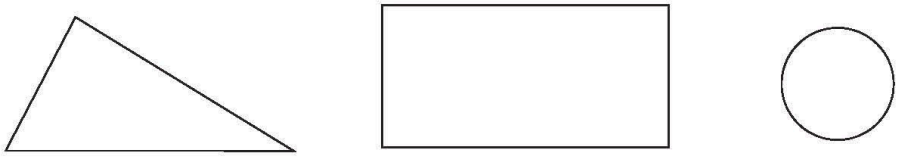
Practice

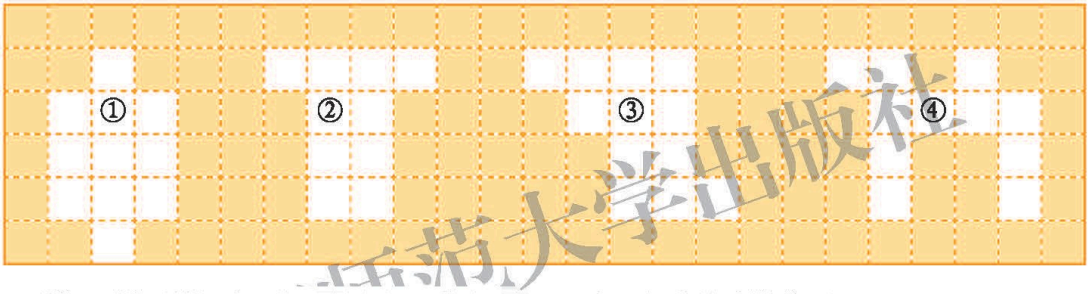
1. Which figure has the largest area? Which has the smallest?



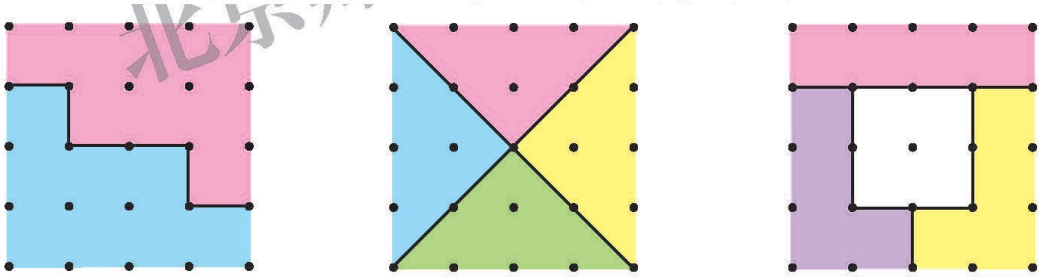
2. Do it yourself.

Cut out Figure 4 in Appendix 2. Which figure has the largest area? Tell your partner how you compared them.

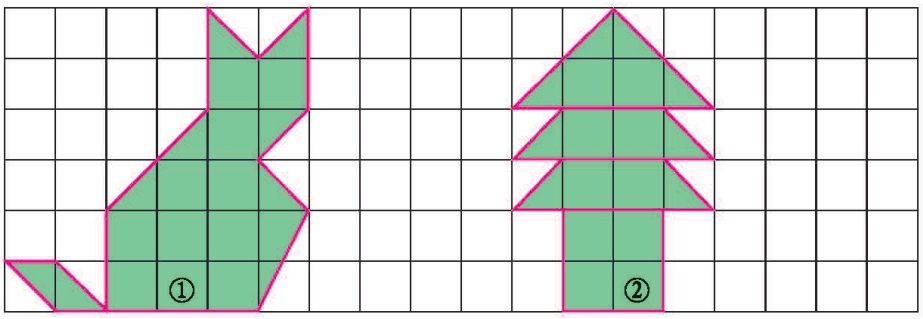
3. Square tiles cover each plot as shown. Which plot needs the fewest tiles?



4. Say how many small squares make up the area of each coloured figure.



5. (1) Count: how many squares make up the area of each pattern below?



(2) Can you design a pattern with the same area as Figure ②? Draw it in Appendix 3, Figure 1.

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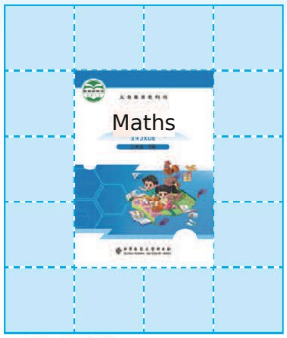
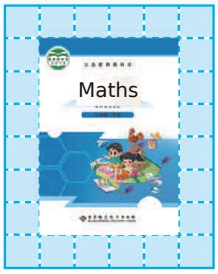
Units of Area





The area of the maths book cover is 6 squares.

I measured 24 squares!

Why are the results different?

The squares they used are not the same size.

We need a standard unit.

What units of area do you know? Let's find out.

This square has sides of 1 cm, so its area is...

A 1 cm square has an area of 1 square centimetre (cm^2).

A 1 dm square has an area of 1 square decimetre (dm^2).

A 1 m square has an area of 1 square metre (m^2).

How big are these units of area? Try it and talk about it.

About 1 cm^2.

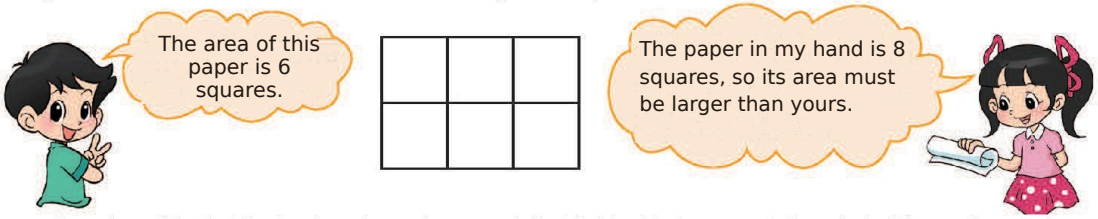
This is 1 square decimetre.

Make a 1 m^2 square out of newspaper.

About 12 pupils can stand on 1 m^2 of ground.

Practice

1. Do you agree with Xiaoxiao? Explain why to a partner.



2. In the grid below, draw a line segment of 1 cm, a square of 1 square centimetre, a line segment of 1 dm and a square of 1 square decimetre.

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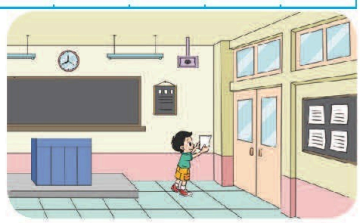
3. Make squares of 1 square centimetre, 1 square decimetre and 1 square metre. Use them as units of area to measure the surfaces of some objects in the classroom.



4. Fill in the appropriate area unit (square metre, square decimetre, square centimetre).

(1) The floor of a room: about 50 ____

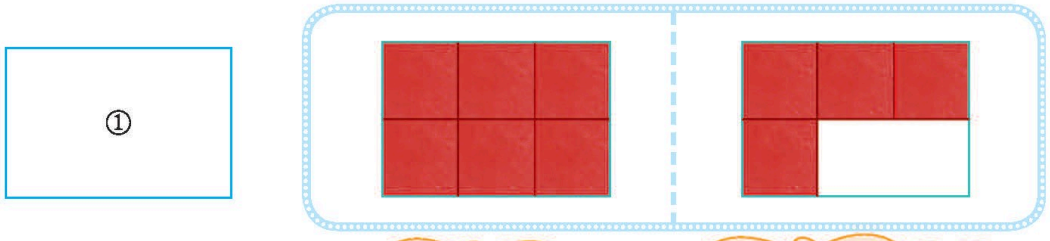
(2) A notebook: about 2 ____

(3) A stamp: about 6 ____

(4) A single bed: about 2 ____

Area of a Rectangle

What is the area of rectangle ①? Cover it with 1 cm² squares.

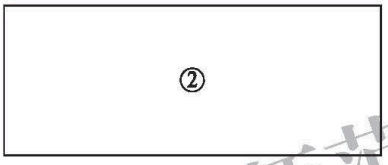


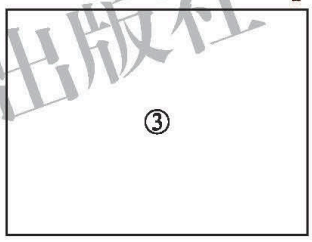
I used 6 small squares in all, so its area is 6 square centimetres.

3 in each row, exactly 2 rows, so the area is:

3×2=6 (cm²)

What is the area of each rectangle below? Use squares.





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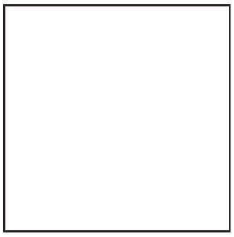
Fill in and think: what do you notice?

	Length/cm	Width/cm	Area/cm²
Figure ①			
Figure ②			
Figure ③			

Area of rectangle =

Think: how do you work out the area of a square?



3 cm

3 cm

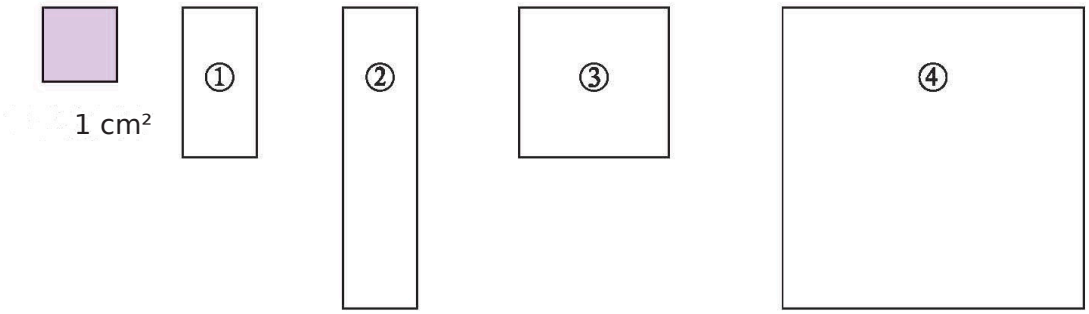
Each row holds 3 squares of 1 square centimetre, and 3 rows fit, so...

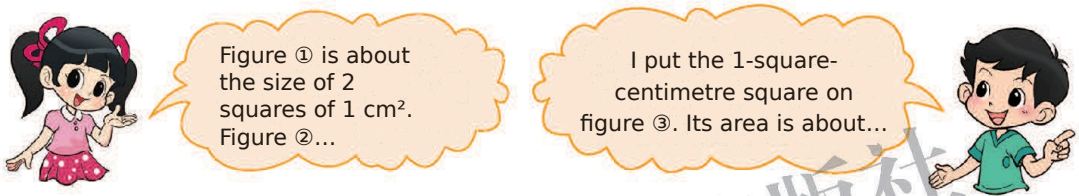
A square is a special rectangle, so just use "side × side".

Area of square =

Try It

Estimate the area of each figure below. Then measure and work it out.





Estimate the area of the classroom.

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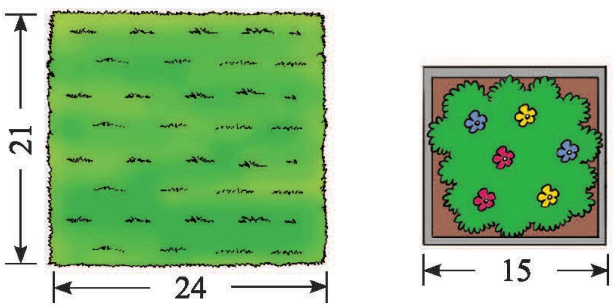
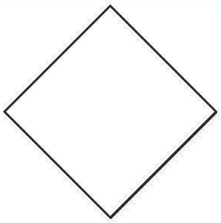
Practice

1. Lay 1-square-centimetre paper squares on figure 2 in Appendix 3 to find its area, and tell your partner your method.

2. Find the areas of the rectangular lawn and the square flower bed. (Unit: m)



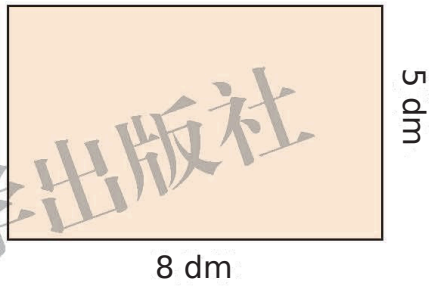
3. Estimate first, then measure and work out the area of each figure below.



4. Estimate first, then measure and calculate.

	Length	Width	Area	Perimeter
Maths book cover				
Classroom floor				
Floor of a room at home				

5. Cut the largest possible square from the rectangular paper on the right. What shape is the part left over? What is its area in square decimetres?



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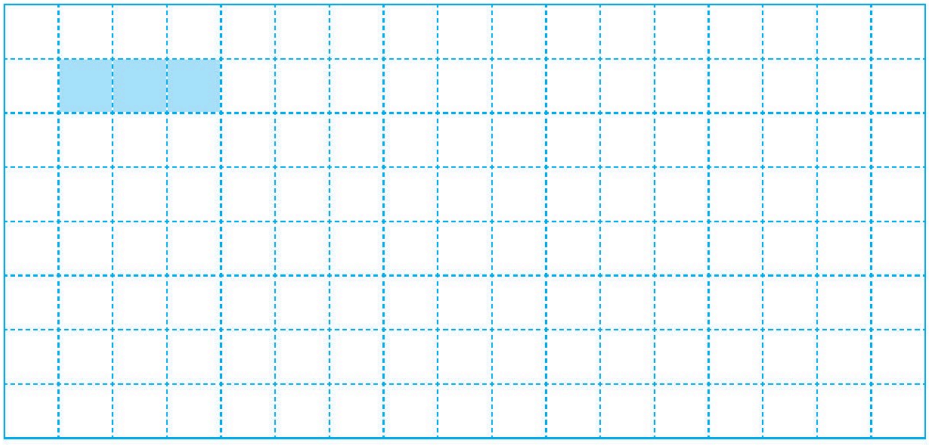
6. A rectangular tablecloth has an area of 72 square decimetres and is 9 decimetres long. How many decimetres wide is it?

7. A square fountain has a perimeter of 20 metres. How many metres long is each side? What is its area in square metres?

8. Think and draw.

1×3 5×5 3×6



Converting Area Units

Use 1-square-centimetre paper squares to cover a 1-square-decimetre square. How many small squares do you need? Draw and explain.

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Each row has 10 squares of 1 square centimetre, and there are ( ) rows.

$1\ dm^2 = (\quad)\ cm^2$

1 square decimetre = 100 square centimetres. So how many square decimetres make 1 square metre?
Explain your thinking.

Using the figure above, think of 1 square decimetre as 1 square metre and 1 square centimetre as 1 square decimetre. Then you can see 1 square metre has...

1 dm = 10 cm, $1\ dm^2 = 100\ cm^2$;
1 m = 10 dm, so I think $1\ m^2$ =...

1 square metre is the area of a square with sides of 1 metre, that is, a square with sides of 10 decimetres, so...

$1\ m^2 = (\quad)\ dm^2$

Practice

1. Tell your partner how you worked out that 1 square metre = 100 square decimetres and 1 square decimetre = 100 square centimetres.

2. Fill in.

3 dm^2 = ( ) cm^2	2 m^2 = ( ) dm^2
500 cm^2 = ( ) dm^2	400 dm^2 = ( ) m^2
20 dm^2 = ( ) cm^2	1000 dm^2 = ( ) m^2

3. Fill in suitable units of area.

(1) The area of the blackboard is about 400 ( ), that is, 4 ( ).

(2) The area of the classroom floor is about 60 ( ), that is, 6000 ( ).

(3) The area of the basketball court is about 420 ( ), that is, 42000 ( ).

4. Who is right?

Did You Know?

In ancient times, people had to measure land to work out harvests and taxes, and so they came to understand area. In ancient China, the characters “幂” (mì) or “积” (jī) were used to stand for area.

幂 (mì): a square cloth for covering things; 积 (jī): to pile up. Can you understand the meanings of these two characters?

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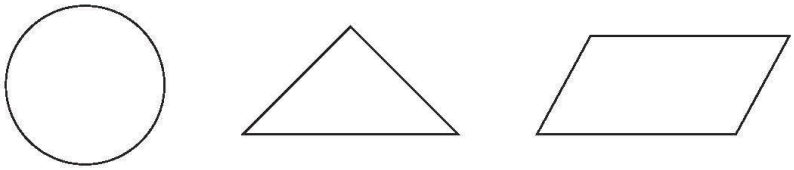
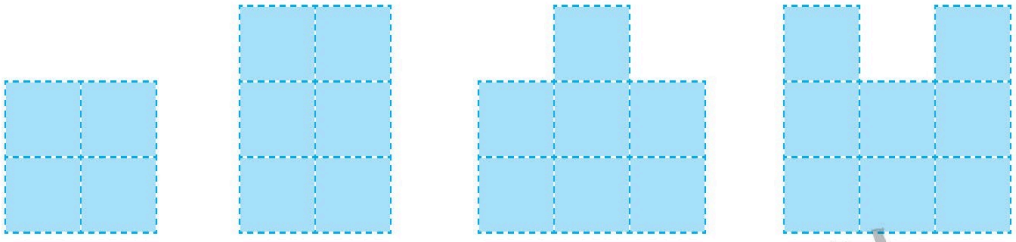
Exercise 4

1. Trace the outline of each shape in red and colour the inside of the shape in blue.

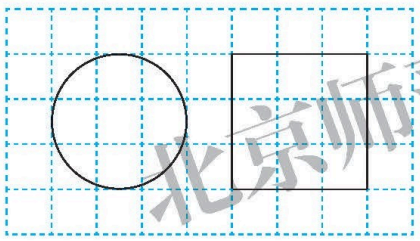


2. Write the area of each figure below. (Each small square has an area of 1 square centimetre.)

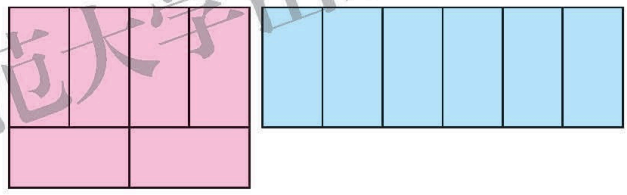


3. Compare the areas of the two figures in each group below and say how you compared them.

(1)



(2)



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4. Fill in suitable units.

(1) A jump rope is about 2 ( ) long.

(2) A bedroom's area is about 22 ( ).

(3) A pencil case lid is about 70 ( ).

(4) A square handkerchief is about 400 ( ).

5. Fill in.

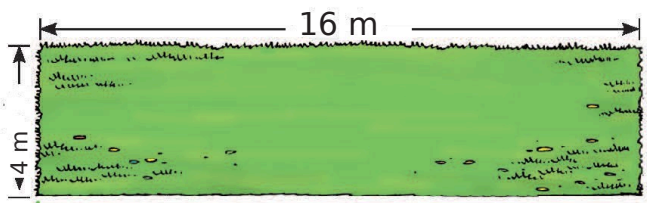
700 dm^2 = ( ) m^2

3000 cm^2 = ( ) dm^2

150 cm = ( ) dm

2 dm^2 = ( ) cm^2

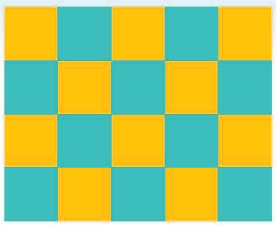
6.



What is the area of this lawn in square metres?

What is the perimeter in metres?

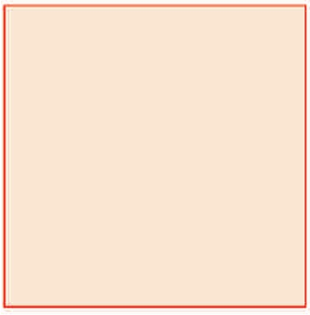
7. On the right is a living room floor laid with square tiles.



(1) How many tiles are there on this floor in total?

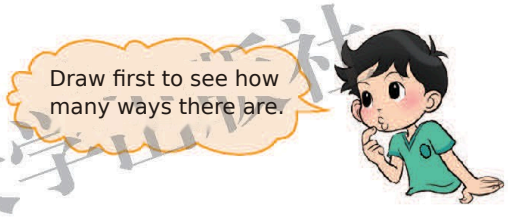
(2) If each tile has sides of 5 decimetres, what is the area of the living room in square metres?

8.

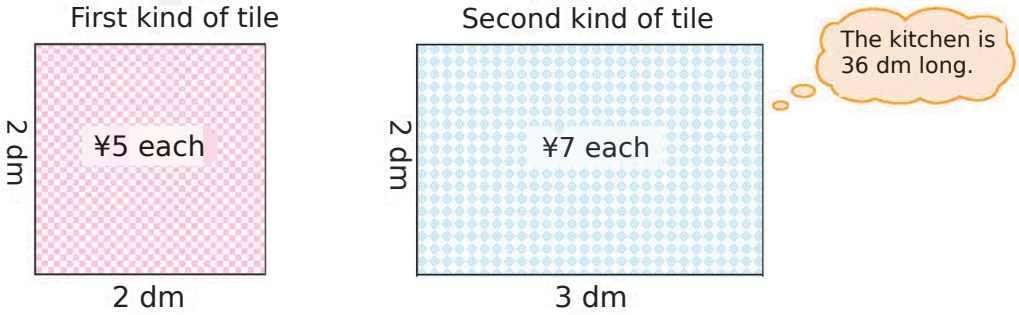


A square has a perimeter of 16 centimetres. What is its area?

9. How can two tables like this be put together? What are the area and perimeter each way?



10. Xiaoxiao's family wants to tile their kitchen. There are two kinds of tiles.



(1) Exactly 180 tiles of the first kind are needed. What is the area of the kitchen?

(2) How many tiles of the second kind are needed?

(3) Which kind of tile is cheaper?

11 Mr Wang has a rectangular sheet of coloured paper, 32 centimetres long and 15 centimetres wide, for the children. What is the largest number of squares with sides of 2 centimetres that he can cut from it?

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12. Comparing perimeter and area.

(1) How many rectangles can you make with 16 cm of wire? Fill in the table.

Length/cm				
Width/cm				
Perimeter/cm				
Area/cm²				

What do you notice?

(2) Measure the side of each square below, work out its perimeter and area, and fill in the table. What do you notice?

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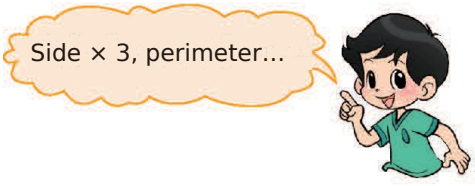
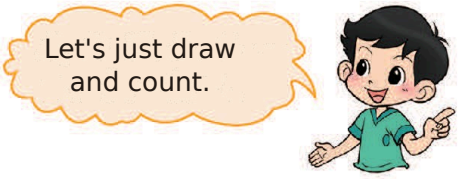
Side length/cm			
Perimeter/cm			
Area/cm²			



(3) Cut a square of coloured paper with sides of 4 cm into small squares with sides of 2 cm. How many can you cut?



Maths Is Fun

Little Designer

Poster

The 5th School Art Festival will be held from 1 to 15 June. The Student Council is collecting festival logo designs from the whole school. If you are interested, please hand in your entry to the Student Council by 25 May. We hope everyone will take part!

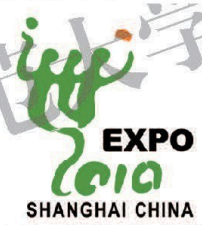
Student Council

Task Design an Art Festival logo.

Discuss

1. Below are patterns and logos we collected. What design features do they have?

Beijing Olympics logo



Shanghai World Expo emblem

School Art Festival logo

The symmetrical five rings show the beauty of harmony; the asymmetrical Chinese seal shows the beauty of movement.

It neatly combines the character "世" (shì) with "2010".

Yang Xizun designed this. The five coloured circles stand for the whole class, and the pattern shows the beauty of symmetry.

2. How can we design a good Art Festival logo?

It should highlight the event's theme.

It should show what is special about our school.

You can use figures with line symmetry.

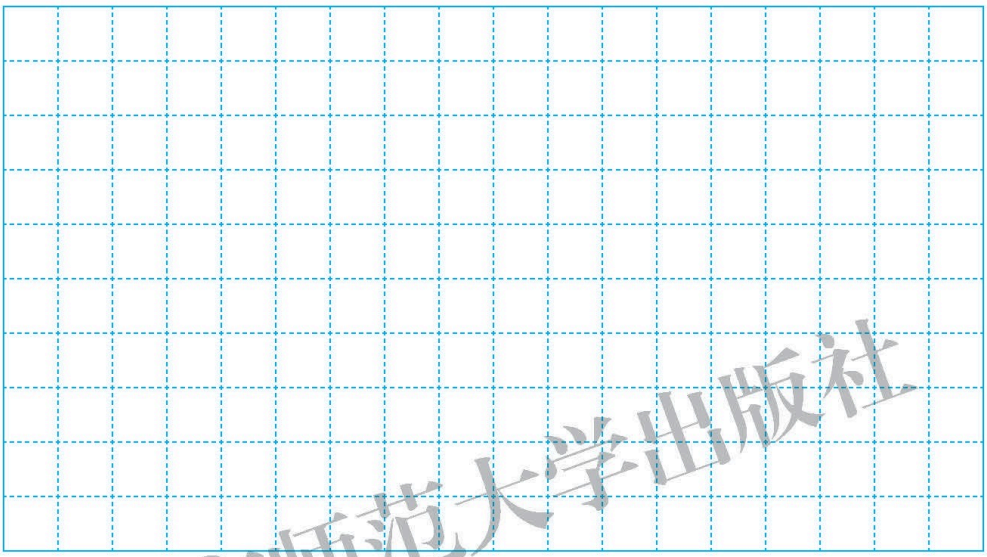
Do It

Design a logo as required below and draw it in the grid.

(1) Give the logo a name.

(2) It should show a theme.

(3) All or part of it can be a figure with line symmetry.



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Think

1. What features does the logo you created have?
2. Which shapes and knowledge that you have learned did you use?
3. What did you gain in the process of designing the logo?

Self-Check

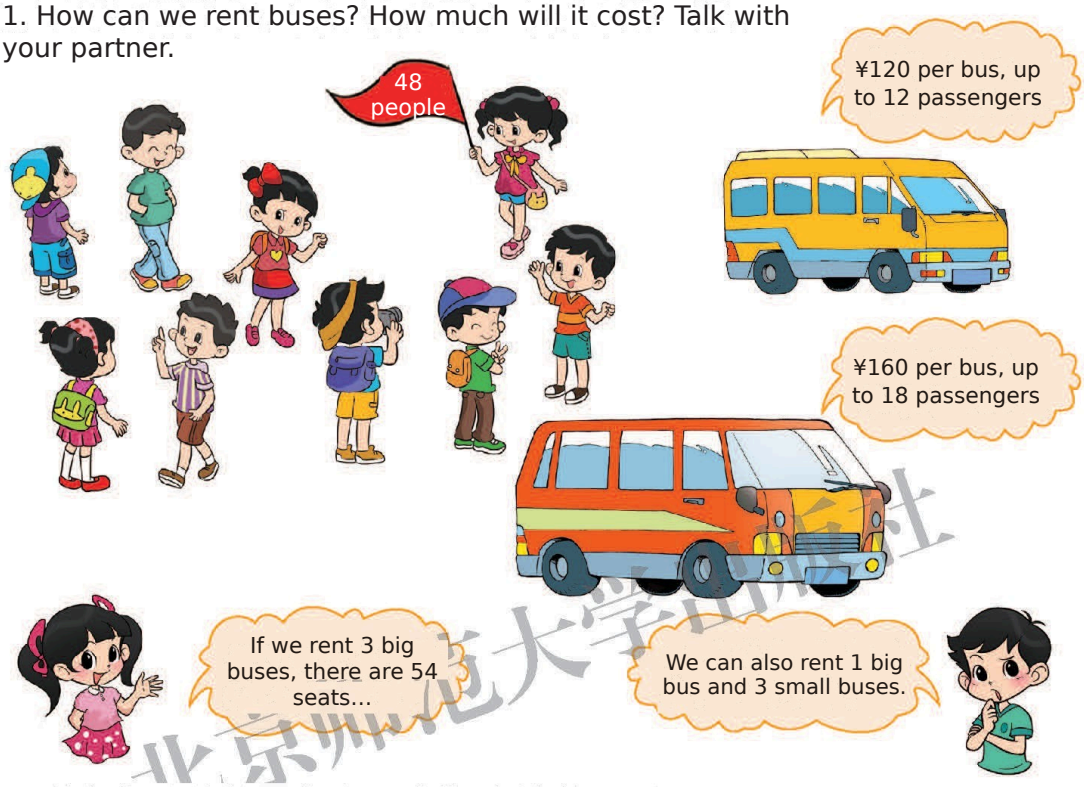
How I did in this activity:

Able to use line-symmetric figures to create patterns.			
My logo has meaning and shows my personality.			
Able to actively communicate with classmates.			
Able to reflect on what I learned.			

Let's Go to the Park

Bus Rental.

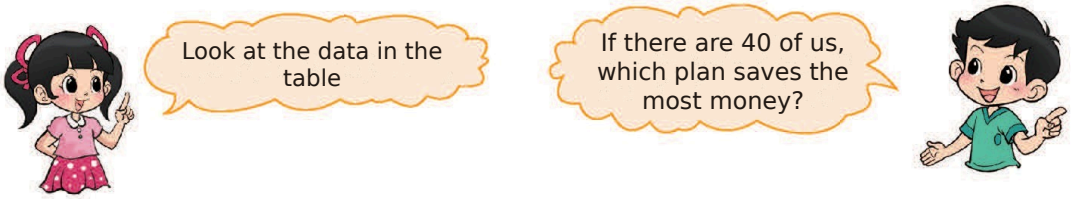
1. How can we rent buses? How much will it cost? Talk with your partner.



2. How many bus rental plans are there in all? Let's sort them out!

	Small buses	Big buses	Seats	Rent (¥)
Plan 1				
Plan 2				

3. Which rental plan saves the most money?



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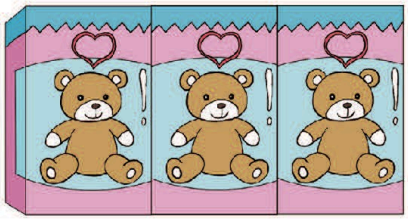
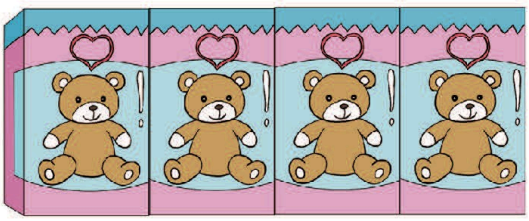
Buying Souvenirs.

1. We want 30 dolls. How can we buy them?



Small pack: 3 in a box, ¥5 a box.



Large pack: 4 in a box, ¥6 a box.

10 small packs make 30 dolls.

Or we could...

2. Write down 3 buying plans. Which one saves the most money?

Let's make a table to compare.

	Small packs	Large packs	Total dolls	Cost (¥)
Plan 1				
Plan 2				
Plan 3				

3. What other problems can you solve by making a table?

What is the cheapest way to buy 50 dolls?

If a big bus seats 8 and a small bus seats 5, and there are 28 of us, which buses should we take to have the fewest empty seats?

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Fun Reasoning

Which interest group is each child in?

The school has football, model plane and computer interest groups. Taoqi, Xiaoxiao and Qisi each joined one of them according to their own interests. No two of them are in the same group.

1. Try it out and discuss with your partner.

Taoqi is not in the computer group, so he could be in the football or model plane group. Xiaoxiao doesn't like football, so she could be... There's too much information. I'm getting muddled.

How should we organise the information?

2. Draw a table to help, and record the information you know.

	Football	Models	Computer
Taoqi			×
Xiaoxiao			
Qisi			

	Football	Models	Computer
Taoqi			×
Xiaoxiao	×		
Qisi	×	✓	×

Taoqi is not in the computer group, so draw "×" in that cell of the table.

Qisi is in the model plane group, so draw "✓". Xiaoxiao and Taoqi can't be in this group, so draw...

3. Finish the table above. Can you work out which group each of them is in?

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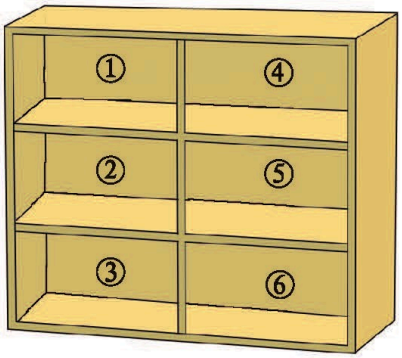
Where is each model plane in the cabinet?

The model plane group has 6 model planes: Taoqi, Qisi, Miaoxiang, Xiaoxiao, Lele and Coach, kept in a cabinet. Use the clues below to find their positions.

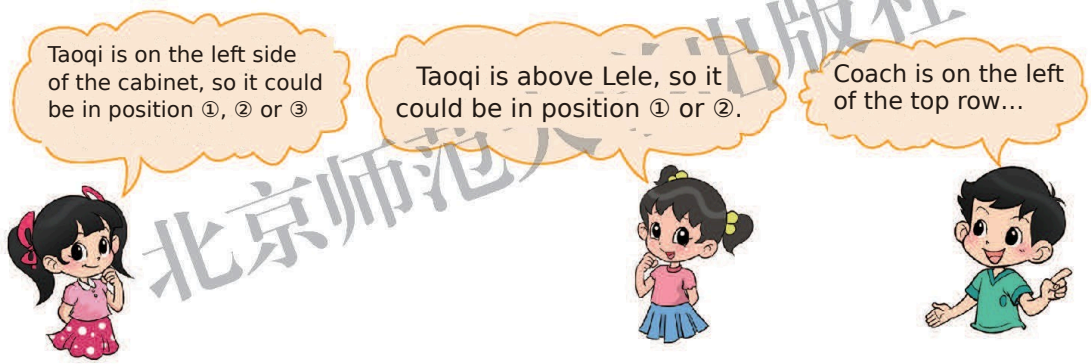


Taoqi and Lele are both placed on the left side of the cabinet, and Taoqi is above Lele.

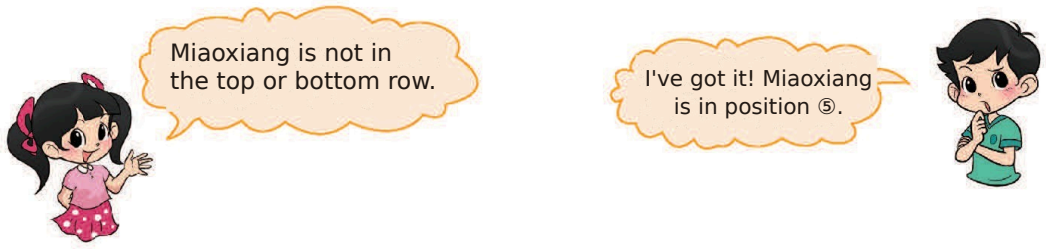
Coach is on the left of the top row. Miaoxiang is not in the top row or the bottom row.

Qisi is not placed next to Coach.

1. Where is Taoqi? Share your ideas with your partner.



2. Where is Miaoxiang? Share your ideas with your partner.



3. Which models are not placed yet? How can we place them? Discuss with your partner.

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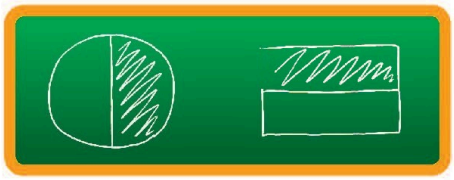
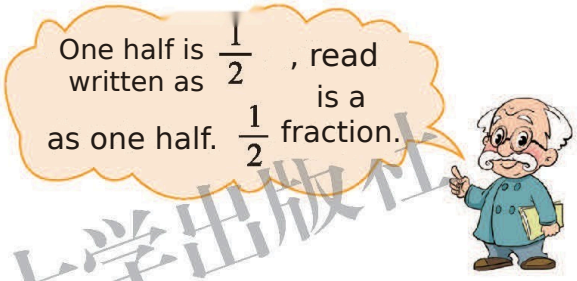
Unit 6 Understanding Fractions

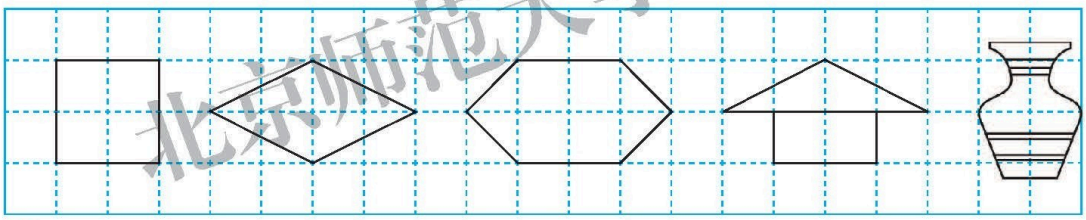
Sharing (1)

How can you show one half?





Colour each shape to show $\frac{1}{2}$



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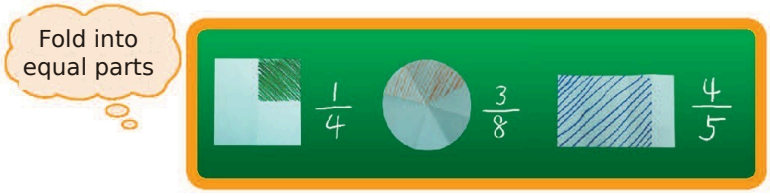
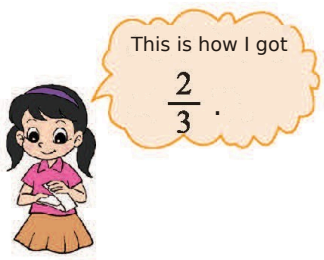
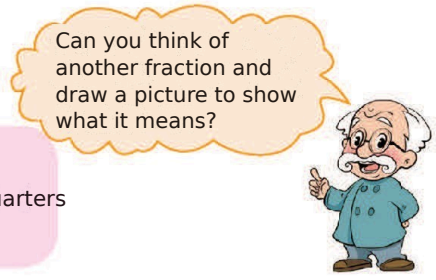
Fold and colour a sheet of paper. What other fractions can you get?





Let's learn.

$\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{2}{3}$, $\frac{4}{5}$ are all fractions.



$\frac{3}{4}$ numerator / fraction line / denominator

Read as: three quarters

Try It

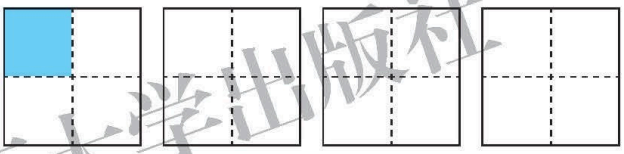
Is each coloured part below $\frac{1}{4}$ of the sheet? Explain why.

In each one, a sheet is divided into 4 equal parts and 1 part is coloured, so...

They are divided equally in different ways, so the coloured parts have different shapes, but they are all...

Divide a square sheet of paper into 4 equal parts. Colour 1 part, 2 parts, 3 parts and 4 parts of it. Fill in the blanks and explain.

Colour 1 of the parts. The coloured part of this sheet is $\frac{1}{4}$.....



$\frac{(\ 1\ )}{(\ 4\ )}$ $\frac{(\ \ )}{(\ \ )}$ $\frac{(\ \ )}{(\ \ )}$ $\frac{(\ \ )}{(\ \ )}$

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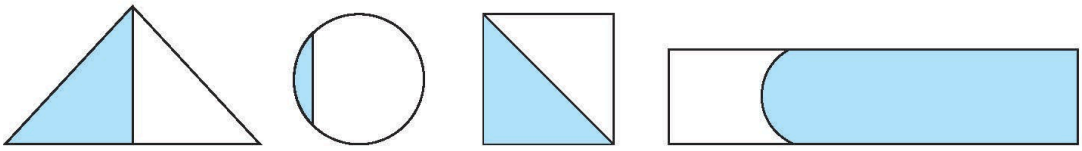
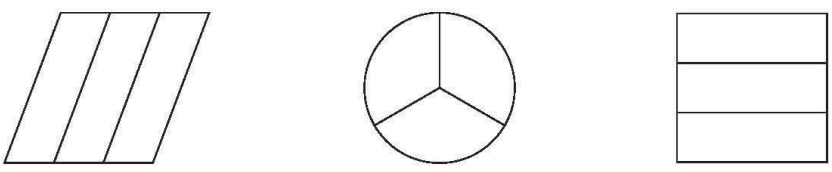
Practice

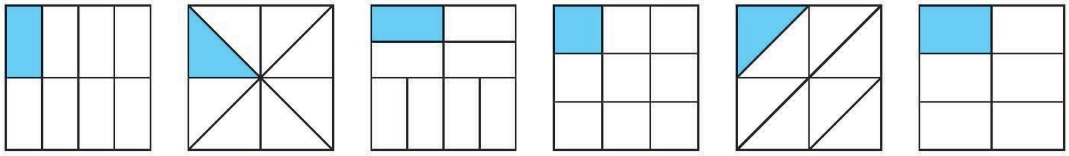
1. Is each blue part $\frac{1}{2}$ of its shape? Explain.



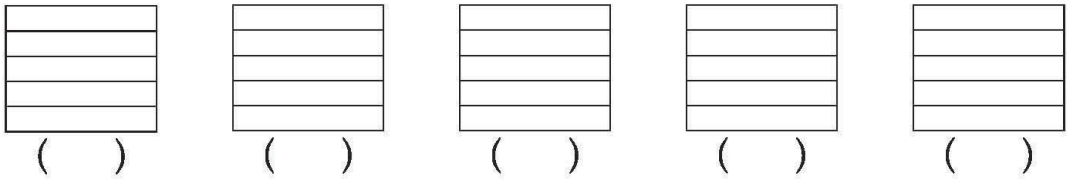
2. Colour each shape to show $\frac{1}{3}$



3. Is each coloured part below $\frac{1}{8}$ of the sheet? Explain why.

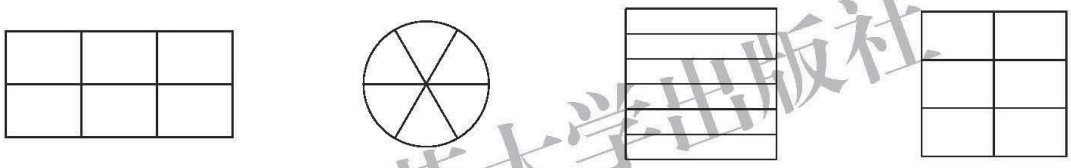


4. Divide a rectangular sheet of paper into 5 equal parts. Colour 1 part, 2 parts, 3 parts, 4 parts and 5 parts of it. Fill in the blanks and explain.



( ) ( ) ( ) ( ) ( )

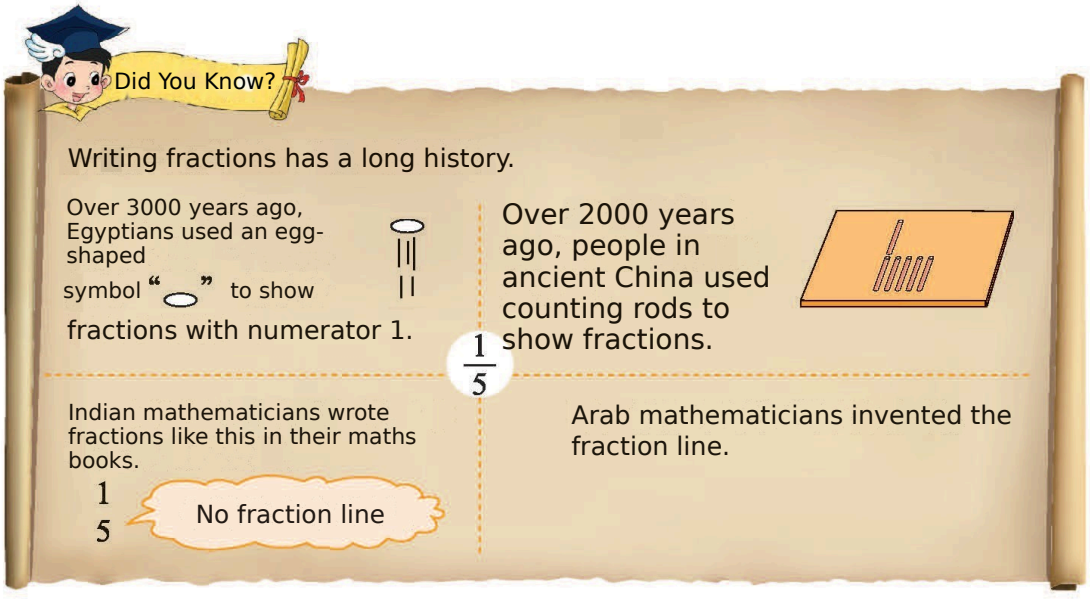
5. Colour each shape to show $\frac{1}{6}$



6. Fold and colour.

(1) Fold a sheet to show $\frac{3}{16}$ and $\frac{2}{3}$

(2) What other fractions of a sheet of paper can you fold?



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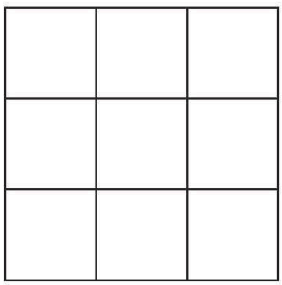
Sharing (2)

Colour the small squares in 3 colours: red, yellow and blue. What fraction of the shape is each colour? Try with Figure 3 in Appendix 3.



The red part is $\frac{(\quad)}{(\quad)}$.

The yellow part is $\frac{(\quad)}{(\quad)}$.

The blue part is $\frac{(\quad)}{(\quad)}$.

I coloured 3 small squares in each colour, so each takes up $\frac{3}{9}$.

I coloured 1 red, and 4 each of yellow and blue

Cut Figure 3 in Appendix 3 into 9 small squares. What fraction of the 9 small squares does each colour take up?

The squares of each colour still take up $\frac{3}{9}$.

1 large square became 9 small squares, so 1 of these small squares is $\frac{1}{9}$; 4 of them are $\frac{4}{9}$.

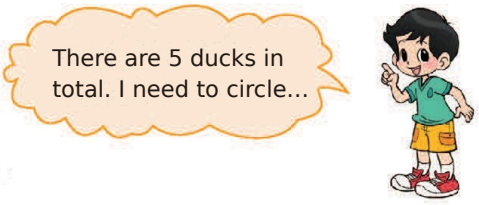
Look at the picture below. What fractions can you find? Tell your partner.

There are 7 butterflies, and 3 are white. The white ones are $\frac{3}{7}$.

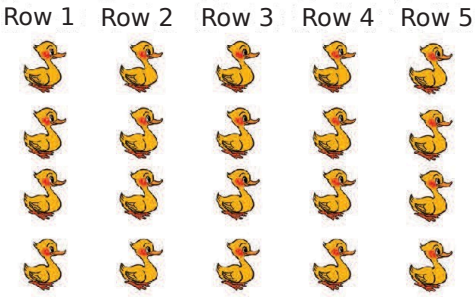
4 girls and 2 boys. Of all the children, the boys are $\frac{2}{6}$.

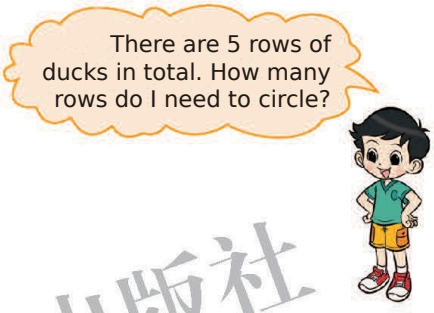
Try It

There are 5 ducklings. Circle $\frac{3}{5}$.



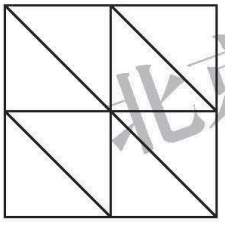
There are 5 rows of ducks. Circle $\frac{3}{5}$.





Practice

1.

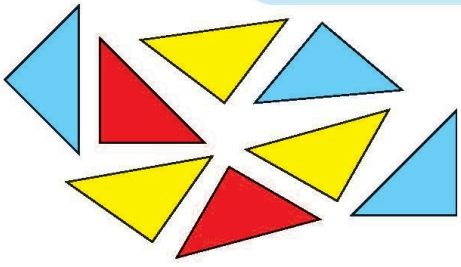


(1) Colour and fill in

Colour 2 triangles red. Red is $\frac{(\ )}{(\ )}$;

Colour 3 triangles yellow. Yellow is $\frac{(\ )}{(\ )}$;

Colour 3 triangles blue. Blue is $\frac{(\ )}{(\ )}$.



(2) What fraction of the 8 small triangles is each colour? Tell your partner.

2. Look at the picture on the right. What fractions can you find? Tell your partner.

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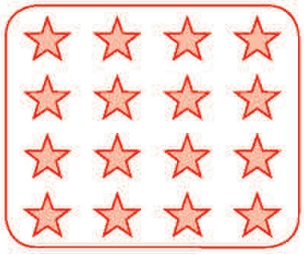
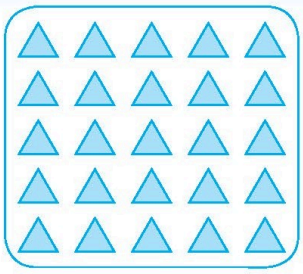
3. Circle and explain.

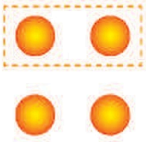
$\frac{3}{4}$



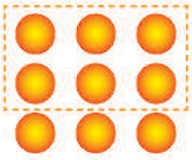
$\frac{2}{5}$



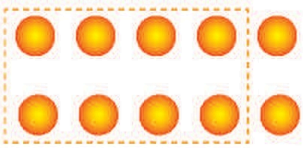
4. What fraction of the total is circled?



$\frac{(\ )}{2}$

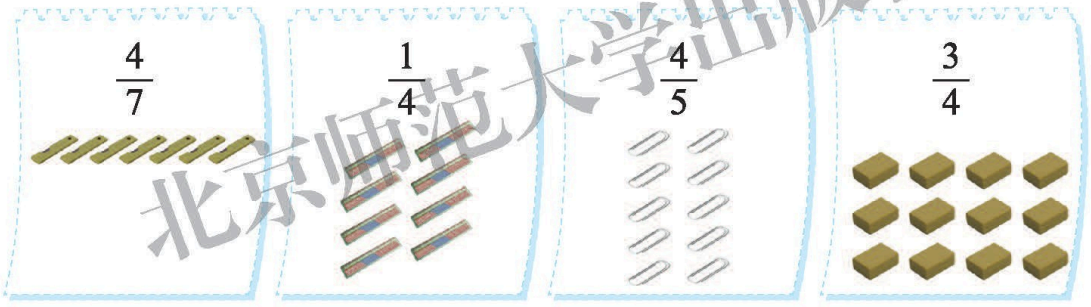


$\frac{(\ )}{3}$



$\frac{(\ )}{5}$

5. Circle according to the fraction.



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6. 12 children have a group photo taken.

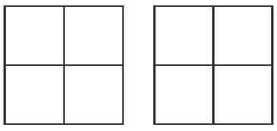
Of all the children, boys are $\frac{(\ )}{(\ )}$.

In this photo, $\frac{(\ )}{(\ )}$ of the children have braids.

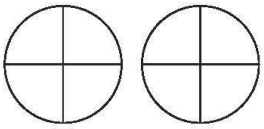
7. Fractions on a clock. Can you show the shaded parts as fractions?

Compare

$\frac{3}{4}$ or $\frac{1}{4}$ – which is larger? Colour and compare.



$\frac{3}{4} \bigcirc \frac{1}{4}$

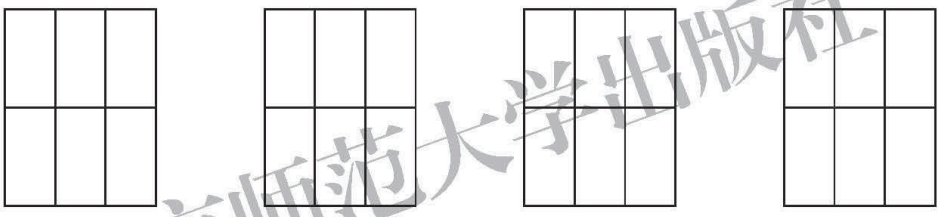


$\frac{3}{4} \bigcirc \frac{1}{4}$

Both divide the square into 4 equal parts; the more parts coloured, the larger the fraction.

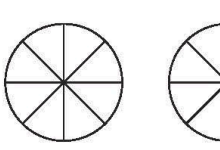
Both fractions have the denominator 4. The larger the numerator...

Find fractions with the same denominator below. Colour and compare.



I'll set them, you compare.

$\frac{5}{8}$ and $\frac{8}{8}$.



$\frac{5}{8} \bigcirc \frac{8}{8}$

$\frac{1}{2}$ and $\frac{1}{4}$.

$\frac{1}{2} \bigcirc \frac{1}{4}$

Draw a picture to compare, and you'll see which is larger.

$\frac{2}{3}$ and $\frac{3}{5}$ which is larger?

Let me try drawing too!

In Grade 5, we will learn more...

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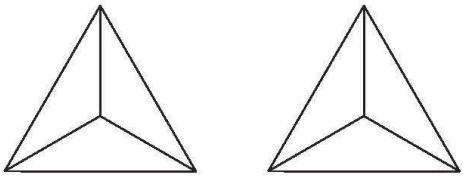
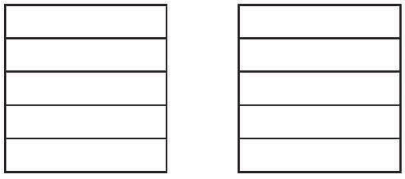
Practice

1. Colour and compare.

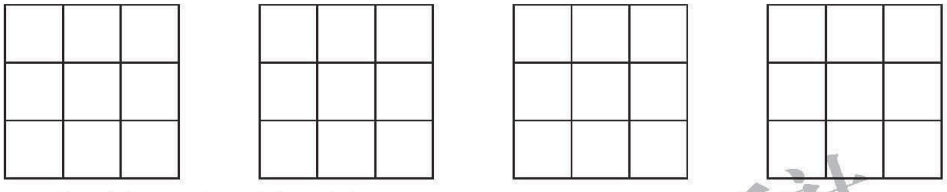


$\frac{2}{3}$ ○ $\frac{1}{3}$



$\frac{4}{5}$ ○ $\frac{5}{5}$

2. Colour a group of fractions with the same denominator, then order them.



3. Draw, colour and compare.

$\frac{3}{8}$ ○ $\frac{7}{8}$

$\frac{1}{4}$ ○ $\frac{1}{8}$

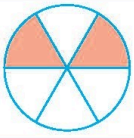
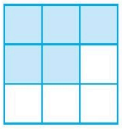
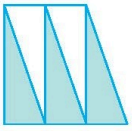
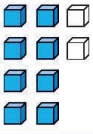
4. Xiaoxiao brought a basket of 7 apples.

Red apples make up $\frac{(\ )}{7}$,

Green apples make up $\frac{(\ )}{7}$。

$\frac{(\ )}{7} > \frac{(\ )}{7}$

5. Write suitable fractions and compare each pair.

				
Coloured part				
Uncoloured part				

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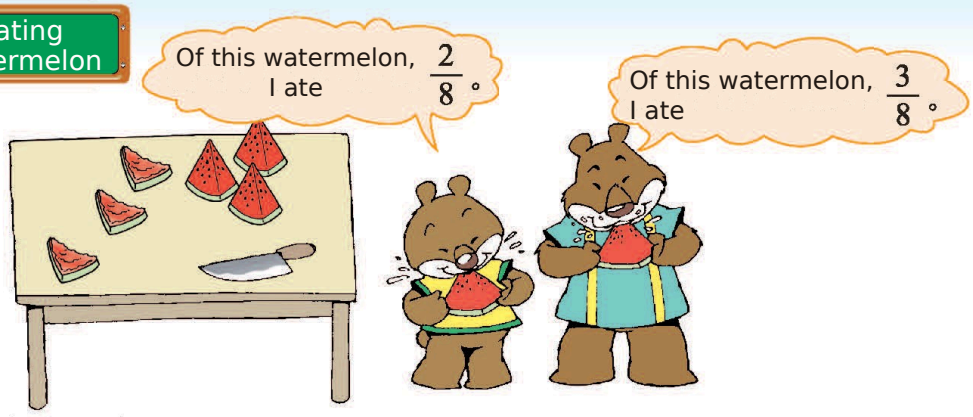
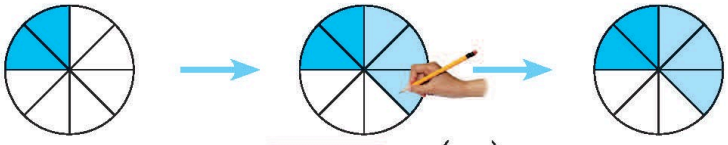
Eating Watermelon

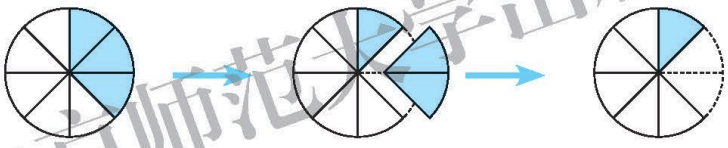


What fraction of the watermelon did Big Bear and Little Bear eat in total?



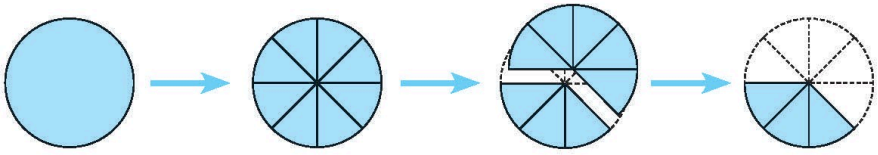
$$\frac{2}{8}+\frac{3}{8}=\frac{2+3}{8}=\frac{(\ )}{8}$$

What fraction more did Big Bear eat than Little Bear?



$$\frac{3}{8}-\frac{2}{8}=\frac{3-2}{8}=\frac{(\ )}{8}$$

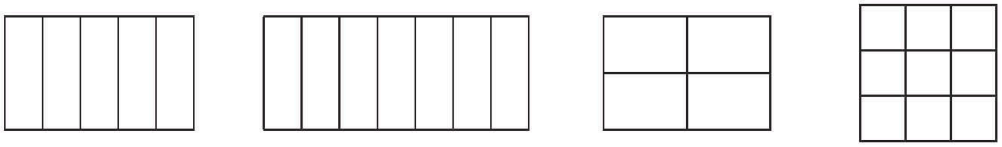
What fraction is left?



$$1-\frac{5}{8}=\frac{(\ )}{8}-\frac{(\ )}{8}=\frac{(\ )}{8}$$

Colour and calculate.

$\frac{3}{5}+\frac{1}{5}=$ $\frac{5}{7}-\frac{2}{7}=$ $1-\frac{1}{4}=$ $\frac{7}{9}+\frac{2}{9}=$



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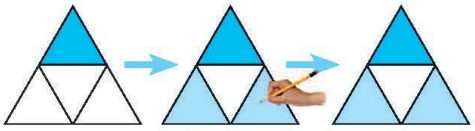
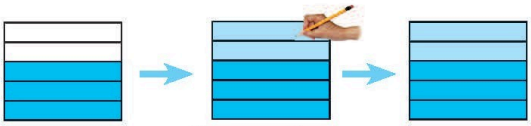
Practice

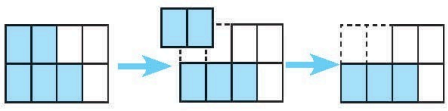
1. Look and fill in.



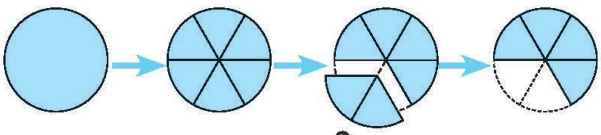
$\frac{1}{4} + \frac{2}{4} =$



$\frac{3}{5} + \frac{2}{5} =$



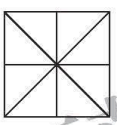
$\frac{5}{8} - \frac{2}{8} =$



$1 - \frac{2}{6} =$

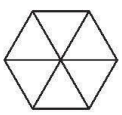
2. Colour and calculate.

$\frac{1}{3} + \frac{1}{3} =$



$\frac{7}{8} - \frac{3}{8} =$

$1 - \frac{7}{10} =$



$\frac{1}{6} + \frac{5}{6} =$

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3.

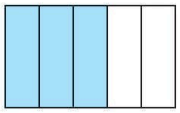
$\frac{1}{9} + \frac{4}{9} =$ $\quad \frac{5}{7} - \frac{2}{7} =$ $\quad 1 - \frac{1}{5} =$

$\frac{5}{8} + \frac{2}{8} =$ $\quad \frac{3}{4} - \frac{3}{4} =$ $\quad 1 - \frac{3}{7} =$

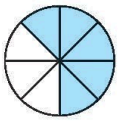
4. What fraction of this paper did they use in total? What fraction is left?

Exercise 5

1. What fraction of each shape is coloured?



( )



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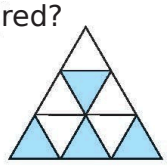

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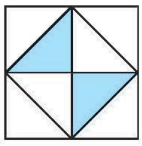
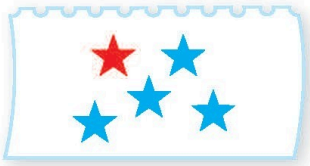


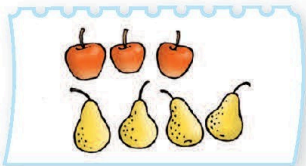
( )

2. Think and fill in.



Red stars make up $\frac{(\ )}{(\ )}$,

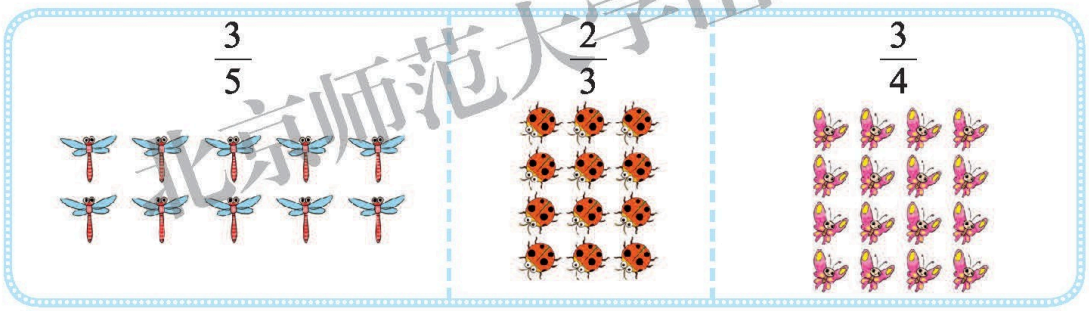
Blue stars make up $\frac{(\ )}{(\ )}$.



Pears are $\frac{(\ )}{(\ )}$,

Apples are $\frac{(\ )}{(\ )}$.

3. Circle according to the fraction.



4. Look and fill in.

(1) In all, ( ) children are skipping rope.

(2) Boys are $\frac{(\ )}{(\ )}$ of the total and girls are $\frac{(\ )}{(\ )}$.

(3) Please ask another maths question and try to answer it.

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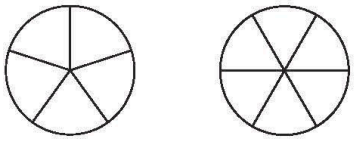
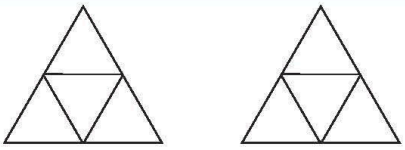
5. Colour and compare.



$\frac{1}{5}$ ○ $\frac{1}{6}$



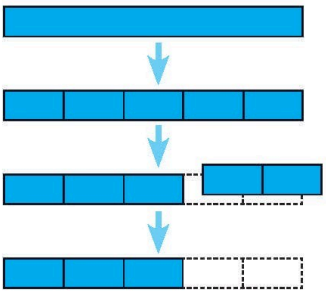
$\frac{2}{4}$ ○ $\frac{3}{4}$

6. Look and fill in.

(1)

$\frac{(\ )}{8} + \frac{(\ )}{8} = \frac{(\ )}{(\ )}$

(2)



$1 - \frac{2}{5} = \frac{(\ )}{5} - \frac{(\ )}{5} = \frac{(\ )}{5}$

7.

$\frac{1}{4} + \frac{2}{4}$ $\quad \frac{7}{10} - \frac{5}{10}$ $\quad \frac{1}{2} + \frac{1}{2}$ $\quad 1 - \frac{2}{9}$

8. A birthday cake is cut into 8 equal pieces. What fraction of the cake did the two children eat altogether? What fraction is left?

9. On the first day, Qisi read $\frac{2}{7}$ of a book. On the second day he read the same amount.

What fraction did he read in the two days? What fraction is left unread?

10. What fraction of each shape is coloured?

( ) ( ) ( )

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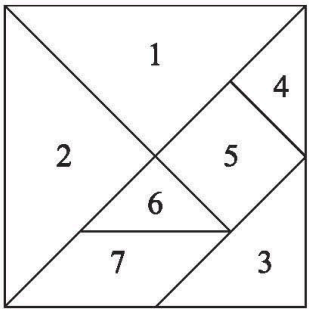
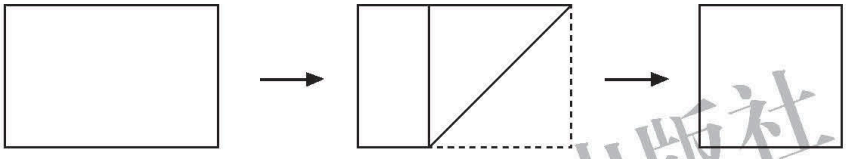
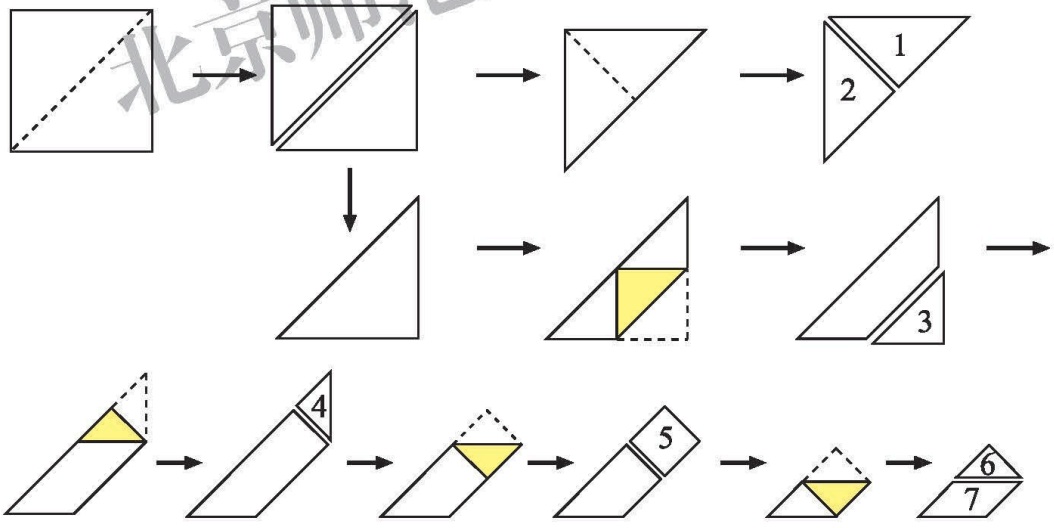
11. Make a tangram.



(1) Take a rectangular piece of paper, fold it to make the largest square you can, and cut it out.



(2) Use the square you cut out to make a tangram in the order below, and colour each piece a different colour.



(3) What fraction of the original square is piece 1? And piece 2? What fraction of the original square are they together?

(4) What fraction of the original square is each of pieces 3, 4, 5, 6 and 7?

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Unit 7 Organising and Presenting Data

Little Shoe Store

Taoqi's class wants to open a pretend shoe store. To "stock" it, they need to do a survey.

Talk about it: how could we do the survey?

We can survey classmates' shoe sizes and stock more of the common ones.

Let's survey the boys and girls separately.

The survey results are below. Organise them and present the results.

Boys' shoe sizes					
33	37	34	34	36	38
36	34	35	34	35	35
37	34	34	38	34	36
35					

Girls' shoe sizes					
32	34	35	34	34	33
35	34	35	34	34	34
34	35	35	33	37	

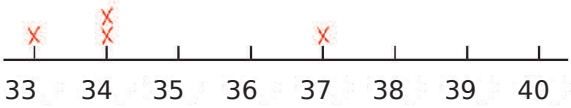
I put the same sizes together...

One girl wears size 32...

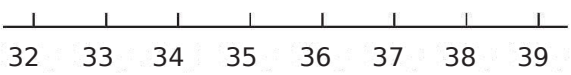
Size 32: 1

This is how Miaoxiang did it. Do you understand? Continue the drawing.

Boys' shoe sizes



Girls' shoe sizes



What do you notice? What would you suggest for "stocking" the shoe store?

The largest boys' shoe size is 38 and the smallest is...

Most girls wear size ..., so we should stock more of those.

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Practice

1. Pretend to open your own shoe store too, and do a "stocking" survey.

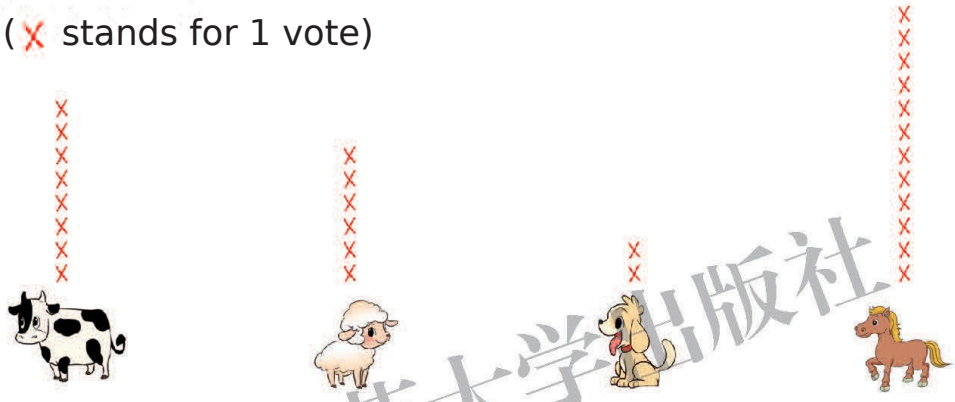
   (1) Discuss: how could you do the survey?

   (2) In pairs, survey your classmates' shoe sizes, then organise and show the results.

   (3) What did you find? What are your ideas for "stocking" the shoe store?

2. Class 3(2) voted for their favourite farm animal. The votes are recorded below.

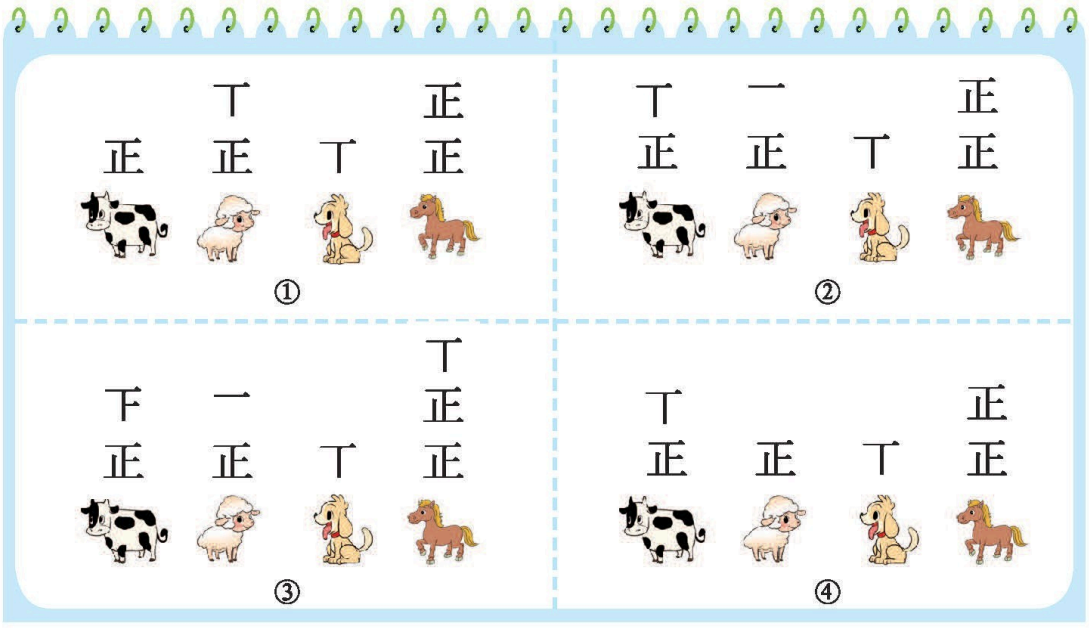
(X stands for 1 vote)



   (1) Which animal is the most popular?

   (2) How many people voted in total?

   (3) Which group below shows exactly the same data as the chart above?



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Happy Growth

On 1/12/2010, the Ministry of Railways changed the height range for children's half-price tickets to 120–150 cm.

Here are the heights of the pupils in Taoqi's class. Do you know how many of them can still buy half-price tickets?

Heights of Taoqi's Classmates

cm

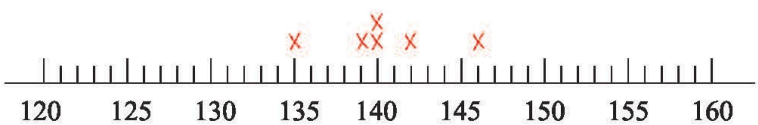
Group 1	139	142	140	140	135	146	Group 4	139	137	136	138	151	142
Group 2	129	140	134	147	134	138	Group 5	135	120	141	143	135	148
Group 3	138	142	129	143	127	146	Group 6	138	132	147	139	148	156

I see that one student in Group 4 is 151 cm tall....

Going group by group, it's easy to miss some! What can we do?

Xiaoxiao thought of a way. She marked the heights of Group 1 on the chart below. Can you go on and mark the heights of the whole class?

I can see how many in our class can buy half-price tickets!



Look at the chart above. What else can you see?

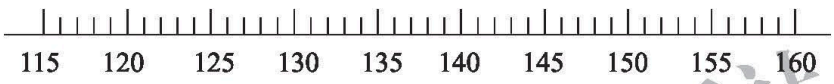
Most pupils in our class are about 140 cm tall.

The tallest is 156 cm and the shortest is 120 cm.

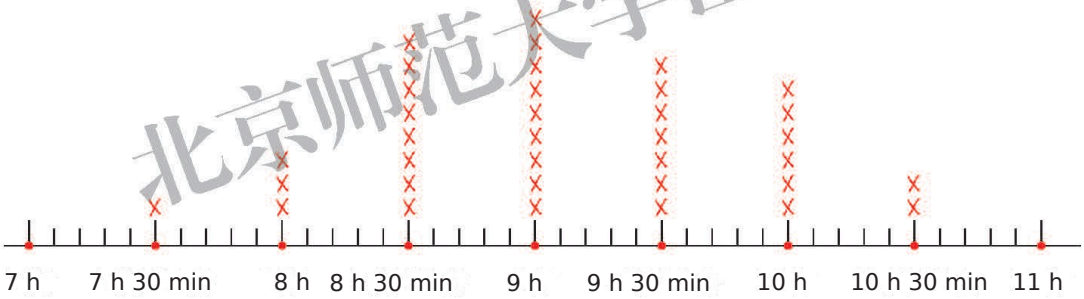
Practice

1. Yao Ming was 147 cm tall when he started primary school. How tall are your classmates? Work in groups: survey first, organise the data in the chart below, then compare with Yao Ming's height when he started school.

( ) Class Heights



2. The chart below shows how long Class 3(2) students sleep.



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(1) Fill in from the chart above.

Sleep time	7 h 30 min	8 h	8 h 30 min	9 h	9 h 30 min	10 h	10 h 30 min
Pupils							

(2) Do you know how many students are in Class 3(2)?

(3) If primary school children should sleep 10 hours a night, what do you think of the survey results?

(4) In groups, survey how long pupils in your class sleep, make a chart and compare it with Class 3(2). How are the two classes' sleep times different?

Final Review

After three years of learning mathematics, we have gained a lot.

We learned about numbers, and shapes too

I have learned some ways to think about and solve problems.

Can you now solve the questions in the question bank? What new questions do you want to explore?

Numbers

Review and Share

1. Talk about numbers in everyday life.

My student number is 7, and I am 129 cm tall...

This book costs ¥12.8 and its cover is 0.23 m long...

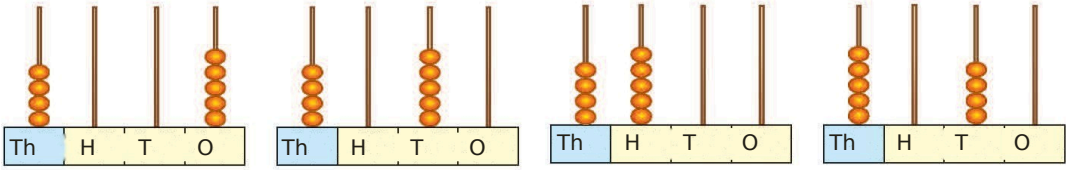
Cut an apple into 4 equal parts. 1 of them is $\frac{1}{4}$.....

2. Do you still remember the place value chart? Think and fill in.

Place value chart

.....	( ) place	( ) place	( ) place	Tens	Ones

Write the following numbers, and explain what "5" means in each number.



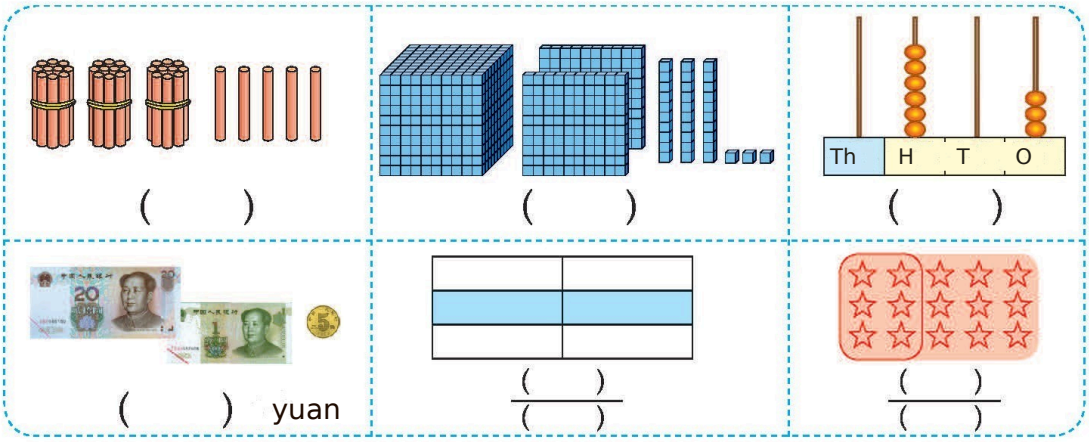
( ) ( ) ( ) ( )

3. Give an example of how to compare numbers.

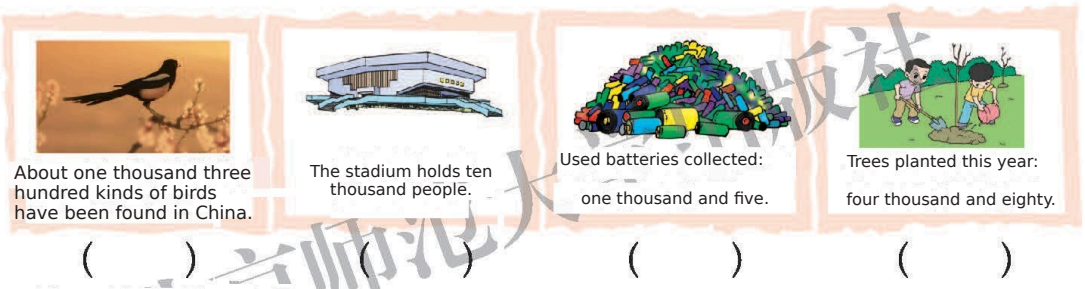
4. Draw pictures of $\frac{1}{8}$ and $\frac{3}{8}$, then explain how to calculate $\frac{3}{8}+\frac{1}{8}$, $\frac{3}{8}-\frac{1}{8}$.

Consolidate and Apply

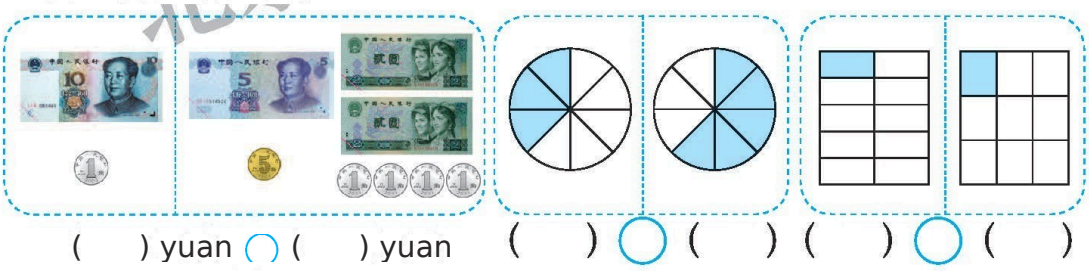
1. Write the numbers.



2. Write the following numbers.



3. Fill in and compare.



4. Find the pattern and fill in.

(1) 2085, 2090, 2095, ( ), ( ).

(2) 1200, 1100, 1000, ( ), ( ).

(3) $\frac{1}{10}$, $\frac{3}{10}$, $\frac{5}{10}$, ( ), ( ).

(4) 0.2, 0.4, 0.6, ( ), ( ).

5. Can you represent the number 127 in different ways?

6. On the tape measure, find how long 0.3 m is. What about 1.05 m?

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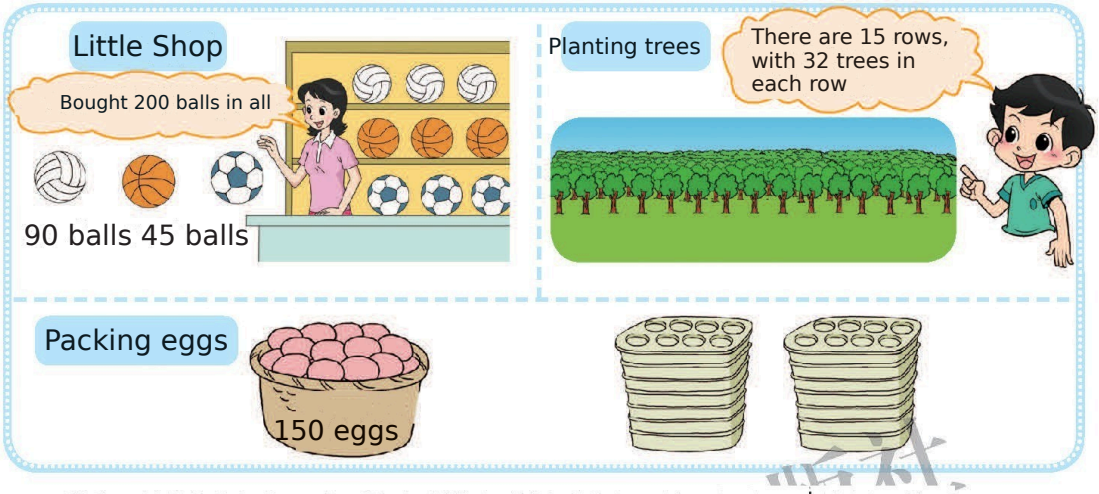
Number Operations

Review and Share

1. Which operations have we learned? Please give examples.

2.

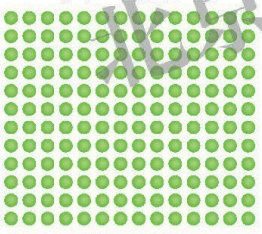


(1) What maths questions can you ask from the picture? Tell a partner.

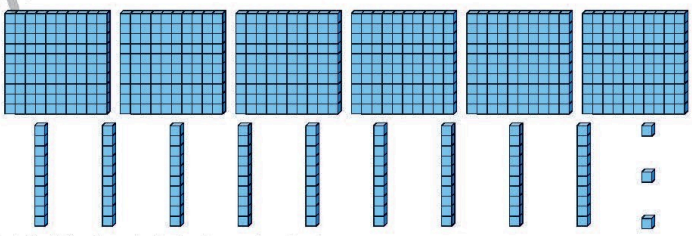
(2) In solving the problems, which operations did you use?

3. (1) Circle, calculate, and explain how you calculated.

$14 \times 12 =$



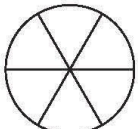
$693 \div 3 =$



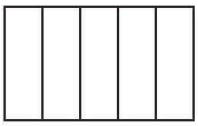
(2) Calculate and explain why the decimal points must be aligned.

yuan jiao	yuan jiao	yuan jiao	yuan jiao
3.2	7.8	19.2	5
+ 4.5	+ 12.6	− 7.2	− 1.9

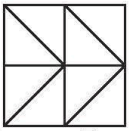
(3) Colour and calculate.



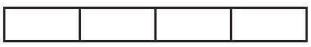
$\frac{1}{6} + \frac{2}{6} =$



$\frac{2}{5} + \frac{3}{5} =$



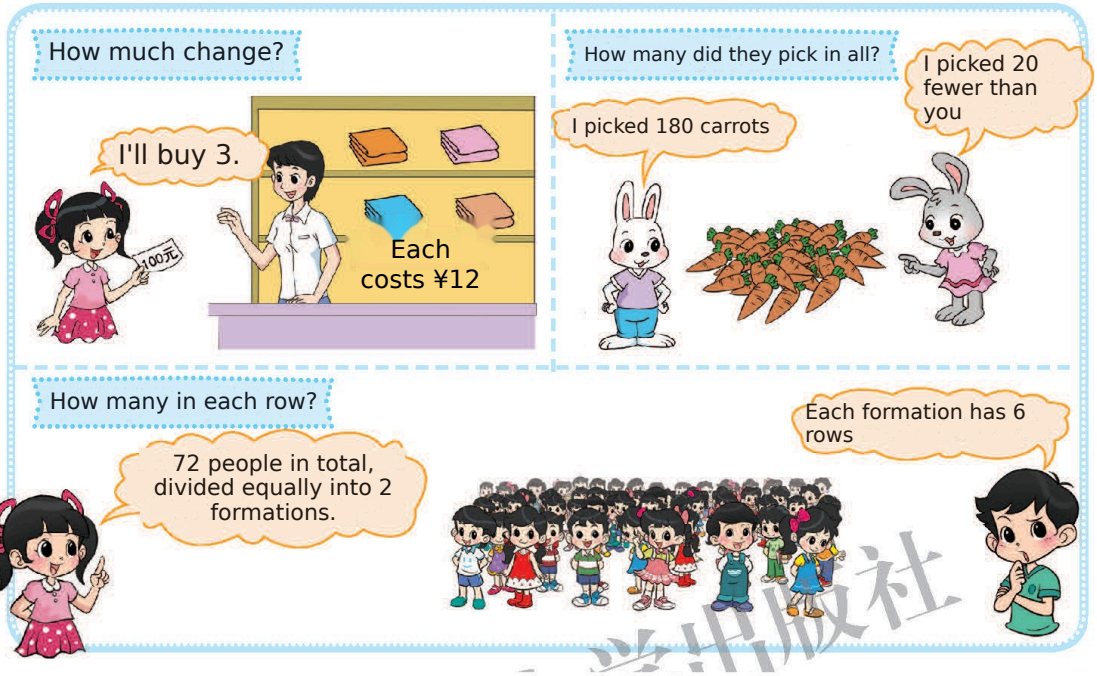
$\frac{7}{8} - \frac{5}{8} =$



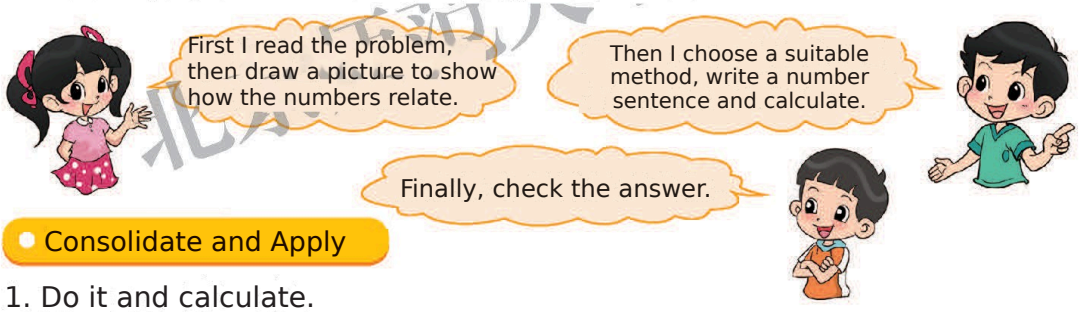
$1 - \frac{3}{4} =$

4. Collect the problems you often get wrong and say how to avoid these mistakes.

5. (1) First draw a diagram to understand the problem, then solve it.



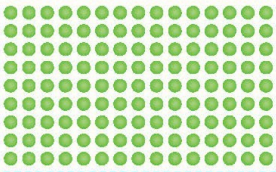
(2) Tell a partner how you solve real-life problems.



Consolidate and Apply

1. Do it and calculate.

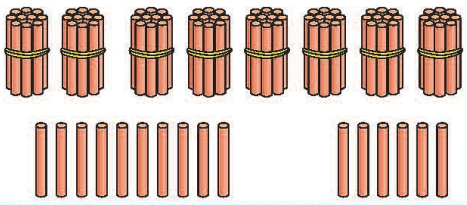
$15 \times 9 =$



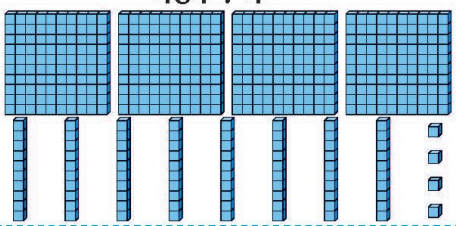
$28 \times 12 =$

×	20	8
10		
2		

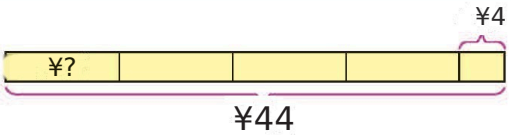
$96 \div 4 =$

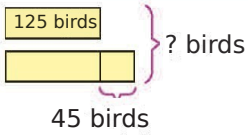


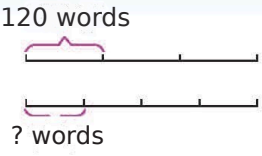
$484 \div 4 =$



2. Talk about it and calculate.







3. Write the answers directly.

$35+17$	20×5	$100-17$	15×5
$40-18$	$2000+400$	102×4	23×5
11×50	15×3	$609\div3$	$48\div3$
$84\div4$	$\frac{1}{8}+\frac{3}{8}$	$\frac{7}{10}-\frac{2}{10}$	$65\div5$

4.

$$\begin{array}{r} 904 \\ -\ 71 \\ \hline \end{array}$$

$$\begin{array}{r} 302 \\ \times\ \ 6 \\ \hline \end{array}$$

$$4\overline{)416}$$

$$\begin{array}{r} 53 \\ +396 \\ \hline \end{array}$$

$$\begin{array}{r} 43 \\ \times\ 18 \\ \hline \end{array}$$

$$6\overline{)312}$$

5.

yuan jiao	yuan jiao	yuan jiao	yuan jiao
$8.1+4.5$	$6.7-1.4$	$2.8+5.6$	$10-3.7$

6.

24×11	43×26	$328\div4$	$215\div5$
25×22	36×34	$432\div6$	$819\div9$
35×13	57×18	$872\div8$	$161\div7$

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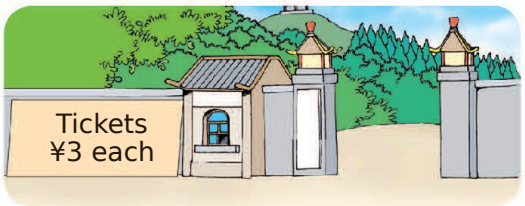
7. Calculate, and say what you work out first and what next.

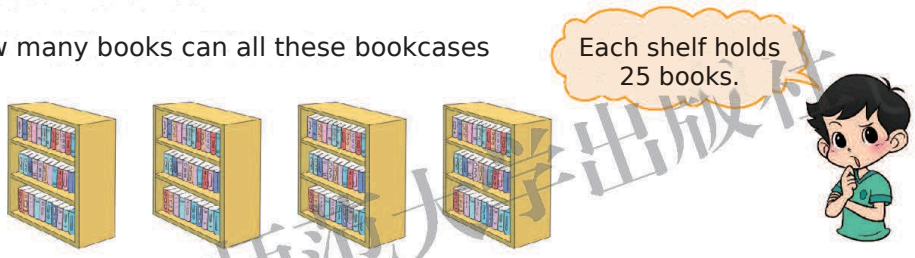
$240 \times 2 \div 6$	$840 \div 4 \times 3$	$128 + 18 \times 6$
$300 - 129 \div 3$	$(128 + 132) \div 5$	$312 \times (300 - 296)$

8. 97 students go to the park with ¥300. Is it enough for the tickets?



9. The school set up a "Red Scarf Charity Station". In the first week the pupils donated 56 storybooks. In the second week they donated 14 more than 2 times the first week's number. How many storybooks did they donate in the second week?

10. How many books can all these bookcases hold?



11. For a yo-yo competition, Mr Wang bought 4 boxes of yo-yos, 2 in each box, for ¥96 in total. On average, how much did each yo-yo cost?

12. How much do 7 pencil boxes cost?

13. Two teachers take 51 students boating in the park. Each big boat seats 6 people and costs ¥60 to rent; each small boat seats 4 people and costs ¥48. Work in groups: write at least 4 boat-rental plans and compare their costs.

	Big boats	Small boats	Seats	Rent (¥)
Plan 1				
Plan 2				
Plan 3				
Plan 4				

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Common Quantities

Review and Share

At the 2012 London Olympics, 16-year-old Ye Shiwen won the women's 400 m individual medley final in 4 min 28.43 s, breaking the world record.

Zhou Lulu broke the world record in the women's over 75 kg weightlifting final with a total of 333 kg, and won the gold medal.

Which quantities are in the information above? Which are units of mass, and which are...

Give examples of about how long 4 minutes is and about how heavy 300 kg is.

What other units of time, money and mass do you know? Give examples.

Consolidate and Apply

1. Write a suitable unit in each ( ).

(1) An elephant weighs about 2 ( ).

(2) A camel weighs about 420 ( ).

(3) An egg weighs about 55 ( ).

(4) Xiaoming's dad works 8 ( ) a day.

(5) Xiaojun ran 60 m in 10 ( ).

(6) A maths lesson lasts 40 ( ).

2. Fill in.

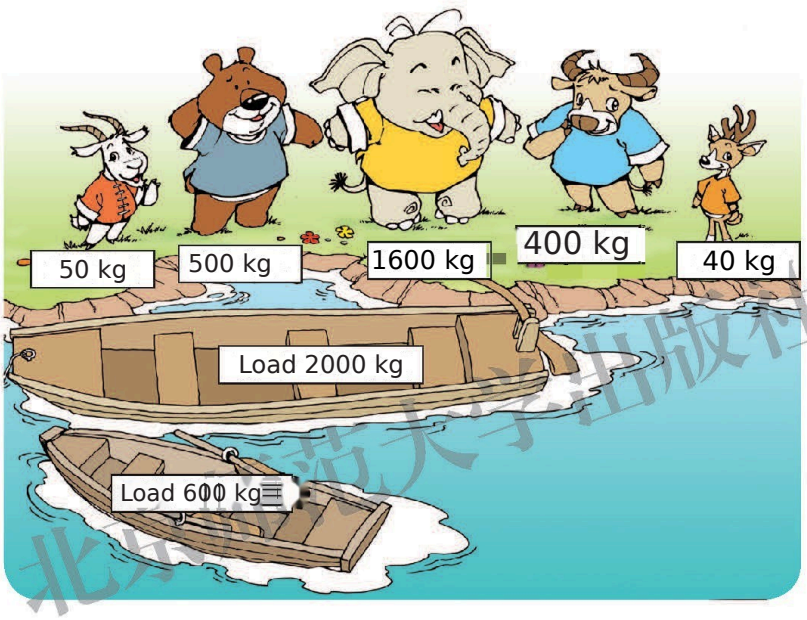
2 h = ( ) min	5 min = ( ) s
3 t = ( ) kg	4000 g = ( ) kg
30 jiao = ( ) yuan	6 jiao = ( ) fen

3. (1) This year, February has ___ days. It is a ___ (common, leap) year.

(2) My birthday is ___ (month) ___ (day). That month has ___ days.

(3) The Children's Day (1 June) show began at 20:00 and lasted 2 h 30 min. It ended at ___ h ___ min.

4. These 5 animals want to cross the river at the same time. Which boat should each take? Draw lines to match.



5. An adult female Tibetan antelope weighs about 25 kg, and a wild yak weighs about 20 times as much. About how many kilograms does a wild yak weigh? How many wild yaks together weigh about 1 tonne?

6. On 15/3/2012, Uncle Wang took a train from Beijing to Shenyang. Of the 3 trains below, he chose the one arriving first. Which did he take?

Train	Train type	From	Departs	To	Time taken
T 157	Express	Beijing	06 : 26	Shenyang North	5 h 55 min
D 21	Bullet train	Beijing	07 : 13	Shenyang North	4 h 48 min
D 29	Bullet train	Beijing	07 : 00	Shenyang North	4 h 46 min

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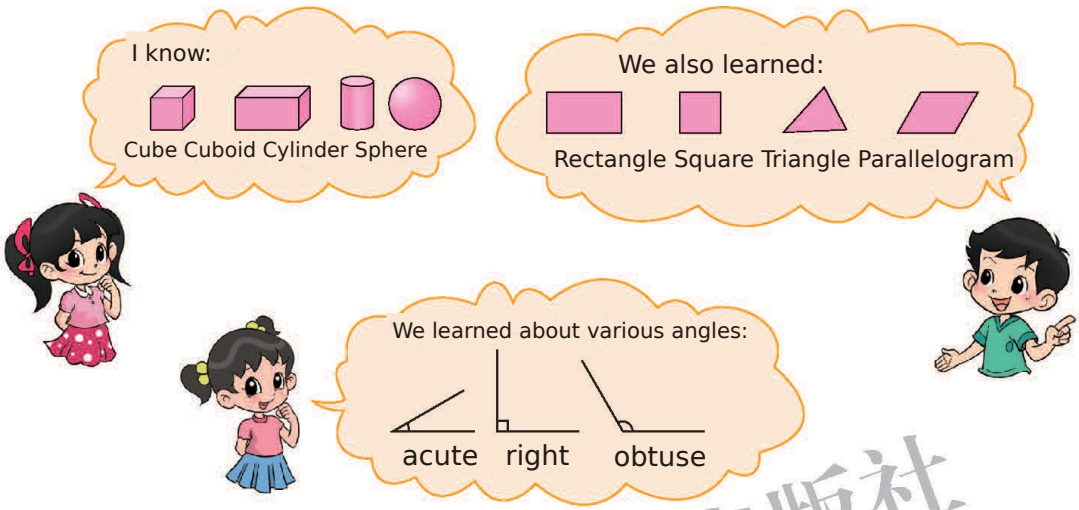
Understanding Shapes

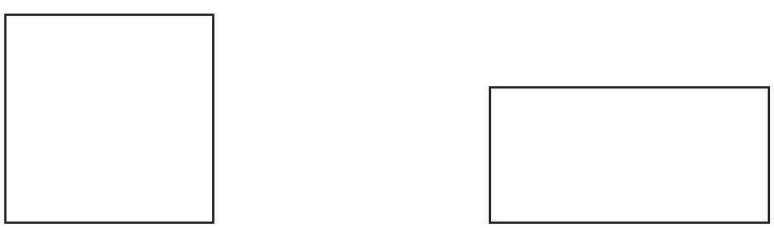
Review and Share

1. What shapes have we learned about? Talk about them with your partner.



2. Fold a right, an obtuse and an acute angle; check with a set square.

3. Which shape is a rectangle? Which is a square? How did you decide?



4. Choose an object, look at it from different positions, and say what you notice.

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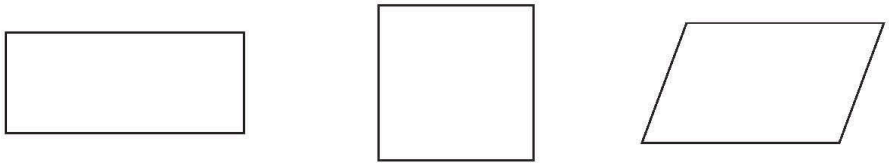
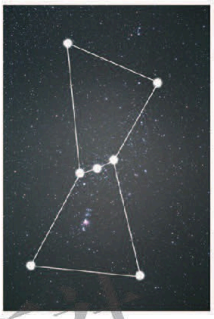
Consolidate and Apply

1. Find a pair of equal sides in each shape below and trace them in colour.



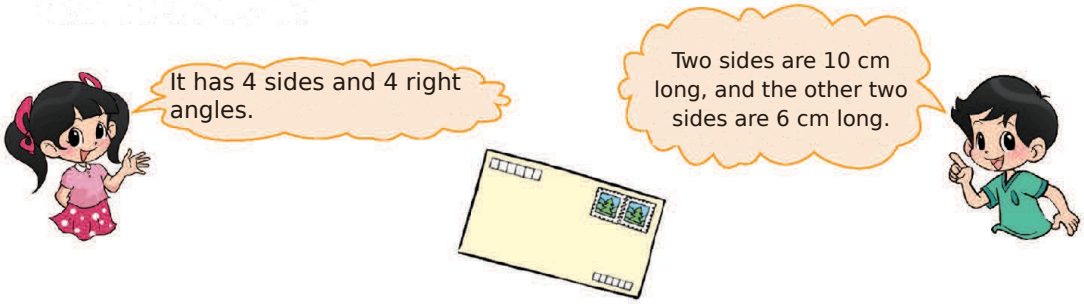
2. In the constellation on the right, trace 5 angles in red, then look for acute and obtuse angles among them.



3. Which chair was each photo below taken from? Draw lines to match.

4. Guess.

Taoqi and Xiaoxiao put a sheet of coloured paper in an envelope. From the dialogue below, can you tell what shape the paper is?



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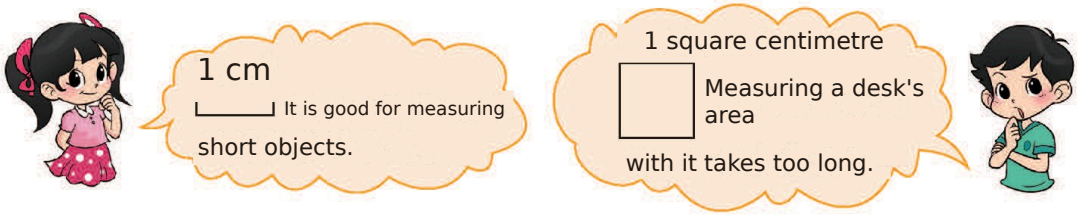
Shapes and Measurement

Review and Share

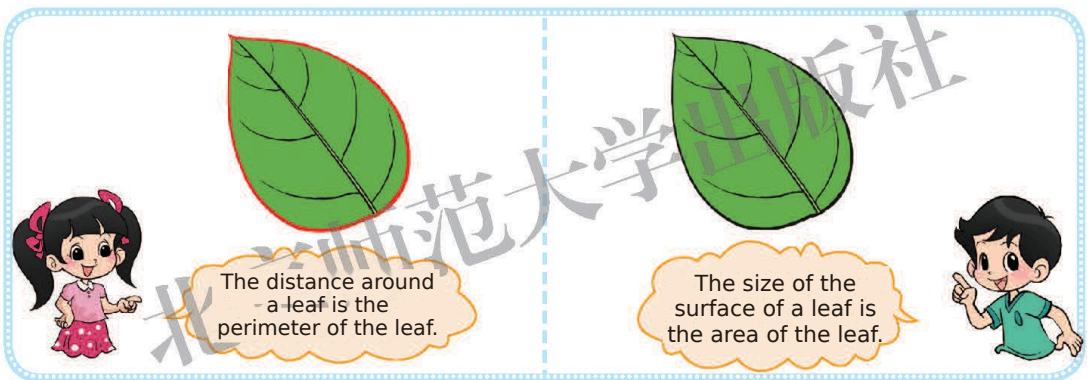
1. Which units of length and which units of area have we learned? Sort them out and talk about them with a partner.

km 1000→ m 10→ dm 10→ cm 10→ mm

m^2 100→ dm^2 100→ cm^2



2. Give examples to explain what perimeter is and what area is.



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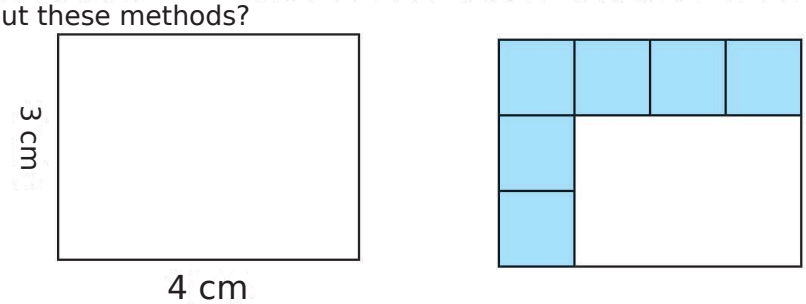
3. How do we find the perimeter and area of a rectangle and a square? How did we work out these methods?



4. Try it: how big is 1 square centimetre? 1 square decimetre? 1 square metre?

Consolidate and Apply

1. Fill in ( ) with a suitable unit.

(1) A big tree is about 8 ( ) tall.

(2) The area of a stamp is 6 ( ).

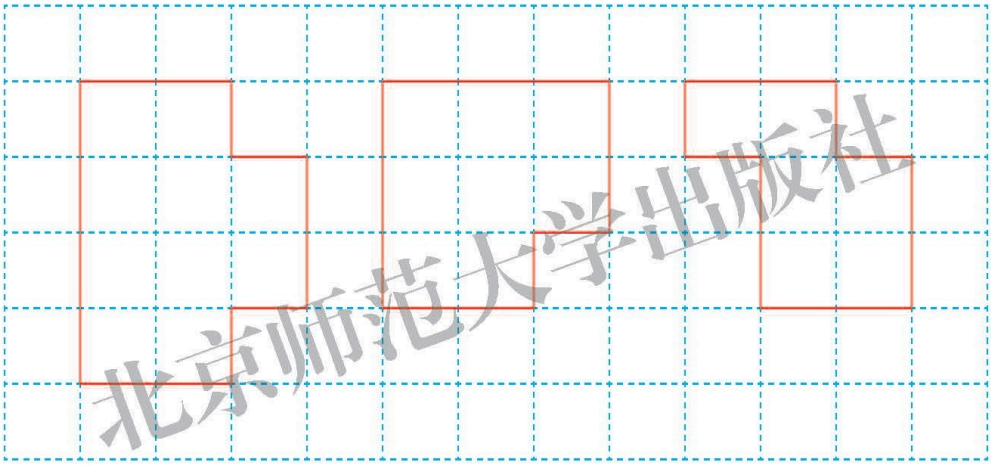
(3) The area of the classroom is 55 ( ).

(4) A pen is 15 ( ) long.

(5) The area of a maths book cover is about 5 ( ).

(6) The railway from Beijing to Xi'an is about 1200 ( ) long.

2. What are the perimeter and area of the figures below? (Each small square has side length 1 cm)



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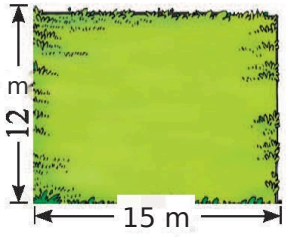
3.

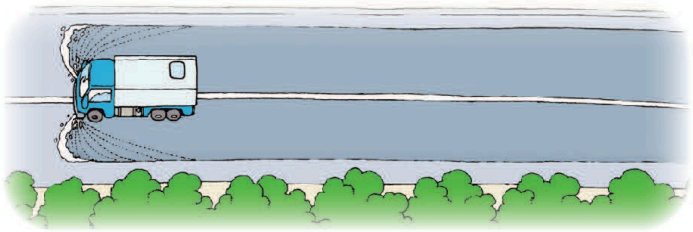
9 m² = ( ) dm²	2000 cm² = ( ) dm²
920 cm = ( ) dm	7 dm² = ( ) cm²
5 m = ( ) dm	400 dm² = ( ) m²

4.



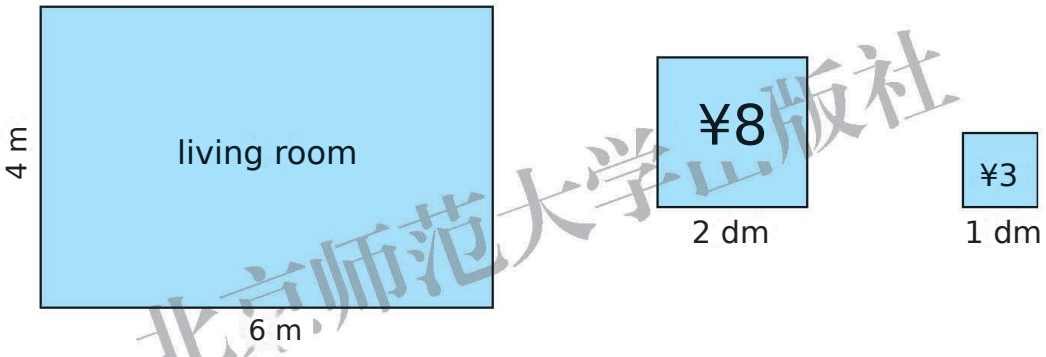
What is the area of this lawn in square metres? A fence is put around the lawn. How many metres long is the fence?

5. A water truck travels 150 m per minute and the sprinkling width is 8 m. If the truck travels for 7 minutes, what area of ground can it sprinkle?



6. A pavement is 27 m long and 3 m wide. Square tiles of area 9 square decimetres are used to pave it. How many such tiles are needed?

7. Taoqi's family is going to tile the living room floor. Which tile is cheaper? How many of those tiles are needed?



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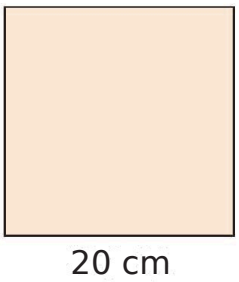
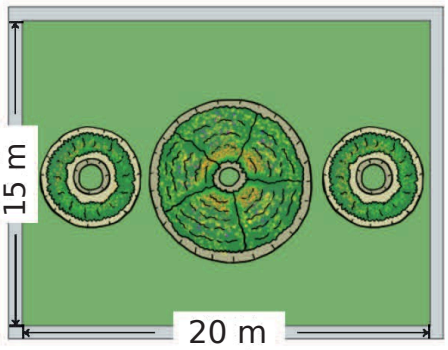
8. Cut a square of paper with side length 20 cm into 4 identical small squares. What are the perimeter and area of each small square?



9. A narrow path 1 m wide is laid around a rectangular flower bed.

(1) How many square metres is the flower bed?

(2) How many square metres is the path?

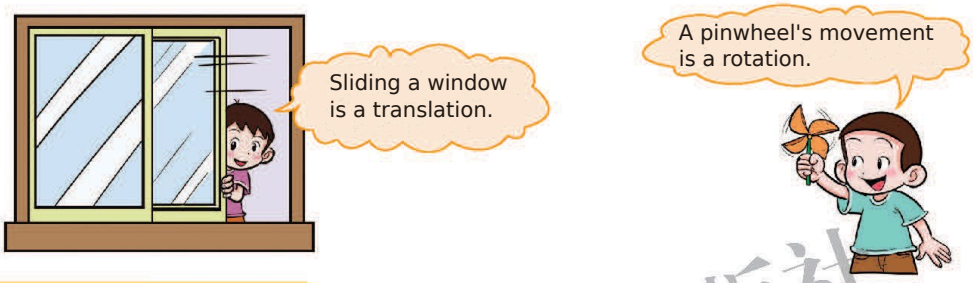


Motion of Shapes

Review and Share

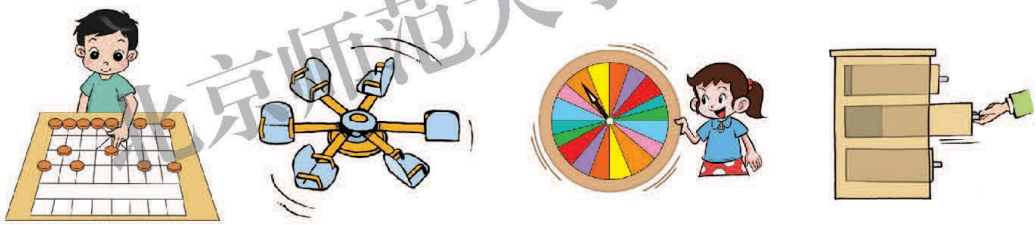
1. Give examples to explain what a symmetrical figure is. Which of the figures below are symmetrical?

2. Give some examples of translations and rotations.

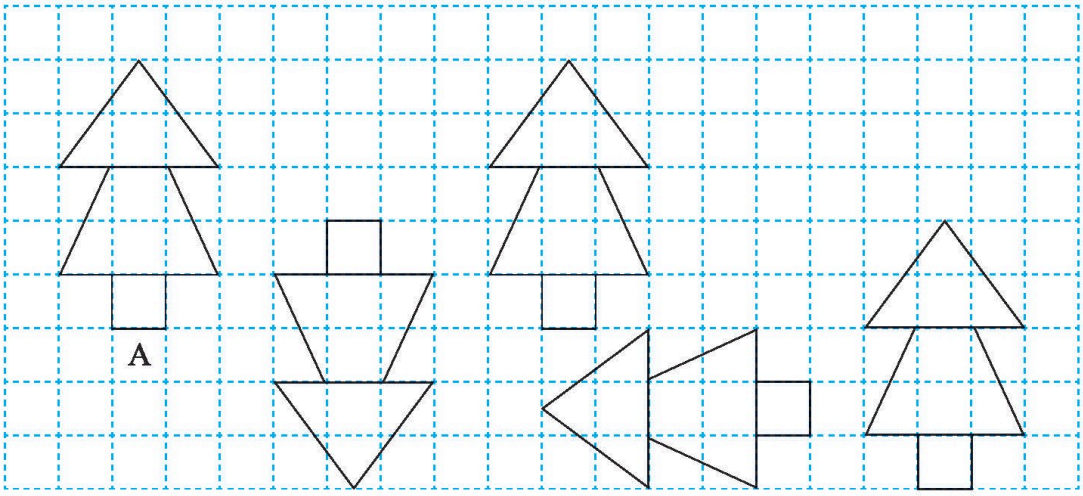


Consolidate and Apply

1. Which movements below are translations, and which are rotations?



2. Which figures can be translated to fit exactly onto figure A? Colour them in any colour you like.



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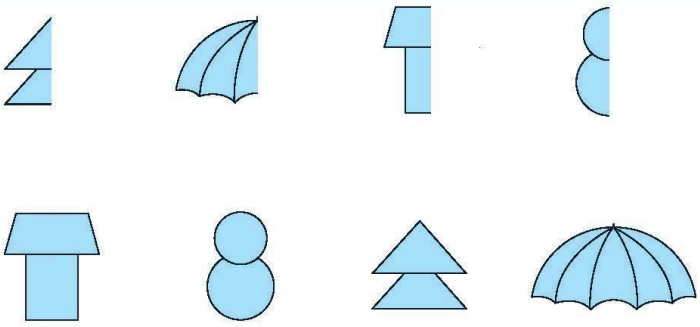
3. Match.

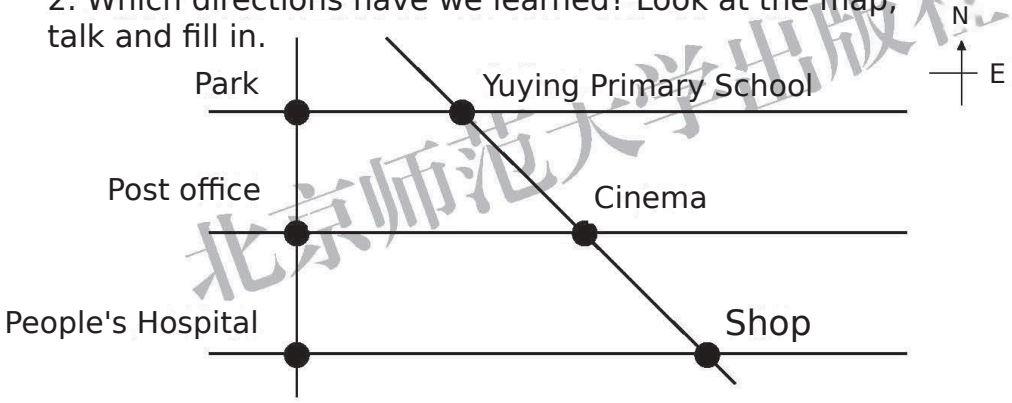


Shapes and Position

Review and Share

1. In the classroom, use above, below, in front, behind, left and right to say where things are in relation to each other. On the playground, find east, south, west and north.

2. Which directions have we learned? Look at the map, talk and fill in.



The park is to the _____ of the post office, the cinema is to the _____ of the shop, and the shop is to the _____ of Yuying Primary School.

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Consolidate and Apply

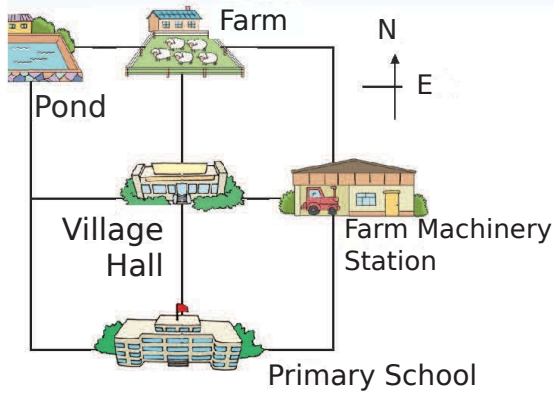
1. Look and fill in.

(1) Taoqi is on Xiaoxiao's ( ) side;
Xiaoxiao is on Taoqi's ( ) side.
(2) The book is ( ) the pencil case.

2. Look and fill in.

(1) The farm is ( ) of the village hall.
(2) The machinery station is ( ) of the village hall.
(3) The fish pond is ( ) of the village hall.

(4) The farm is ( ) of the primary school.



Statistics

Review and Share

1. Choose different ways to divide the whole class into two groups, and say how you divided them.

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2. In groups, choose one way to sort the class and record the results.

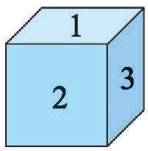
Jan–Jun	○○○○○○○○○○○○
Jul–Dec	○○○○○○○○

≥ 140 cm	正正正
< 140 cm	正丅

3. Say what you found out from the results.

Consolidate and Apply

1. The cube on the right has 6 faces, marked 1, 2, 3, 4, 5 and 6.



(1) Toss it 20 times and record the number that faces up each time.

Toss	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Number																				

(2) Summarise the results recorded by the whole class.

Number	1	2	3	4	5	6
Times						

(3) Observe how many times each number appears in the table above. What do you find?

2. These are the scores of the boys in Class 3 (1) in a 1-minute skipping test.

No.	Score	No.	Score	No.	Score	No.	Score	No.	Score
1	186	6	161	11	135	16	116	21	109
2	157	7	114	12	186	17	85	22	175
3	160	8	91	13	180	18	120	23	95
4	107	9	91	14	89	19	149	24	119
5	126	10	184	15	185	20	120	25	131

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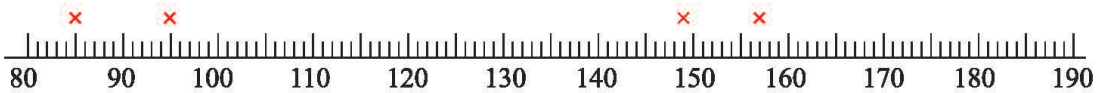
(1) Complete the chart below using the table above.



(2) Which boy skipped the most in 1 minute? How many skips?

(3) The school's pass mark for the 1-minute skipping test is 110 skips.

( ) boys passed, or $\frac{(\ )}{(\ )}$ of all the boys.

What Did You Learn This Term?

1. _____

2. _____

3. _____

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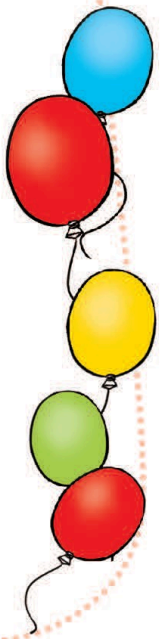

Problem Bank

What maths problems have you come across in everyday life and at school? Write them down. Can you solve them?

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Appendix 1

Figure 1

Figure 2

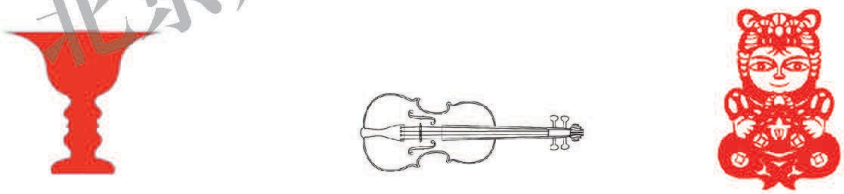

Figure 3

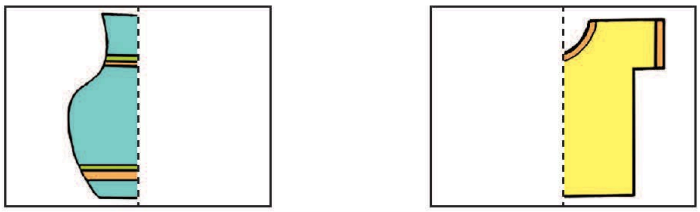

Figure 4

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Appendix 2

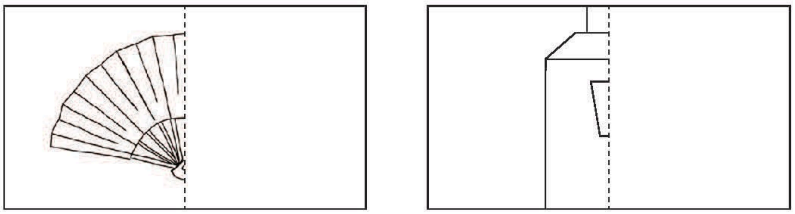

Figure 1

Figure 2

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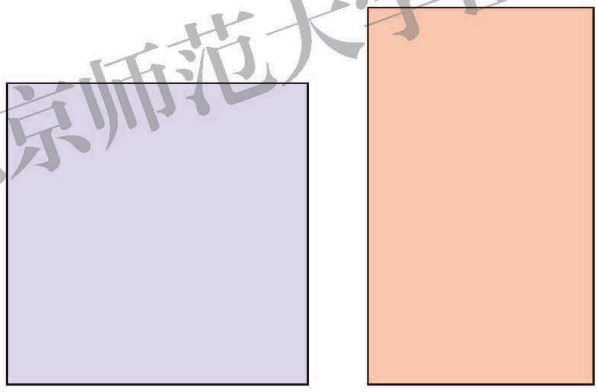

Figure 3

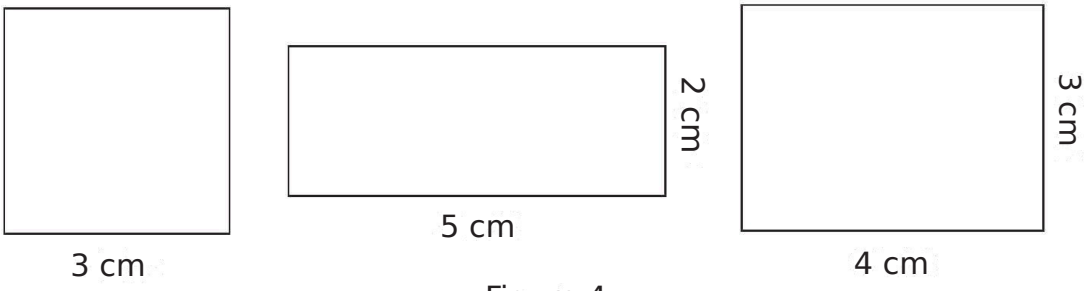

Figure 4

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Appendix 3

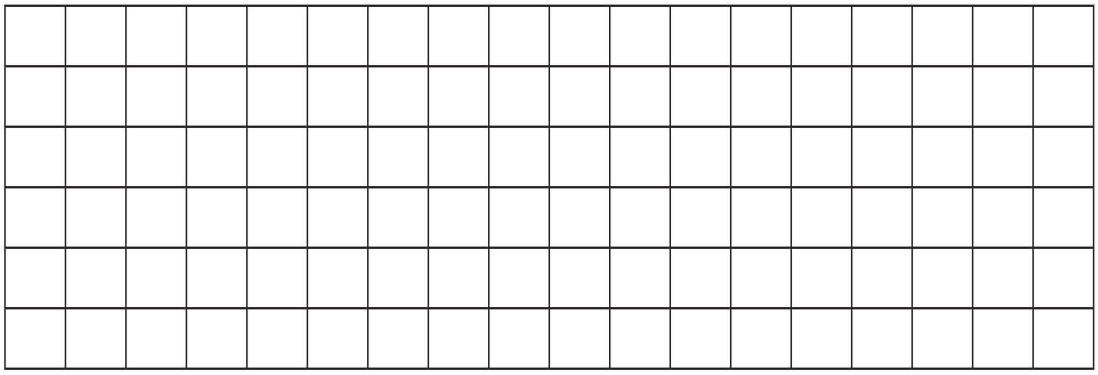

Figure 1

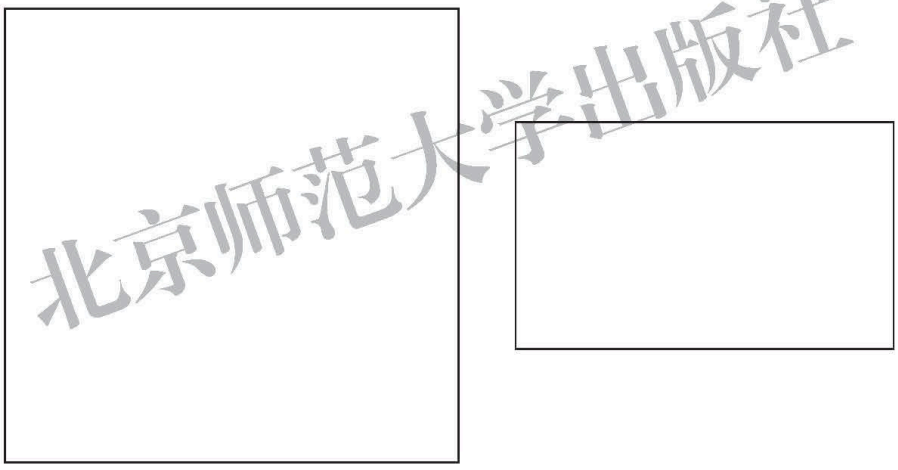

Figure 2

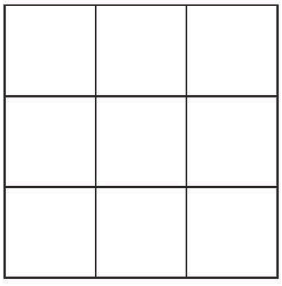

Figure 3

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Afterword

The Compulsory Education Textbooks (Beijing Normal University Edition) were written by many heads and core members of the groups that drew up the national curriculum standards for basic education, together with subject experts, education experts, psychologists and master teachers. They rest on solid research and advanced educational ideas, and offer high-quality writing and professional service. The textbooks strive to reflect the spirit of the national curriculum standards for basic education, make use of many kinds of information resources, show the latest progress in each subject where appropriate, and stress the use of knowledge, skills and ways of thinking in real life. They stay close to students' lives, pay attention to how students learn, meet their different learning needs and promote the all-round development of every student.

The Compulsory Education Textbooks (Beijing Normal University Edition) · Mathematics (Grades 1–6), also known as the New Century Primary School Mathematics Textbooks, fully embody the basic ideas of the new century's reform of the mathematics curriculum, with the healthy growth of children as their fundamental aim. This is the 4th edition of the New Century Primary School Mathematics Textbooks. Building on the strengths of the first three editions, it has been fully revised in accordance with the Mathematics Curriculum Standards for Compulsory Education (2011 Edition).

The new edition has the following distinctive features. Its basic form of presentation is "a situation plus a series of questions", which seeks to make the unfolding of the content, the students' learning, the teachers' teaching and the reaching of the curriculum goals one whole, good for learning and for teaching, so that students keep going through the process of thinking a problem through "from beginning to end". It seeks to give students realistic, interesting and challenging learning materials, to give them time and space to explore and to share their ideas, to show how mathematical knowledge is formed and used, and to meet the needs of different students.

The aim is for students to: appreciate the close links between mathematics, nature and human society; gain the basic knowledge, basic skills, basic ideas and basic experience of activities that they need and that suit their age; develop the ability to find and pose problems and to analyse and solve them; understand the value of mathematics, become more interested in learning it and more confident of learning it well; and gain a first sense of innovation and a scientific attitude.

The chief editors of the New Century Primary School Mathematics Textbooks are Liu Jian, Kong Qiping and Zhang Dan. The other members of the writing group are Hu Guangti, Liu Keqin, Wang Yong, Tao Wenzhong, Wang Mingming, Zhu Dejiang, Chen Xiaomei, Zhang Hong, Qian Shouwang, He Fengbo, Hua Yinglong, Huang Lihua, Zhao Yanhui, Li Tie'an, Wei Huinü, Zhu Yuhong, Ren Jingye and Hou Huiying.

The chief editors of this volume are He Fengbo and Chen Xiaomei. The main people who took part in revising and finalising it were Hou Huiying, Zhu Dejiang, Huang Lihua, Wang Mingming and Xiao Xiaoyu; the final text was compiled and approved by Liu Jian and Zhang Dan. Many local teaching researchers and front-line teachers gave valuable advice on this new round of revision.

We hope that teachers and students will send us their valuable comments as they use the book, so that we can revise and improve it further.

You are welcome to contact us by phone or letter: Primary School Mathematics Editorial Office, Beijing Normal University Press (100088), (010) 58802836, xxsx@bnupg.com; Mathematics Studio, Basic Education Curriculum Research Centre, Beijing Normal University (100082), (010) 58435911, xsjmath@126.com.

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